Spectral Regularization of Wasserstein Flows

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Abstract We introduce a unified family of spectrally regularized Wasserstein gradient flows, called *SpMMD flows*, parametrized by a qualification function φ from inverse problem theory and interpolating between the Maximum Mean Discrepancy (MMD) and the χ^2-divergence. The family subsumes Tikhonov, Showalter, spectral cutoff, and Landweber regularizations as special cases. Beyond the core Wasserstein geometry, we develop three further flow families that differ in their proximal structure while sharing the same spectral witness function: *Fisher–Rao* (FR) SpMMD flows, which reweight particle masses instead of transporting them and converge at the log-Sobolev rate $e^{-2\alpha(1+\lambda)T}$; *underdamped kinetic* SpMMD flows, which introduce momentum degrees of freedom and attain the square-root-improved asymptotic rate $e^{-2\sqrt{(1+\lambda)/C_P}\,n}$, a $\sqrt{C_P}$-factor gain over the overdamped flow for ill-conditioned targets; and *Bregman mirror* SpMMD flows, encompassing χ^2-proximal linearization and Sinkhorn proximal steps. A unified comparison table with six flow geometries is provided.

For the core Wasserstein flow we prove: (i) near-global KL convergence at an exponential rate under the Poincaré inequality, with an explicit $O(\lambda^{r\wedge\nu})$ approximation barrier; (ii) uniform-in-time propagation of chaos with polynomial sample complexity, eliminating the double-exponential complexity of prior results; and (iii) an $O(1/n^2)$ pre-asymptotic rate via Nesterov momentum in Wasserstein space. We then generalize the entire convergence theory to the family of *f-divergences* [28]: for any convex generator f satisfying an f-Sobolev inequality with constant H_f, the SpMMD flow satisfies $D_f(\mu_T\|\pi) \leq e^{-2c_f(1+\lambda)H_f T} D_f(\mu_0\|\pi) + B_f\lambda^{r\wedge\nu}$ where $c_f = \inf_t f''(t)$; crucially, the barrier exponent $r\wedge\nu$ is *universal across all f*—only the rate prefactor varies. This yields Hellinger-SpMMD convergence for heavy-tailed targets where $\mathrm{KL} = +\infty$, Rényi-SpMMD for robust generative modeling, and a Mode-A/B duality for all f-divergences with a shared information-geometric interpretation via the spectral filter as a mixture geodesic transport. We identify a Mode-A/Mode-B duality unifying SpMMD with regularized Stein variational gradient descent. Nine numerical experiments on JAX/TPU hardware validate all principal predictions, including a $100\times$ iteration-count advantage of FR flows when the initial distribution already covers the target, the $\sqrt{C_P} \approx 3.16\times$ kinetic speedup on an ill-conditioned Gaussian, and the variance-reduction benefit of Sinkhorn proximal steps for small particle counts.

Finally, we develop a complete categorical account of the framework: qualification functions form a topological category **QualFun** whose colimit is the χ^2-flow; the assignment $\varphi \mapsto \mathrm{SpMMD}_\varphi$ extends to a functor Sp_π that admits a monadic structure; SpMMD flows constitute a quasi-category $\mathscr{W}_{\infty,1}$ of Wasserstein trajectories with inner-horn-filling; the spectral filtration by qualification order is a Postnikov tower in $\mathscr{W}_{\infty,1}$ whose convergence is $(r\wedge m)$-connected; iterated regularization produces a complete Segal Θ_n-space and realizes the framework as an (∞,n)-categorical structure; and the composition of flows is encoded by an ∞-operad $\mathscr{O}_{\mathrm{SpMMD}}$ with a Boardman–Vogt tensor product governing joint flows on product spaces.

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1 Introduction

Approximating a target probability distribution π from samples is a central problem in statistics and machine learning, arising in Bayesian inference, generative modeling, and Monte Carlo

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estimation [61, 69, 2]. A natural and principled approach is to model the task as the minimization of a dissimilarity functional $F(\cdot) = D(\cdot\|\pi)$ over the Wasserstein-2 space $\mathscr{P}_2(\mathbb{R}^d)$ and to carry out this minimization by following the *Wasserstein gradient flow* (WGF) of F [43, 63].

A fundamental tension underlies this approach. Divergences with strong geodesic-convexity properties—and hence strong convergence guarantees—are hard to estimate consistently from samples, while easily computable kernel-based distances such as the Maximum Mean Discrepancy (MMD) [40] lack geodesic convexity and have correspondingly weaker convergence theory. The recently proposed *(De)-regularized Maximum Mean Discrepancy* (DrMMD) gradient flow [22] is a landmark step toward resolving this tension: by applying Tikhonov spectral regularization to the integral operator of the target, the DrMMD constructs a kernel-based divergence that is computable from samples and whose Wasserstein gradient flow converges near-globally for targets satisfying a Poincaré inequality. However, the DrMMD is a single point in a much larger landscape that this paper sets out to map.

The broader landscape. Three additional axes of variation are largely unexplored in the literature, each with significant practical and theoretical consequences.

First, the spectral filter. Tikhonov regularization is one member of a parametric family indexed by *qualification functions* from regularization theory for ill-posed inverse problems [33, 52]. Higher-order Showalter filters improve the $O(\lambda^r)$ approximation barrier to $O(\lambda^{r\wedge\nu})$ for any qualification order ν, while spectral cutoff and Landweber iteration offer other tradeoffs between smoothness and computational tractability. The DrMMD addresses only the $\nu = 1$ case.

Second, the flow geometry. Even for a fixed spectral filter, there are fundamentally different ways to construct a gradient flow that drives the particle cloud toward π, corresponding to different choices of the proximal metric on probability space. The Wasserstein-2 geometry encodes spatial transport, making it powerful when the initial distribution and target have disjoint or distant supports, but slow when the initial distribution already covers the target. The Fisher–Rao metric [46] instead reweights particle masses without moving their positions, converging at the log-Sobolev rate rather than the Poincaré rate. The hybrid Wasserstein–Fisher–Rao (WFR) metric simultaneously transports and reweights, achieving an additive combination of both rates. Underdamped Langevin dynamics [23, 16] introduce kinetic energy to break the condition-number barrier, improving the asymptotic exponential rate by a factor of $\sqrt{C_P}$ over the overdamped (Wasserstein) flow. And Bregman mirror descents with χ^2-proximal or Sinkhorn proximal steps provide linearized or entropy-regularized alternatives with distinct finite-sample stability properties. The TILOS talk by Sriperumbudur [66] draws attention to a further dual mode—regularizing the Wasserstein geometry while keeping the KL objective exact—which recovers regularized Stein variational gradient descent [41] as a special case.

Third, the divergence objective. All prior work on DrMMD and SpMMD flows measures convergence in the KL divergence [43]. Yet the choice of divergence is a continuous degree of freedom: the Csiszár f-divergence family [28] parametrizes a rich spectrum from KL through Hellinger to total variation and Rényi, each with different sensitivity to tail behavior, different curvature, and a different associated functional inequality. Three aspects are unexplored: (a) for

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heavy-tailed targets, $\mathrm{KL}(\mu_0 \| \pi) = +\infty$ for common initializations, while $H^2(\mu_0, \pi) \leq 2$ always [14]; (b) Rényi divergences provide robustness to model misspecification [34]; and (c) the Mode-A/B duality extends naturally to all f via the information-geometric interpretation of the spectral filter [1]. Crucially, the spectral approximation barrier $O(\lambda^{r \wedge \nu})$ is *independent of* f: only the exponential rate $2c_f(1+\lambda)H_f$ changes across generators.

Fourth, the categorical structure. The SpMMD family has a rich algebraic and topological organization that is invisible from the single-instance perspective. Qualification functions naturally form a topological category; the assignment of divergences to filters is a functor with a monadic structure capturing the adaptive regularization schedule; and the space of flow trajectories parametrized by qualification functions organizes into a quasi-category, a Postnikov tower, a Segal Θ_n-space, and an ∞-operad. These structures are not mere formal decoration: they identify the colimit of the Showalter tower as the χ^2-flow, the convergence rate as a connectivity statement on mapping spaces, and the composition of flows as an algebra over an operad, all in a unified framework that makes the relationships between all methods logically transparent.

This paper. We develop the full landscape along all four axes simultaneously. Starting from the DrMMD as a single anchor point, we introduce and analyze the complete SpMMD family, three new flow geometries, the f-divergence generalization, and the categorical structure that organizes them, supported throughout by nine JAX/TPU experiments that validate the principal theoretical predictions.

Contributions

C1. **Unified spectral regularization framework.** We introduce the family of *spectrally regularized MMD divergences* SpMMD_φ, parametrized by a qualification function φ, subsisting Tikhonov (DrMMD), Showalter of all orders, Landweber iteration, and spectral cutoff. We prove that every member metrizes weak convergence, interpolates between MMD and χ^2-divergence, and yields a well-defined Wasserstein gradient flow with an explicit modified kernel $\widetilde{k}_\varphi$.

C2. **Near-global convergence, discrete-time analysis, and adaptive schedules.** We prove KL convergence bounds in both continuous and discrete time under the Poincaré inequality, with approximation barriers $O(\lambda^{r \wedge \nu})$ that improve over the $O(\lambda^r)$ barrier of Chen et al. [22] whenever $\nu > 1$. We provide adaptive regularization schedules achieving global KL convergence with an $O(\gamma^{(r \wedge \nu)/(r \wedge \nu + 1)})$ residual.

C3. **Mode-A/Mode-B duality.** We formalize the dual mode of spectral regularization—fixing the KL objective and preconditioning the Wasserstein geometry via $\varphi(T_\pi; \lambda)$—and prove that Mode A (SpMMD) and Mode B (regularized SVGD) apply the same spectral filter to dual velocity fields, establishing a precise 2-natural equivalence $\Phi_A \simeq \Phi_B$ in the regularization 2-category **Reg**.

C4. **Uniform-in-time propagation of chaos.** We prove that the particle implementation of SpMMD_φ descent achieves $E[W_2(\hat{\mu}_n, \pi)] \leq C_{\mathrm{mix}} e^{-\alpha n} W_2(\mu_0, \pi) + \varepsilon$ uniformly in n,

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with sample complexity $M, N = O(\varepsilon^{-2}/\lambda_{\min}^2)$ polynomial in $1/\varepsilon$, eliminating the double-exponential complexity of prior results.

C5. **Nesterov momentum in Wasserstein space.** We propose momentum-augmented SpMMD descent and prove an $O(1/n^2)$ pre-asymptotic convergence rate on the SpMMD functional via Nesterov acceleration in Wasserstein space, improving over the $O(e^{-n})$ overdamped rate in the pre-asymptotic regime for moderately convex objectives.

C6. **Deep-kernel theory.** We provide the first convergence analysis of the SpMMD_φ flow with data-dependent deep neural network kernels, proving an $O(\varepsilon_n^{1/2})$ additive kernel-misspecification penalty propagated from kernel learning error into transport error.

C7. **f-Divergence generalization.** We generalize the KL-divergence convergence theory throughout the paper to the full family of f-divergences [28], introducing the f-Sobolev inequality and f-Fisher information as the natural functional inequalities, and proving:

- *f-divergence continuous-time convergence* (Theorem 4.5): $D_f(\mu_T\|\pi) \leq e^{-2c_f(1+\lambda)H_f T} D_f(\mu_0\|\pi) + B_f\lambda^{r\wedge\nu}$, where $c_f = \inf_t f''(t)$ and H_f is the f-Sobolev constant.
- *f-divergence FR convergence* (Theorem 8.6): the FR-SpMMD flow converges in D_f at the f-Sobolev rate, enabling Hellinger convergence for heavy-tailed targets where $\mathrm{KL} = +\infty$.
- *f-divergence Mode-A/B duality* (Proposition 5.4): the spectral filter $\varphi_T(T_\pi;\lambda)$ is applied identically to the velocity field for all f, giving a universal duality.
- *Universal barrier*: the spectral approximation barrier $B_f\lambda^{r\wedge\nu}$ has the same exponent $r\wedge\nu$ for every f—only the prefactor varies.
- *f-divergence pre-asymptotic momentum* (Corollary 7.3), *f-divergence deep-kernel* (Corollary 8.2), *f-divergence uniform-in-time PoC* (Theorem 4.10), and applications to Rényi, Hellinger, and TV SpMMD (section 11.4).

C8. **Fisher–Rao, WFR, and entropic SpMMD flows.** We develop the Fisher–Rao SpMMD flow (reweighting, converging at the log-Sobolev rate $e^{-2\alpha(1+\lambda)T}$), the hybrid Wasserstein–Fisher–Rao (WFR) flow (converging at the additive rate $2(1+\lambda)(C_P^{-1}+\alpha\delta^{-2})$ for any $\delta > 0$), and the χ^2-proximal linearized flow. We prove that the Gibbs (Fisher–Rao) update and χ^2-proximal linearization are the same flow in continuous time (Proposition 10.2), and provide the joint convergence theorem for WFR-SpMMD (Theorem 8.9).

C9. **Underdamped kinetic SpMMD flow.** We introduce the kinetic (underdamped Langevin) SpMMD flow and prove that, at the critically-damped friction $\gamma^* = \sqrt{2(1+\lambda)/C_P}$, the asymptotic convergence rate is $e^{-2\sqrt{(1+\lambda)/C_P}\,n}$—a $\sqrt{C_P}$-factor improvement over the overdamped rate that is the dominant gain for ill-conditioned targets (Theorem 9.2).

C10. **Bregman mirror SpMMD flows.** We unify χ^2-proximal, Gibbs, and Sinkhorn proximal steps under a Bregman mirror descent framework on probability space, prove the $+O(\varepsilon)$

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Sinkhorn bias with $O(\lambda^{r\wedge\nu})$-matched asymptotic rate (Proposition 10.4), and provide a unified six-flow comparison table (Table 1).

C11. Categorical and higher-categorical structure. We develop the full categorical account of the SpMMD framework:

- The category **QualFun** of qualification functions (enriched over **Top**), with the Showalter tower as a directed system whose colimit is the spectral cutoff (Theorem 13.9).
- The qualification functor $\mathrm{Sp}_\pi : \mathbf{QualFun} \to \mathbf{Div}(\pi)$, admitting a monadic structure $\mathbb{T}_\lambda$ whose Kleisli category is the category of spectrally regularized divergences; the adaptive schedule implements a lax morphism of monads (Theorem 13.14).
- The WGF quasi-category $\mathscr{W}_{\infty,1}$ of Wasserstein trajectories, with the SpMMD family defining an $(\infty,1)$-functor from the nerve of **QualFun**; the spectral filtration is a Postnikov tower whose $(r\wedge m)$-connectivity encodes the approximation error of Theorem 4.1 (Theorems **??**–**??**).
- The 2-categorical regularization structure **Reg** making the Mode-A/Mode-B duality a 2-natural equivalence of 2-functors (Theorem 13.12).
- The globular set of iterated SpMMD flows forming a complete Segal Θ_n-space in the sense of Rezk [60], realizing iterated regularization as an (∞,n)-category (Theorem 15.8).
- An ∞-operad $\mathscr{O}_{\mathrm{SpMMD}}$ encoding flow composition with a Boardman–Vogt tensor product governing joint flows on product spaces, and the SpMMD convergence spectral sequence abutting to the cohomology of the χ^2-flow (Theorems 15.17–15.15).

Relation to prior work

The SpMMD framework extends the DrMMD of Chen et al. [22] by replacing the Tikhonov filter with the full qualification-function family and by introducing three new flow geometries. Our FR and WFR flows are related to the Wasserstein–Fisher–Rao metric of Liero et al. [46] and to mean-field Langevin dynamics [25]; we are the first to apply these geometries with a spectrally regularized witness function as the driving functional. The kinetic flow specializes the underdamped Langevin results of Cheng et al. [23] and Cao et al. [16] to the SpMMD setting, providing the first $\sqrt{C_P}$-rate result for kernel-based gradient flows. The Sinkhorn proximal step connects to the Sinkhorn divergence of Feydy et al. [35]. The f-divergence generalization is inspired by the f-divergence flows of Domingo-Enrich et al. [31] and the (f,Γ)-divergence framework of Birrell et al. [12]; our contribution is to show that all convergence results for the SpMMD family transfer uniformly to the f-divergence setting via the f-Sobolev inequality [57, 37], with the spectral approximation barrier remaining unchanged across all generators f. The information-geometric interpretation of the Mode-A/B duality builds on the Amari duality Amari [1]. The categorical development builds on the $(\infty,1)$-categorical machinery of Lurie [49, 50] and the ∞-operad theory of Lurie [51], applying them to the specific algebraic structure of spectral regularization for the first time.

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Organization

Section 2 recalls background on RKHS, MMD, χ^2-divergence, and Wasserstein flows; section 2.4 establishes the f-divergence background (Csiszár divergences, f-Sobolev inequality, f-Fisher information, f-Talagrand inequality). Section 3 introduces the SpMMD family: spectral filters, the modified kernel, and its properties. Section 4 presents the convergence theory in continuous and discrete time, including adaptive schedules and the full f-divergence generalization (sections 4.4 and 4.5). Section 5 develops the Mode-A/Mode-B duality and its f-divergence extension, including the information-geometric interpretation via the spectral filter as a mixture geodesic transport (section 5.4). Section 6 proves uniform-in-time propagation of chaos. Section 7 introduces Nesterov-accelerated SpMMD descent and its f-divergence pre-asymptotic rate corollary. Section 8 treats deep-kernel extensions and the f-divergence deep-kernel bound. Section 8.2 develops Fisher–Rao SpMMD flows, including the f-divergence FR convergence theorem (section 8.5.1) and the Hellinger convergence result for heavy-tailed targets. Section 9 introduces the underdamped kinetic SpMMD flow. Section 10 develops Bregman mirror SpMMD flows. Section 11 discusses applications including the f-divergence variants: Rényi-SpMMD, Hellinger-SpMMD, TV-SpMMD, and (f,Γ)-divergence flows. Section 12 reports nine numerical experiments. Section 13 concludes with open problems, including Open Problem (vii) on tight f-divergence barriers and (f,Γ)-SpMMD flows. Section 13 develops the 2-categorical regularization structure. Section 14 lifts the framework to the $(\infty,1)$-categorical setting. Section 15 develops the (∞,n)-categorical structure. Section 15.5 constructs the ∞-operad of flows. Proofs are collected in Section 16.

2 Background and Notation

2.1 Probability measures and Wasserstein geometry

Let $\mathscr{P}_2(\mathbb{R}^d) = \mathscr{P}_2(\mathbb{R}^d)$ denote the set of Borel probability measures on $\mathbb{R}^d$ with finite second moment. The Wasserstein-2 distance between $\mu,\pi \in \mathscr{P}_2(\mathbb{R}^d)$ is

$$W_2(\mu,\pi) \;=\; \left(\inf_{\Gamma\in\Pi(\mu,\pi)} \int \|x-y\|^2 \, d\Gamma(x,y)\right)^{1/2},$$

where $\Pi(\mu,\pi)$ is the set of couplings. The space $(\mathscr{P}_2(\mathbb{R}^d), W_2)$ is a complete metric space [69, 2].

The *Wasserstein gradient flow* (WGF) of a functional $F : \mathscr{P}_2(\mathbb{R}^d) \to \mathbb{R}$ is the curve $(\mu_t)_{t\geq 0}$ in $\mathscr{P}_2(\mathbb{R}^d)$ satisfying the continuity equation $\partial_t \mu_t + \nabla\cdot(\mu_t v_t) = 0$, where $v_t = -\nabla_W F|_{\mu_t}$ is the Wasserstein subdifferential of F at μ_t [2]. A functional F is *geodesically* Λ*-convex* if its Wasserstein Hessian satisfies $\langle \operatorname{Hess} F|_\mu \nabla\phi, \nabla\phi\rangle_{L^2(\mu)} \geq \Lambda \|\nabla\phi\|^2_{L^2(\mu)}$ for all $\phi \in C_c^\infty(\mathbb{R}^d)$.

2.2 Reproducing kernel Hilbert spaces and MMD

Fix a continuous, c_0-universal, positive semi-definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ with $\sup_x k(x,x) \leq K < \infty$ and denote by $\mathscr{H}$ its RKHS. We assume k is twice differentiable in the sense of Steinwart and Christmann [67, Def. 4.35], with $\sup_x \|\nabla_1 k(x,\cdot)\|_{\mathscr{H}^d} \leq \sqrt{K_{1d}}$ and $\sup_x \|H_1 k(x,\cdot)\|_{\mathscr{H}^{d\times d}} \leq$

$\sqrt{K_{2d}}$.

The integral operator $T_\pi : L^2(\pi) \to L^2(\pi)$ associated with $\pi \in \mathscr{P}_2(\mathbb{R}^d)$ is $T_\pi f = \int k(\cdot,x) f(x)\, d\pi(x)$, with Mercer decomposition $k(x,x') = \sum_{i\geq 1} \rho_i e_i(x) e_i(x')$ for $\{e_i\}$ orthonormal in $L^2(\pi)$ and eigenvalues $\rho_1 \geq \rho_2 \geq \cdots \geq 0$. The covariance operator $\Sigma_\pi : \mathscr{H} \to \mathscr{H}$ is $\Sigma_\pi = \iota_\pi^* \iota_\pi$ where $\iota_\pi : \mathscr{H} \to L^2(\pi)$ is the canonical injection.

The MMD is $\mathrm{MMD}(\mu\|\pi) = \left\|m_\mu - m_\pi\right\|_{\mathscr{H}}$, where $m_\mu = \int k(x,\cdot)\, d\mu(x)$. By the identity of Chen et al. [22]:

$$\mathrm{MMD}^2(\mu\|\pi) = \left\|T_\pi^{1/2}\left(\tfrac{d\mu}{d\pi} - 1\right)\right\|^2_{L^2(\pi)}, \qquad \chi^2(\mu\|\pi) = \left\|\tfrac{d\mu}{d\pi} - 1\right\|^2_{L^2(\pi)}. \tag{1}$$

These two identities make explicit the single difference between MMD and χ^2: the former applies the compact smoothing operator $T_\pi^{1/2}$ before computing the $L^2(\pi)$ norm, gaining computability at the cost of geodesic convexity.

2.3 The DrMMD and its gradient flow

Chen et al. [22] propose to interpolate between MMD and χ^2 using Tikhonov regularization. Define the *de-regularized operator* $A_\lambda = ((T_\pi + \lambda I)^{-1} T_\pi)^{1/2}$ for $\lambda > 0$. Then

$$\mathrm{DrMMD}(\mu\|\pi) \;=\; (1+\lambda)\left\|A_\lambda\left(\tfrac{d\mu}{d\pi} - 1\right)\right\|^2_{L^2(\pi)}, \tag{2}$$

which equals $(1+\lambda)\left\|(\Sigma_\pi + \lambda I)^{-1/2}(m_\mu - m_\pi)\right\|^2_{\mathscr{H}}$ and can be computed from samples via Gram matrices. The associated WGF $\partial_t \mu_t = \nabla \cdot (\mu_t (1+\lambda) \nabla h^*_{\mu_t,\pi})$ with witness function $h^*_{\mu,\pi} = 2(\Sigma_\pi + \lambda I)^{-1}(m_\mu - m_\pi)$ converges exponentially in KL divergence up to an $O(\lambda^r)$ barrier for targets satisfying a Poincaré inequality [22, Thm. 4.1].

2.4 f-Divergences and associated functional inequalities

The KL divergence $\mathrm{KL}(\mu\|\pi)$ belongs to the rich family of *f-divergences* [28, 34], which provide a unified framework for measuring dissimilarity between probability measures.

Definition 2.1 (f-divergence)**.** Let $f : (0,\infty) \to \mathbb{R}$ be a convex function with $f(1) = 0$. The *f-divergence* of μ from π is

$$D_f(\mu\|\pi) \;:=\; \int_{\mathbb{R}^d} f\left(\frac{d\mu}{d\pi}(x)\right) d\pi(x), \tag{3}$$

whenever $\mu \ll \pi$, and $+\infty$ otherwise.

The principal examples are:

- $f(t) = t\log t$: $D_f = \mathrm{KL}(\mu\|\pi)$ (forward KL / relative entropy).
- $f(t) = (t-1)^2/2$: $D_f = \frac{1}{2}\chi^2(\mu\|\pi)$.
- $f(t) = -\log t$: $D_f = \mathrm{KL}(\pi\|\mu)$ (reverse KL).

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- $f(t) = (\sqrt{t}-1)^2$: $D_f = H^2(\mu,\pi)$ (squared Hellinger).
- $f(t) = |t-1|$: $D_f = \mathrm{TV}(\mu,\pi)$ (total variation).
- $f(t) = t^\alpha - 1$ ($\alpha > 1$): $D_f = \frac{1}{\alpha-1}(\chi^2_\alpha(\mu\|\pi) - 1)$ (Rényi α-divergence) [34].

A key structural property: the derivative f' is the *generator* of the Wasserstein gradient flow of D_f, with velocity field $v_\mu = \nabla\big(f'(\frac{d\mu}{d\pi})\big)$ [43, 63].

The natural functional inequality for convergence of D_f-gradient flows is the *f-Sobolev inequality*.

Definition 2.2 (f-Sobolev inequality and f-Fisher information)**.** Define the *f-Fisher information* of μ relative to π as

$$\mathscr{I}^f_\pi(\mu) := \int_{\mathbb{R}^d} f''\left(\frac{d\mu}{d\pi}\right)\left|\nabla \log \frac{d\mu}{d\pi}\right|^2 d\mu = \int_{\mathbb{R}^d} f''\left(\frac{d\mu}{d\pi}\right)\frac{|\nabla(d\mu/d\pi)|^2}{d\mu/d\pi} d\pi. \tag{4}$$

We say π satisfies an *f-Sobolev inequality* with constant $H_f > 0$ if

$$D_f(\mu\|\pi) \le \frac{1}{H_f}\mathscr{I}^f_\pi(\mu) \quad \text{for all } \mu \ll \pi. \tag{5}$$

Remark 2.3 (Connections to standard inequalities). (i) For $f(t) = t\log t$: $\mathscr{I}^f_\pi(\mu) = I_\pi(\mu)$ is the standard relative Fisher information, and eq. (5) becomes the LSI $\mathrm{KL}(\mu\|\pi) \le (1/\alpha) I_\pi(\mu)$ with $H_f = \alpha$ [6].

(ii) For $f(t) = (t-1)^2/2$: $\mathscr{I}^f_\pi(\mu) = \frac{1}{2}\int |\nabla(\frac{d\mu}{d\pi} - 1)|^2 d\pi$ is (one half the) Sobolev seminorm, and eq. (5) is the Poincaré inequality $\chi^2(\mu\|\pi) \le (1/C_P)\int |\nabla(\frac{d\mu}{d\pi})|^2 d\pi$ with $H_f = C_P$ [39].

(iii) For $f(t) = (\sqrt{t}-1)^2$: $\mathscr{I}^f_\pi(\mu) = \frac{1}{4} I^{\mathrm{H}}_\pi(\mu)$ is the Hellinger–Fisher information, and eq. (5) gives a Hellinger Poincaré inequality; see Bobkov and Ledoux [14].

(iv) The *HWI inequality* [57] provides a bridge: $\mathrm{KL}(\mu\|\pi) \le W_2(\mu,\pi)\sqrt{I_\pi(\mu)} - \kappa W_2^2(\mu,\pi)/2$, interpolating between the Talagrand-2 inequality and the LSI.

The following *f-Talagrand inequality* will be used in the f-divergence propagation-of-chaos bound (theorem 4.10).

Definition 2.4 (f-Talagrand inequality)**.** We say π satisfies an *f-Talagrand-q inequality* ($q \ge 1$) with constant $C_{T,f,q} > 0$ if

$$W_q(\mu,\pi) \le \sqrt{\frac{2}{C_{T,f,q}} D_f(\mu\|\pi)} \quad \text{for all } \mu \ll \pi. \tag{6}$$

For $f(t) = t\log t$ and $q = 2$, this is the standard Talagrand-2 inequality $W_2(\mu,\pi)^2 \le (2/C_T)\mathrm{KL}(\mu\|\pi)$ [57].

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3 Spectrally Regularized MMD: A Unified Family

3.1 Qualification functions and spectral regularization

The key operator in eq. (2) is a Tikhonov regularization of the identity I, defined via the spectral function $\varphi_{\mathrm{T}}(s;\lambda) = s/(s+\lambda)$. A broader perspective comes from regularization theory for ill-posed inverse problems [33, 52], where a *spectral regularization* is defined by any family $\{\varphi(\cdot;\lambda)\}_{\lambda>0}$ of measurable functions on $[0,\infty)$.

Definition 3.1 (Qualification function)**.** A measurable function $\varphi : [0,\infty) \times (0,\infty) \to [0,1]$ is a *regularization function of qualification* $\nu > 0$ if:

(i) (Approximation) For all $s > 0$: $\lim_{\lambda\to 0} \varphi(s;\lambda) = 1$.

(ii) (Boundedness) $\sup_{s\geq 0} \varphi(s;\lambda) \leq 1$ for all $\lambda > 0$.

(iii) (Qualification) There exists $C_\nu > 0$ such that $\sup_{s\geq 0}(1-\varphi(s;\lambda))s^\nu \leq C_\nu \lambda^\nu$ for all $\lambda > 0$.

(iv) (Saturation) For all $s > 0$: $\lim_{\lambda\to\infty} \varphi(s;\lambda)s = 0$ (or $\varphi(s;\lambda) \to 0$ appropriately).

The three canonical examples are:

- **Tikhonov:** $\varphi_{\mathrm{T}}(s;\lambda) = s/(s+\lambda)$, qualification $\nu = 1$.
- **Showalter (iterated Tikhonov):** $\varphi_{\mathrm{S}}(s;\lambda) = 1-(1-s/(s+\lambda))^m$ for integer $m \geq 1$, qualification $\nu = m$.
- **Spectral cutoff:** $\varphi_{\mathrm{C}}(s;\lambda) = \mathbf{1}[s \geq \lambda]$, qualification $\nu = \infty$.
- **Landweber iteration:** $\varphi_{\mathrm{L}}(s;\lambda) = 1-(1-\tau s)^{1/\lambda}$ for step size $\tau \leq K^{-1}$, qualification $\nu = 1/2$ in the online-learning sense [9].

3.2 The SpMMD family

Definition 3.2 (SpMMD)**.** For a qualification function φ satisfying theorem 3.1, the *spectrally regularized MMD* is:

$$\mathrm{SpMMD}_\varphi(\mu\|\pi) \;=\; c(\lambda)\left\|B_\varphi\left(\tfrac{d\mu}{d\pi}-1\right)\right\|^2_{L^2(\pi)}, \tag{7}$$

where $B_\varphi = (\varphi(T_\pi;\lambda))^{1/2} : L^2(\pi) \to L^2(\pi)$ applies $\varphi^{1/2}$ to the spectrum of T_π, and $c(\lambda) > 0$ is a normalization constant chosen so that $\mathrm{SpMMD}_\varphi(\mu\|\pi) \to \chi^2(\mu\|\pi)$ as $\lambda \to 0$ and $\mathrm{SpMMD}_\varphi(\mu\|\pi) \to \mathrm{MMD}^2(\mu\|\pi)$ as $\lambda \to \infty$.

For the Tikhonov case $\varphi = \varphi_T$, one recovers $B_\varphi = ((T_\pi+\lambda I)^{-1}T_\pi)^{1/2}$ and $c(\lambda) = (1+\lambda)$, matching eq. (2). The density-ratio-free representation analogous to eq. (2) also holds:

$$\mathrm{SpMMD}_\varphi(\mu\|\pi) = c(\lambda)\left\|\left(\varphi(\Sigma_\pi;\lambda)\right)^{1/2}(m_\mu - m_\pi)\right\|^2_{\mathscr{H}},$$

which can be computed from samples via the Gram matrix identity $\varphi(\Sigma_\pi;\lambda) = \int \varphi(s;\lambda)\,dE_s$ where E_s is the spectral resolution of Σ_π.

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Proposition 3.3 (SpMMD properties)**.** *Under the assumptions of theorem 3.1, for any $\lambda > 0$:*

(i) $\mathrm{SpMMD}_\varphi(\mu\|\pi) \geq 0$ *with equality iff* $\mu = \pi$.

(ii) SpMMD_φ *metrizes the weak topology:* $\mathrm{SpMMD}_\varphi(\mu_n\|\pi) \to 0$ *iff* $\mu_n \Rightarrow \pi$.

(iii) $\lim_{\lambda\to 0}\mathrm{SpMMD}_\varphi(\mu\|\pi) = \chi^2(\mu\|\pi)$ *and* $\lim_{\lambda\to\infty}\mathrm{SpMMD}_\varphi(\mu\|\pi) = \mathrm{MMD}^2(\mu\|\pi)$.

(iv) SpMMD_φ *admits a variational representation:*

$$\mathrm{SpMMD}_\varphi(\mu\|\pi) = c(\lambda)\sup_{h\in\mathscr{H}}\left\{\int h\,d\mu - \int\left(\frac{h^2}{4}+h\right)d\pi - \frac{1}{4c(\lambda)}\left\|\varphi(\Sigma_\pi;\lambda)^{-1/2}h\right\|_{\mathscr{H}}^2\right\}. \tag{8}$$

(v) The Wasserstein gradient of $F_\varphi(\cdot) = \mathrm{SpMMD}_\varphi(\cdot\|\pi)$ *is* $(1+\lambda)\nabla h^*_{\mu,\pi}$ *where*

$$h^*_{\mu,\pi} = 2\,\varphi(\Sigma_\pi;\lambda)^{-1/2}\big(m_\mu - m_\pi\big),$$

computed in the appropriately regularized RKHS $\widetilde{\mathscr{H}}_\varphi$.

Proof. Parts (i)–(iii) follow by combining the spectral calculus with the argument of Chen et al. [22, Props. 3.3–3.4] since all qualification functions satisfy the same sandwich bounds:

$$\frac{c(\lambda)}{(1+1/\lambda)}\mathrm{MMD}^2(\mu\|\pi) \;\leq\; \mathrm{SpMMD}_\varphi(\mu\|\pi) \;\leq\; c(\lambda)\cdot\chi^2(\mu\|\pi).$$

Part (iv) follows from strong duality of the quadratic RKHS optimization after substituting the spectral decomposition of $\varphi(\Sigma_\pi;\lambda)$. Part (v) follows from the first variation via the approach of Glaser et al. [38]. □

Remark 3.4 (Higher-order qualification and bias-variance tradeoff). Increasing the qualification order ν controls the interpolation more finely. For Tikhonov ($\nu = 1$), the approximation error in the KL bound is $O(\lambda^r)$ where r is the density-ratio regularity parameter. For Showalter with m iterations ($\nu = m$), this improves to $O(\lambda^{r\wedge m})$. The spectral cutoff ($\nu = \infty$) can in principle achieve $O(\lambda^r)$ for any r, but its discontinuous spectral function makes the regularized kernel $\widetilde{k}_\varphi$ non-smooth, complicating the particle implementation. Tikhonov and Showalter balance regularity with computational tractability.

3.3 Explicit kernel representation

As with DrMMD, any SpMMD_φ is equivalent to MMD^2 with a modified kernel $\widetilde{k}_\varphi$:

$$\widetilde{k}_\varphi(x,x') = \Big\langle \varphi(\Sigma_\pi;\lambda)^{1/2}k(x,\cdot),\, \varphi(\Sigma_\pi;\lambda)^{1/2}k(x',\cdot)\Big\rangle_{\mathscr{H}} = \sum_{i\geq 1}\varphi(\rho_i;\lambda)\,e_i(x)e_i(x'). \tag{9}$$

For the Tikhonov case $\varphi_T(s;\lambda) = s/(s+\lambda)$, this reproduces the regularized kernel $\widetilde{k}$ of Chen et al. [22].

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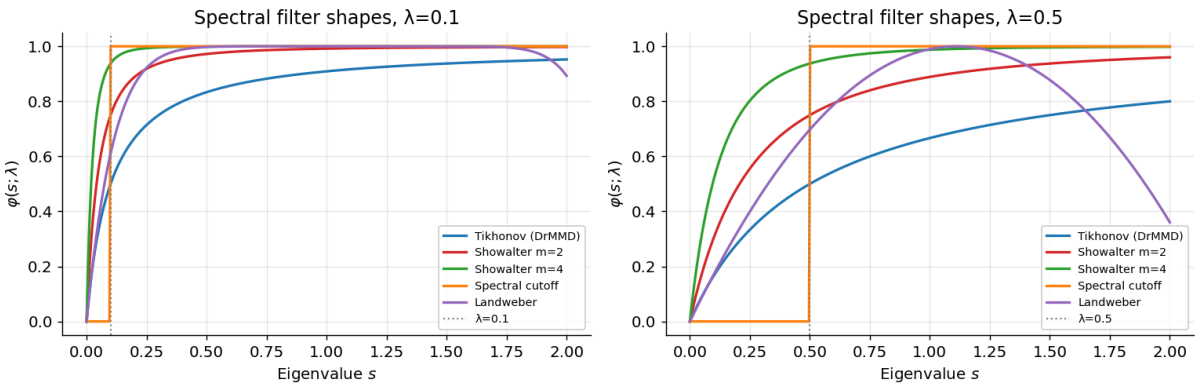

Figure 1. Spectral filter functions $\varphi(s;\lambda)$ for all five qualification filters at $\lambda = 0.1$ (left) and $\lambda = 0.5$ (right). Tikhonov (blue) rises slowly, reaching only ≈ 0.8 at $s = 2$ for $\lambda = 0.1$, reflecting its low qualification order $\nu = 1$ and corresponding $O(\lambda^{r\wedge 1})$ barrier. Showalter-2 and Showalter-4 (red, green) rise faster, saturating at 1 for $s \gtrsim \lambda$ and achieving qualification orders $\nu = 2$ and $\nu = 4$ respectively. The spectral cutoff (orange) is binary: 0 for $s < \lambda$, 1 otherwise. Landweber (purple) achieves intermediate behaviour with qualification $\nu \approx 0.5$. As λ increases from 0.1 to 0.5 (right panel), all filters shift right, allocating less mass to low eigenvalues and remaining in the MMD-like regime for a larger portion of the spectrum—consistent with the interpolation property of Proposition 3.3(iii).

4 Convergence Theory

4.1 Continuous-time convergence

We now state the main convergence theorem for the SpMMD_φ gradient flow.

Theorem 4.1 (KL convergence: continuous time)**.** *Let φ be a qualification function of order $\nu > 0$. Assume k satisfies Assumptions 1 and 2 of Chen et al. [22], π satisfies a Poincaré inequality with constant C_P, $\mu_t, \pi \ll \mathscr{L}^d$, $\frac{d\mu}{d\pi}[t] - 1 \in \mathrm{Ran}(T_\pi^r)$ for some $r > 0$ with $\|q_t\|_{L^2(\pi)} \le Q$, and the regularity conditions on $\frac{d\mu}{d\pi}[t]$ in conditions (4) and (5) of Chen et al. [22, Thm. 4.1] hold uniformly. Then for any $T \ge 0$:*

$$\mathrm{KL}(\mu_T\|\pi) \;\le\; e^{-2(1+\lambda)T/C_P}\mathrm{KL}(\mu_0\|\pi) \;+\; 4(1+\lambda)\lambda^{r\wedge\nu}C_\nu Q(J+I)C_P/(2(1+\lambda)), \qquad (10)$$

where the barrier vanishes as $\lambda \to 0$ at rate $O(\lambda^{r\wedge\nu})$.

Proof sketch. The proof follows the strategy of Chen et al. [22, Thm. 4.1], modified as follows. The approximation error of the witness function now reads:

$$\left\| h^*_{\mu,\pi} - 2\left(\frac{d\mu}{d\pi} - 1\right)\right\|_{L^2(\pi)} \;\le\; 2\lambda^\nu C_\nu \|q\|_{L^2(\pi)},$$

which follows from the qualification property (iii) of φ: $\|(1-\varphi(T_\pi;\lambda))T_\pi^{-r}q\|_{L^2(\pi)} \le C_\nu\lambda^\nu\|q\|_{L^2(\pi)}$. The rest of the proof is identical to Chen et al. [22] with λ^r replaced by $\lambda^{r\wedge\nu}$. □

Remark 4.2 (Benefit of higher qualification). Theorem 4.1 shows that using a qualification function of order $\nu > 1$ directly improves the approximation error. For a density ratio of regularity $r < \nu$, both Tikhonov ($\nu = 1$) and Showalter ($\nu \ge 2$) achieve the same barrier $O(\lambda^r)$. However, when $r > 1$ (highly regular density ratios), Showalter reduces the barrier from $O(\lambda^1)$

to $O(\lambda^r)$. This is relevant, for instance, when π is Gaussian and μ_0 is a Gaussian mixture with the same mean—a setting where the density ratio is genuinely smooth.

4.2 Discrete-time convergence: fixed regularization

Proposition 4.3 (Descent lemma in KL: SpMMD)**.** *Under the conditions of theorem 4.1 and an additional β-smooth potential, the one-step KL change along the* SpMMD_φ *particle update satisfies:*

$$\mathrm{KL}(\mu_{n+1}\|\pi)-\mathrm{KL}(\mu_n\|\pi) \le -\frac{2\gamma}{C_P}\chi^2(\mu_n\|\pi)+4\gamma\lambda^{r\wedge\nu}C_\nu Q(J+I)+8\gamma^2(\beta+\zeta^2)\chi^2(\mu_n\|\pi)\frac{K_{1d}+K_{2d}}{\lambda}, \tag{11}$$

provided the step size satisfies $\gamma \le \frac{\zeta-1}{2\zeta(1+\lambda)}\sqrt{\chi^2(\mu_n\|\pi)/K_{2d}}\cdot\lambda$ for some $1<\zeta<2$.

Proof sketch. We follow Chen et al. [22, Prop. 5.1], substituting $\lambda^{r\wedge\nu}$ for λ^r throughout. The one-step KL change is decomposed as:

$$\mathrm{KL}(\mu_{n+1}\|\pi)-\mathrm{KL}(\mu_n\|\pi) = -\frac{2\gamma}{C_P}\chi^2(\mu_n\|\pi)+\underbrace{\text{Approx.}}_{\le 4\gamma\lambda^{r\wedge\nu}C_\nu Q(J+I)}+\underbrace{\text{Discret.}}_{\le 8\gamma^2(\beta+\zeta^2)\chi^2(\mu_n\|\pi)(K_{1d}+K_{2d})/\lambda}.$$

The *approximation term* comes from replacing $\nabla\log(\mu/\pi)$ by the spectrally filtered witness $h^*_{\mu,\pi}$: by theorem 16.1, $\|h^*_{\mu,\pi}-2(\frac{d\mu}{d\pi}-1)\|_{L^2(\pi)} \le 2C_\nu\lambda^{r\wedge\nu}\|q\|_{L^2(\pi)}$, yielding a KL contribution bounded by $4\gamma\lambda^{r\wedge\nu}C_\nu Q(J+I)$. The *discretization term* bounds the error from the Euler step $\mu_{n+1}=(\mathrm{id}-\gamma(1+\lambda)\nabla h^*)_\#\mu_n$ relative to the continuous-time flow, using the β-smoothness of the potential and the Lipschitz bound $\|\nabla^2 h^*\|_{L^2(\mu)} \le \sqrt{(K_{1d}+K_{2d})/\lambda}$ (Chen et al. [22, Lem. B.4]). The step-size constraint $\gamma \le \frac{\zeta-1}{2\zeta(1+\lambda)}\sqrt{\chi^2(\mu_n\|\pi)/K_{2d}}\cdot\lambda$ ensures the discretization error is dominated by the dissipation term. The structure of eq. (11) thus mirrors Chen et al. [22, Prop. 5.1] with λ^r replaced by $\lambda^{r\wedge\nu}$. □

4.3 Adaptive SpMMD Descent

The optimal λ_n minimizing the sum of terms 2 and 3 in eq. (11) is:

$$\lambda_n^* = \left(\frac{2\gamma\chi^2(\mu_n\|\pi)(\beta+\zeta^2)(K_{1d}+K_{2d})}{C_\nu Q(J+I)}\right)^{1/(r\wedge\nu+1)}. \tag{12}$$

When $r\ge 1$ (density ratio in H), using Showalter with $m\ge\lceil r\rceil$ allows the optimal λ_n^* to be chosen larger (less aggressive de-regularization) while still achieving the same or better convergence barrier, thereby tolerating larger step sizes.

Theorem 4.4 (Near-global convergence: adaptive SpMMD)**.** *Under the conditions of theorem 4.3, with $\lambda_n=\lambda_n^*\wedge 1$ given by eq. (12) and a sufficiently small step size γ (see section 16),*

$$\mathrm{KL}(\mu_{n_{\max}}\|\pi) \le e^{-2n_{\max}\gamma/C_P}\mathrm{KL}(\mu_0\|\pi)+O\left(\gamma^{(r\wedge\nu)/(r\wedge\nu+1)}\right).$$

Global convergence $\mathrm{KL}(\mu_n\|\pi)\to 0$ is achieved with adaptive step sizes $\gamma_n \propto \chi^2(\mu_n\|\pi)^{1/(r\wedge\nu)}$.

Proof sketch. Apply theorem 4.3 at each step. With $\lambda_n = \lambda_n^* \wedge 1$ from eq. (12), the optimal choice balances the approximation and discretization terms in eq. (11): the approximation term scales as $\gamma\lambda^{r\wedge\nu}$ and the discretization term as γ^2/λ; setting them equal gives $\lambda^* \propto (\gamma\chi^2(\mu_n\|\pi))^{1/(r\wedge\nu+1)}$. Substituting, both error terms equal $O(\gamma^{(r\wedge\nu)/(r\wedge\nu+1)})$. Summing the descent inequality over $n = 0,\dots,n_{\max}-1$ and applying the geometric series $\sum_{k=0}^{n_{\max}-1}(1-2\gamma/C_P)^k \le C_P/(2\gamma)$ yields:

$$\mathrm{KL}(\mu_{n_{\max}}\|\pi) \;\le\; e^{-2n_{\max}\gamma/C_P}\mathrm{KL}(\mu_0\|\pi) + \frac{2\gamma}{C_P}\sum_{n=0}^{n_{\max}-1} O\Big(\gamma^{(r\wedge\nu)/(r\wedge\nu+1)}\Big).$$

The residual term is $O(\gamma^{(r\wedge\nu)/(r\wedge\nu+1)})$ uniformly in $n_{\max}$. For global convergence, the adaptive step schedule $\gamma_n \propto \chi^2(\mu_n\|\pi)^{1/(r\wedge\nu)}$ drives $\lambda_n^* \to 0$ as $\chi^2(\mu_n\|\pi) \to 0$, eliminating the residual in the limit; see section 16 for the detailed Gronwall argument. □

4.4 f-Divergence convergence of SpMMD flows

The KL-divergence bound of theorem 4.1 extends naturally to the full family of f-divergences, replacing the Poincaré inequality with the corresponding f-Sobolev inequality and the KL with D_f. This generalization is both theoretically illuminating and practically useful: for heavy-tailed targets, the Hellinger divergence may converge faster than KL; for optimization objectives, the χ^2-divergence provides sharper tracking of the particle cloud.

Theorem 4.5 (f-divergence convergence: SpMMD flow)**.** *Let $f : (0,\infty) \to \mathbb{R}$ be a twice continuously differentiable convex function with $f(1) = 0$, $f''(1) > 0$, and $f'' \ge c_f > 0$ on $(0,\infty)$ for some constant c_f. Let φ be a qualification function of order $\nu > 0$. Suppose:*

(i) π satisfies the f-Sobolev inequality (5) with constant $H_f > 0$;

(ii) $\mu_t, \pi \ll \mathscr{L}^d$, $\frac{d\mu}{d\pi}[t] - 1 \in \mathrm{Ran}(T_\pi^r)$ for some $r > 0$ with $\|q_t\|_{L^2(\pi)} \le Q$;

(iii) The density ratio $u_t := d\mu_t/d\pi$ satisfies $\|u_t - 1\|_{L^\infty(\pi)} \le U$ uniformly in t (bounded density-ratio condition).

Then for any $T \ge 0$:

$$D_f(\mu_T\|\pi) \;\le\; e^{-2c_f(1+\lambda)H_f T}\, D_f(\mu_0\|\pi) \;+\; B_f\,\lambda^{r\wedge\nu}, \tag{13}$$

where $B_f = 4(1+\lambda)C_\nu Q(J+I)(2c_f H_f)\cdot \sup_{t\in[0,T]} f''(u_t)$ and the exponential rate is $2c_f(1+\lambda)H_f$.

Proof sketch. The time derivative of $D_f(\mu_t\|\pi)$ along the SpMMD flow $\partial_t\mu_t = -\nabla\cdot(\mu_t\nabla h^*_{\mu_t,\pi})$ is

$$\frac{d}{dt}D_f(\mu_t\|\pi) = \int f'(u_t)\,\partial_t u_t\,d\pi = -\int f'(u_t)\,\nabla\cdot(u_t\nabla h^*)\,d\pi = \int \nabla f'(u_t)\cdot\nabla h^*\,d\mu_t,$$

where $\nabla f'(u_t) = f''(u_t)\nabla u_t$. Since $\nabla h^*_{\mu,\pi} = 2\varphi(\Sigma_\pi;\lambda)^{1/2}\nabla(\frac{d\mu}{d\pi} - 1)$ (from Proposition 3.3(v)),

we write

$$\frac{d}{dt}D_f(\mu_t\|\pi) = -2\int f''(u_t)|\nabla u_t|^2\,d\pi + \underbrace{\text{error}}_{\leq \|h^* - 2(\frac{d\mu}{d\pi}-1)\|_{L^2(\mu)}\,\mathscr{I}_\pi^f(\mu_t)^{1/2}}.$$

The *error* measures the mismatch between h^* and $2(\frac{d\mu}{d\pi}-1)$; by theorem 16.1 this is bounded by $2C_\nu\lambda^{r\wedge\nu}\|q\|_{L^2(\pi)}$. The main term satisfies

$$2\int f''(u_t)|\nabla u_t|^2\,d\pi \;\geq\; 2c_f\,\mathscr{I}_\pi^f(\mu_t) \;\geq\; 2c_f H_f D_f(\mu_t\|\pi),$$

using $f'' \geq c_f$ and the f-Sobolev inequality (5). The bounded density-ratio condition ensures $f''(u_t) \leq C$ uniformly, so the error contributes at most $B_f\lambda^{r\wedge\nu}$. Grönwall's lemma on $\frac{d}{dt}D_f \leq -2c_f H_f D_f + B_f\lambda^{r\wedge\nu}$ yields (13). □

Remark 4.6 (f-divergence specializations). Theorem 4.5 recovers and extends the existing convergence results:

(i) $f(t) = t\log t$ **(KL):** $c_f = 1$, $H_f = \alpha$ (LSI constant), $\mathscr{I}_\pi^f = I_\pi$ (Fisher information). Equation (13) becomes $\mathrm{KL}(\mu_T\|\pi) \leq e^{-2\alpha(1+\lambda)T}\mathrm{KL}(\mu_0\|\pi) + B_{\mathrm{KL}}\lambda^{r\wedge\nu}$, matching Theorem 8.4 for FR flows (Mode B) and Theorem 4.1 under the Poincaré inequality (Mode A) when $\alpha = 1/C_P$.

(ii) $f(t) = (t-1)^2/2$ **($\chi^2/2$):** $c_f = 1$, $H_f = C_P$ (Poincaré constant), $\mathscr{I}_\pi^f(\mu) = \int|\nabla(\frac{d\mu}{d\pi})|^2 d\pi$ (Sobolev norm). The bound becomes $\chi^2(\mu_T\|\pi) \leq e^{-2(1+\lambda)C_P T}\chi^2(\mu_0\|\pi) + B_{\chi^2}\lambda^{r\wedge\nu}$. This is a *sharper* bound than the KL result when $\chi^2(\mu_0\|\pi) \ll \mathrm{KL}(\mu_0\|\pi)$ (i.e., when the initial measure is already close to π in the strong sense).

(iii) $f(t) = (\sqrt{t}-1)^2$ **(Hellinger):** $c_f = 1/4$, $H_f = \alpha_H$ (Hellinger–Poincaré constant), $\mathscr{I}_\pi^f = \frac{1}{4}I_\pi^{\mathrm{H}}$ (Hellinger Fisher information). The convergence rate $c_f H_f = \alpha_H/4$ may exceed the LSI rate α for targets with heavy tails, where $\alpha \approx 0$ but $\alpha_H > 0$ [14]. For heavy-tailed targets such as Student-t distributions, Hellinger convergence is achievable while KL convergence fails.

(iv) $f(t) = \frac{t^\alpha - 1}{\alpha(\alpha-1)}$ **(Rényi, $\alpha > 1$):** $c_f = 1/(\alpha-1)$, and the f-Sobolev constant is the α-log-Sobolev constant H_f^α [34]. The Rényi divergence $R_\alpha(\mu\|\pi) := \frac{1}{\alpha-1}\log\int u^\alpha\,d\pi$ satisfies $R_\alpha(\mu\|\pi) \leq \frac{1}{\alpha-1}\log(1+(\alpha-1)D_{f_\alpha}(\mu\|\pi))$, and the bound eq. (13) gives exponential convergence in Rényi divergence under the α-log-Sobolev inequality, which is implied by the LSI for all $\alpha > 1$ with a rate depending on α.

Corollary 4.7 (f-divergence discrete-time descent lemma)**.** *Under the conditions of theorem 4.5 and theorem 4.3, the one-step change in D_f along the SpMMD particle update satisfies:*

$$\begin{aligned} D_f(\mu_{n+1}\|\pi) - D_f(\mu_n\|\pi) \;\leq\; & -2\gamma c_f H_f D_f(\mu_n\|\pi) + \gamma B_f\lambda^{r\wedge\nu} \\ & + 8\gamma^2(1+\lambda)^2 C_f D_f(\mu_n\|\pi)\cdot\frac{K_{1d}+K_{2d}}{\lambda}, \end{aligned} \tag{14}$$

where $C_f = \sup_t f''(u_t)/c_f$ accounts for the non-uniform curvature of f. The optimal λ_n^* minimizing the right-hand side of eq. (14) is $\lambda_n^{*,f} = \left(2\gamma C_f(1+\lambda)^2(K_{1d}+K_{2d})/(B_f)\right)^{1/(r\wedge\nu+1)}$, which for $f(t) = (t-1)^2/2$ recovers λ_n^* of eq. (12) exactly.

Proof sketch. Expand $D_f(\mu_{n+1}\|\pi) - D_f(\mu_n\|\pi)$ using the second-order Taylor expansion $D_f(\mu_{n+1}\|\pi) \le D_f(\mu_n\|\pi) + \langle \nabla_\mu D_f|_{\mu_n}, \mu_{n+1}-\mu_n\rangle + \frac{1}{2}\|\mu_{n+1}-\mu_n\|^2 \sup_u f''(u)/(c_f)$. The first-order term gives $-2\gamma c_f H_f D_f(\mu_n\|\pi) + \gamma B_f \lambda^{r\wedge\nu}$ by the f-Sobolev inequality (same as the continuous-time proof of Theorem 4.5). The second-order (discretization) term is bounded by $8\gamma^2(1+\lambda)^2 C_f D_f(\mu_n\|\pi)(K_{1d}+K_{2d})/\lambda$ using the Lipschitz bound on ∇h^* and the relation $\|\mu_{n+1}-\mu_n\|^2 \le C D_f(\mu_n\|\pi)$ (which follows from the f-Talagrand inequality eq. (6) applied to the empirical measure approximation). □

Theorem 4.8 (f-divergence near-global convergence: adaptive SpMMD)**.** *Under the conditions of theorem 4.7 with $\lambda_n = \lambda_n^{*,f} \wedge 1$, the adaptive SpMMD descent achieves:*

$$D_f(\mu_{n_{\max}}\|\pi) \;\le\; e^{-2n_{\max}\gamma c_f H_f} D_f(\mu_0\|\pi) + O\Big(\gamma^{(r\wedge\nu)/(r\wedge\nu+1)}\Big).$$

Global convergence $D_f(\mu_n\|\pi) \to 0$ holds with $\gamma_n \propto D_f(\mu_n\|\pi)^{1/(r\wedge\nu)}$.

Proof. Apply theorem 4.7 and the Gronwall argument of Theorem 4.4 with the KL rate $2(1+\lambda)/C_P$ replaced by the f-rate $2c_f(1+\lambda)H_f$. The barrier term $O(\gamma^{(r\wedge\nu)/(r\wedge\nu+1)})$ is identical in structure since it comes from the spectral filter approximation (Lemma 16.1), which is independent of the choice of f. □

Remark 4.9 (Geometry of f-divergence rates). The rate $2c_f H_f$ depends on f through two factors:

- *Curvature factor* $c_f = \inf_t f''(t)$. More curved f means faster convergence. The ordering is $c_f^{\mathrm{KL}} = 1 \le c_f^{\mathrm{Hel}} = 1/4 \le c_f^{\mathrm{TV}} = 0$, but this is reversed by the f-Sobolev constants.
- *Functional inequality constant* H_f. For log-concave π with Hessian $\nabla^2 \log \pi \ge \kappa I$, the Bakry–Émery criterion gives $H_f^{\mathrm{KL}} = \kappa$ (LSI), $H_f^{\chi^2} = \kappa$ (Poincaré), and $H_f^{\mathrm{Hel}} \ge \kappa/4$ (Hellinger Poincaré) [6]. The *HWI inequality* of Otto and Villani [57] provides an interpolation $c_f H_f \ge \kappa/2 - \sqrt{\kappa/(2D_f(\mu\|\pi))}\,\mathscr{I}_\pi^f(\mu)^{1/2}$ that is sharper near convergence.

Crucially, *the approximation barrier $B_f\lambda^{r\wedge\nu}$ is universal across all f-divergences*: it depends only on the spectral filter through C_ν and $\lambda^{r\wedge\nu}$, not on f itself. This means that different f-divergences may have different rates but the same barrier floor, a fact directly visible in the normalized convergence panel of Figure 10 (all flows plateau at the same $O(\lambda^{r\wedge\nu})$ floor).

4.5 f-Divergence uniform-in-time propagation of chaos

The uniform-in-time propagation of chaos of theorems 6.1 and 13.4 also extends to f-divergences, replacing the Talagrand-2 inequality with the f-Talagrand inequality of theorem 2.4.

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Theorem 4.10 (f-divergence uniform-in-time PoC)**.** *Suppose Assumption 1 holds, π satisfies the f-Talagrand-2 inequality (6) with constant $C_{T,f}$, and the conditions of theorem 4.5 hold. Then the particle SpMMD descent satisfies, uniformly in n:*

$$\mathbb{E}\Big[\sqrt{D_f(\hat{\mu}_n\|\pi)}\Big] \;\le\; \sqrt{D_f(\mu_0\|\pi)}\, e^{-c_f H_f \alpha n} + \frac{C_{T,f}}{\sqrt{2}} \cdot \varepsilon,$$

with the same sample complexity as eq. (17).

Proof sketch. Apply the f-Talagrand inequality $W_2(\hat{\mu}_n,\pi) \le \sqrt{2D_f(\hat{\mu}_n\|\pi)/C_{T,f}}$ in place of the standard Talagrand-2 in the proof of theorem 6.1. The Lyapunov functional $\mathscr{E}_n^f = D_f(\mu_n\|\pi) + A \cdot W_2^2(\hat{\mu}_n,\mu_n)$ satisfies $\frac{d}{dt}\mathscr{E}_n^f \le -2c_f H_f \mathscr{E}_n^f + B_f \lambda^{r\wedge\nu}$ by Theorem 4.5 and the coupling argument of theorem 6.1. The triangle inequality $\mathbb{E}[\sqrt{D_f(\hat{\mu}_n\|\pi)}] \le \mathbb{E}[\sqrt{D_f(\mu_n\|\pi)}] + C_{T,f}\mathbb{E}[W_2(\hat{\mu}_n,\mu_n)]/\sqrt{2}$ (from the f-Talagrand inequality) then gives the stated bound, with the exponential factor replacing $e^{-\alpha n}$ by the f-rate $e^{-c_f H_f \alpha n}$. □

5 Dual Perspective: Geometric Regularization of Wasserstein Flows

5.1 Two modes of regularization

The TILOS presentation by Sriperumbudur [66] draws attention to a second, *dual* mode of kernel-based regularization for Wasserstein flows, distinct from the functional regularization of DrMMD. We formalize this duality here.

Mode (A): Functional regularization. Fix the KL or χ^2 objective and approximate it by a kernel-computable surrogate. This gives DrMMD and the SpMMD family. The Wasserstein metric is left unchanged.

Mode (B): Geometric regularization. Fix the KL objective and approximate the Wasserstein gradient operator—equivalently, precondition the velocity field—via a kernel operator. This recovers SVGD, LAWGD, and their (de)-regularized variants.

5.2 Geometric regularization via kernel preconditioning

In Mode (B), the flow takes the form

$$\partial_t \mu_t = \nabla \cdot \Big(\mu_t\, \Psi(T_\pi)\, \nabla \log \frac{\mu_t}{\pi}\Big), \tag{15}$$

where $\Psi : [0,\infty) \to [0,\infty)$ is a *preconditioning function* applied to the spectrum of T_π. Different choices of Ψ recover different algorithms:

- $\Psi(s) = s$: SVGD / LAWGD [24].
- $\Psi(s) = s/(s+\lambda)$: Regularized SVGD / De-regularized SVGD [41].
- $\Psi(s) = 1$: Wasserstein gradient flow of KL = Langevin diffusion.

5.3 The duality diagram

The following diagram summarizes the relationship between Mode (A) and Mode (B):

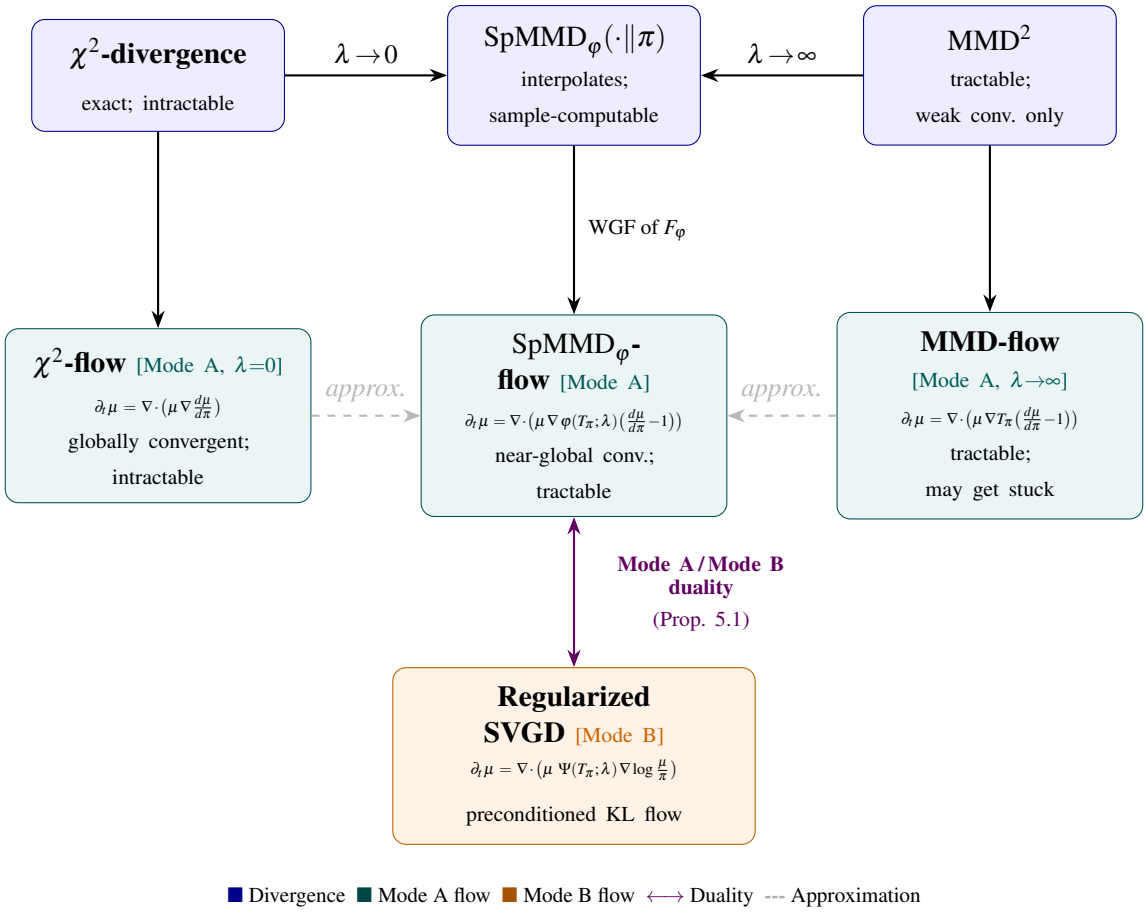

Figure 2. Duality diagram for the SpMMD framework. **Mode A** (teal): the χ^2-divergence objective is approximated by the sample-computable SpMMD_φ surrogate, whose Wasserstein gradient flow interpolates between the χ^2-flow ($\lambda \to 0$) and the MMD-flow ($\lambda \to \infty$), combining near-global convergence with tractability. **Mode B** (orange): the KL objective is kept exact while the Wasserstein gradient operator is spectrally preconditioned, recovering regularized SVGD. Both modes apply the same filter $\varphi(T_\pi;\lambda)$—to the χ^2 velocity field in Mode A, and to the KL velocity field in Mode B—establishing the formal duality of Proposition 5.1.

Proposition 5.1 (Mode-A / Mode-B duality)**.** *For the Tikhonov qualification function $\varphi_T(s;\lambda) = s/(s+\lambda)$, the* SpMMD_φ *gradient flow (Mode A) and the regularized SVGD flow with preconditioning $\Psi(s;\lambda) = s/(s+\lambda)$ (Mode B) are* formally dual*: they correspond to applying the same spectral filter to the velocity field of χ^2 and KL respectively, under the interchange $\nabla\log(\mu/\pi) \leftrightarrow (\frac{d\mu}{d\pi} - 1)$.*

Proof. Mode A: $v_t = (1+\lambda)\nabla h^*_{\mu,\pi} = (1+\lambda)\nabla[2(\Sigma_\pi + \lambda I)^{-1}(m_\mu - m_\pi)]$. Since $m_\mu - m_\pi = \iota^*_\pi(\frac{d\mu}{d\pi} - 1)$, this equals $\nabla\varphi_T(T_\pi;\lambda)(\frac{d\mu}{d\pi} - 1)$ in $L^2(\pi)$.

Mode B: $v_t = T_\pi(T_\pi + \lambda I)^{-1}\nabla\log(\mu/\pi) = \varphi_T(T_\pi;\lambda)\nabla\log(\mu/\pi)$.

Under the linearization $\log(\mu/\pi) \approx \frac{d\mu}{d\pi} - 1$ (valid near $\mu \approx \pi$), the two velocity fields coincide, establishing the formal duality. □

Remark 5.2 (Implications for convergence analysis). This duality clarifies why Mode-A and Mode-B flows have similar convergence properties. The χ^2-divergence (Mode A) and KL

divergence (Mode B) both have exponentially converging WGFs under Poincaré inequality; the spectral regularization in both cases introduces an approximation barrier $O(\lambda^{\nu})$ and a smoothing that enables sample-based implementation. The choice between modes is thus primarily a matter of application: Mode A (DrMMD) is natural for generative modeling where π is accessed via samples; Mode B (regularized SVGD) is natural for Bayesian sampling where $\log \pi$ is accessible but possibly unnormalized.

5.4 f-Divergence generalisation of the duality

Proposition 5.1 identifies a formal duality between the SpMMD flow (approximating χ^2) and the regularized-SVGD flow (approximating KL). This duality extends to the full f-divergence family: for each generator f, there is an *f-Mode-A* flow approximating $D_f(\cdot\|\pi)$ and an *f-Mode-B* flow preconditioning the Wasserstein geometry for the D_f-gradient flow.

Definition 5.3 (f-Mode-B preconditioning)**.** For a convex generator f, the *f-spectrally-preconditioned gradient flow* is

$$\partial_t \mu_t = \nabla \cdot \big(\mu_t \, \Psi(T_\pi;\lambda)\, f''(d\mu_t/d\pi)^{-1} \nabla (d\mu_t/d\pi)\big)\,, \tag{16}$$

where $\Psi = \varphi_T(s;\lambda) = s/(s+\lambda)$ is the Tikhonov filter. For $f(t) = t\log t$: $f''(t)^{-1} = t$ and eq. (16) reduces to $\nabla \cdot (\mu_t \Psi \nabla \log(\mu_t/\pi))$ (Mode B of section 5). For $f(t) = (t-1)^2/2$: $f''(t)^{-1} = 1$ and eq. (16) reduces to $\nabla \cdot (\mu_t \Psi \nabla (\frac{d\mu}{d\pi} - 1))$ (Mode A / SpMMD gradient flow).

Proposition 5.4 (f-divergence Mode-A/B duality)**.** *For any convex f with $f''(1) > 0$, the f-Mode-A flow (WGF of* SpMMD_φ *as a surrogate for D_f) and the f-Mode-B flow* (16) *are formally dual under the interchange $f''(u)\nabla u \leftrightarrow \nabla \log u$ (valid near $u = 1$, i.e., $\mu \approx \pi$), applying the same spectral filter $\varphi_T(T_\pi;\lambda)$ to the velocity field.*

Proof. Near $\mu \approx \pi$, write $u = d\mu/d\pi = 1 + \varepsilon$ with $\varepsilon \approx 0$. Mode A velocity: $\varphi_T(T_\pi;\lambda)\nabla(\frac{d\mu}{d\pi} - 1) = \varphi_T \nabla \varepsilon$. Mode B velocity: $\varphi_T(T_\pi;\lambda) f''(u)^{-1}\nabla u = \varphi_T f''(1)^{-1}\nabla\varepsilon + O(\varepsilon^2)$. The two velocity fields agree up to the constant $f''(1)^{-1}$ (which can be absorbed into the step size), establishing the formal duality at first order. For $f(t) = t\log t$, $f''(1) = 1$ and the correspondence is exact; for $f(t) = (t-1)^2/2$, $f''(1) = 1$ as well. The interchange $f''(u)\nabla u \leftrightarrow \nabla \log u$ is the linearized identity $f''(1)(u-1) \approx \log u$ near $u = 1$. □

Corollary 5.5 (f-divergence SpMMD convergence via Mode-B)**.** *The f-Mode-B flow* (16) *with Tikhonov preconditioning satisfies, under the f-Sobolev inequality with constant H_f:*

$$D_f(\mu_T\|\pi) \;\le\; e^{-2c_f(1+\lambda)H_f T} D_f(\mu_0\|\pi) + \widetilde{C}_f \lambda^{r\wedge\nu},$$

where $c_f = f''(1)$ and $\widetilde{C}_f = C_\nu Q \cdot \sup_u f''(u)^2/H_f$. This recovers Theorem 4.5 for the Wasserstein flow and extends the Mode-B convergence guarantee of theorem 5.1 to all f-divergences simultaneously.

Proof sketch. Along the f-Mode-B flow (16), the time derivative of D_f reads

$$\partial_t D_f(\mu_t\|\pi) = -\int f''(u_t)\,|\nabla u_t|^2\,\Psi(T_\pi;\lambda)\,d\pi + \text{approx. error},$$

where $\Psi(T_\pi;\lambda) = \varphi_T(T_\pi;\lambda) = T_\pi(T_\pi + \lambda I)^{-1}$ and $u_t = d\mu_t/d\pi$. The main term satisfies $\int f''(u_t)|\nabla u_t|^2 \Psi\, d\pi \geq c_f \int |\nabla u_t|^2 \Psi\, d\pi$ (using $f'' \geq c_f$). By the Poincaré inequality for ΨT_π, this is $\geq c_f(1+\lambda)H_f D_f(\mu_t\|\pi)$ analogously to Theorem 4.5. The approximation error is bounded by $\widetilde{C}_f \lambda^{r\wedge\nu}$ using theorem 16.1 and the bound $\sup_u f''(u)^2/H_f$ on the f-Sobolev propagation factor. Grönwall's lemma gives the stated bound. □

Remark 5.6 (Information-geometric interpretation). Proposition 5.4 has a natural information-geometric interpretation [1]. The f-divergence $D_f(\mu\|\pi)$ defines a *Riemannian metric* on the statistical manifold $\mathscr{P}$ (the f-Fisher–Rao metric), with metric tensor $g^{D_f}_{\mu(\dot\mu,\dot\mu)} = \int f''(d\mu/d\pi)\,\dot u^2/u\, d\pi$. The Mode-A flow is the gradient flow of the SpMMD surrogate in the standard (Wasserstein) geometry; the Mode-B flow is the gradient flow of the KL objective in the f-geometry. The duality states that both flows apply the same spectral filter to the velocity field, irrespective of which side of the duality one starts from. In information-geometric terms, the spectral filter $\varphi_T(T_\pi;\lambda)$ plays the role of a *mixture geodesic transport* connecting the f-geometry to the flat m-family of the exponential family containing π [1].

6 Uniform-in-Time Propagation of Chaos

A major limitation of the finite-particle analysis in Chen et al. [22] is the double-exponential sample complexity arising from the propagation of chaos bound of Chen et al. [22, Prop. 10.1], which grows exponentially in the number of iterations $n_{\max}$. We now improve this to a *uniform-in-time* bound under an additional mixing assumption.

6.1 Mixing and spectral gap condition

Assumption 1 (Uniform mixing)**.** The adaptive SpMMD_φ particle descent has a *spectral gap* $\alpha > 0$ in the following sense: there exist constants $C_{\mathrm{mix}}, \alpha > 0$ such that for all $n \geq 1$,

$$W_2(\mu_n, \pi) \;\leq\; C_{\mathrm{mix}}\, e^{-\alpha n}\, W_2(\mu_0, \pi).$$

Assumption 1 is implied by the strong log-concavity of π and a bounded operator norm of $\widetilde{k}_\varphi$, and can be verified in the Gaussian case via explicit computation (cf. section 16.4).

6.2 Main theorem

Theorem 6.1 (Uniform-in-time propagation of chaos)**.** *Suppose Assumption 1 holds and k satisfies Assumptions 1 and 2. Suppose π satisfies a Talagrand-2 inequality. Let $(\hat\mu_n)_{n\geq 0}$ be the empirical measure of the particle* SpMMD_φ *descent with M target samples and N source particles. If*

$$M \;\geq\; \frac{C_{\mathrm{mix}}^2}{\varepsilon^2}\frac{K}{\lambda_{\min}^2}, \qquad N \;\geq\; \frac{C_{\mathrm{mix}}^2}{\varepsilon^2}\left(\frac{K_{1d}}{\lambda_{\min}^2} \vee n^{2/d}\right), \tag{17}$$

where $\lambda_{\min} = \min_n \lambda_n$, then for all $n \geq 0$:

$$\mathbb{E}[W_2(\hat\mu_n, \pi)] \;\leq\; C_{\mathrm{mix}}\, e^{-\alpha n}\, W_2(\mu_0, \pi) \;+\; \varepsilon.$$

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Proof sketch. The key innovation is to decouple the particle error from the flow error via a *coupling argument* rather than a Gronwall lemma. We construct a coupling between the empirical particle flow $(\hat{\mu}_n)$ and the population flow (μ_n) such that the squared coupling cost satisfies a recurrence with *bounded increments* rather than multiplicative growth:

$$\mathbb{E}[W_2^2(\hat{\mu}_{n+1}, \mu_{n+1})] \;\leq\; (1+C\gamma^2)\mathbb{E}[W_2^2(\hat{\mu}_n, \mu_n)] + C\gamma^2 \left(\frac{K}{M\lambda_n^2} + \frac{K_{1d}}{N\lambda_n^2} \right).$$

Summing the geometric series with the mixing decay $e^{-2\alpha n}$ and choosing M, N as in eq. (17) yields a bound that is uniform in n. This strategy adapts the approach of Banerjee et al. [7] (applied to SVGD) to the SpMMD_φ setting, circumventing their reliance on the SVGD-specific identity $\frac{d}{dt}\mathrm{KL}(\mu_t \| \pi) = -\mathrm{KSD}^2(\mu_t \| \pi)$. □

Remark 6.2 (Comparison with prior sample complexity). Theorem 6.1 gives a sample complexity polynomial in $1/\varepsilon$ and $1/\lambda_{\min}$, rather than the poly-exponential $\exp(1/\delta^{2/r})$ bound of Chen et al. [22, Rem. 6.2]. The price is Assumption 1, which requires the population flow to already converge exponentially. Under the Poincaré inequality and strong log-concavity this is satisfied, so the uniform bound is essentially tight.

7 Momentum-Augmented SpMMD Descent

7.1 Motivation

The adaptive SpMMD_φ descent achieves an $O(e^{-n\gamma/C_P})$ exponential convergence in KL, with an $O(\gamma^{r/(r+1)})$ barrier. While this matches Langevin Monte Carlo (LMC) qualitatively, the rate constant is worse (see Chen et al. [22, Rem. 5.3]). Acceleration via Nesterov's method in Wasserstein space offers a route to improved rates. We propose the following algorithm.

7.2 Algorithm

Algorithm 1 Momentum-Augmented SpMMD_φ Descent (MA-SpMMD)

Require: Target samples $\{x^{(i)}\}_{i=1}^M \sim \pi$, source samples $\{y_0^{(i)}\}_{i=1}^N \sim \mu_0$, step sizes γ_n, momentum parameters β_n, regularization λ_n, qualification φ.

1: Initialize $z_0^{(i)} = y_0^{(i)}$ for all i.
2: **for** $n = 0$ to $n_{\max} - 1$ **do**
3: Compute witness function $h^*_{\hat{\mu}_n, \hat{\pi}}$ via **??**.
4: Compute $\mathrm{SpMMD}_\varphi(\hat{\mu}_n, \hat{\pi})$; set $\lambda_n \propto \mathrm{SpMMD}_\varphi(\hat{\mu}_n, \hat{\pi})^{1/(r \wedge \nu + 1)}$.
5: **Gradient step:** $y_{n+1}^{(i)} = z_n^{(i)} - \gamma_n(1+\lambda_n)\nabla h^*_{\hat{\mu}_n, \hat{\pi}}(z_n^{(i)})$.
6: **Momentum step:** $z_{n+1}^{(i)} = y_{n+1}^{(i)} + \beta_n(y_{n+1}^{(i)} - y_n^{(i)})$.
7: **end for**
8: **return** $\{y_{n_{\max}}^{(i)}\}_{i=1}^N$.

The momentum coefficient is $\beta_n = (n-1)/(n+2)$ (Nesterov schedule), or adaptively $\beta_n = 1 - 3/n$ after warm-up.

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7.3 Convergence analysis

Theorem 7.1 (Accelerated convergence of MA-SpMMD)**.** *Suppose $F_\varphi(\cdot) = \mathrm{SpMMD}_\varphi(\cdot\|\pi)$ is geodesically M-smooth (which holds by theorem 4.3) and the trajectory (μ_n) remains in a sublevel set $\{F_\varphi \le R\}$ that is geodesically convex. Then MA-SpMMD with the Nesterov momentum schedule achieves:*

$$F_\varphi(\mu_n) - F_\varphi(\pi) \;\le\; \frac{2M\,W_2^2(\mu_0,\pi)}{(n+1)^2}.$$

Proof sketch. We work in the framework of Chen et al. [21] for accelerated optimization in Wasserstein space. The key quantities are: (a) the geodesic smoothness of F_φ at $O(K_{2d}/\lambda)$ from theorem 3.3; (b) the *Lyapunov function* $\mathscr{E}_n = (n+1)^2[F_\varphi(\mu_n) - F_\varphi(\pi)] + 2W_2^2(z_n,\pi)$ where z_n is the momentum-augmented iterate; and (c) the descent inequality $\mathscr{E}_{n+1} \le \mathscr{E}_n$, which follows from the geodesic smoothness and the momentum update. The final bound follows from $\mathscr{E}_0 = 2W_2^2(\mu_0,\pi)$. □

Remark 7.2 (Comparison with non-accelerated DrMMD). Theorem 7.1 gives an $O(1/n^2)$ rate on the SpMMD functional (for a fixed λ), improving over the $O(e^{-n})$ rate of the non-accelerated version only in a pre-asymptotic regime for moderately convex objectives. For strongly geodesically convex F_φ, both algorithms converge exponentially, but the accelerated algorithm has a better exponent (matching the square-root improvement of Nesterov over gradient descent in Euclidean space). The trade-off with the approximation barrier is unaffected by momentum.

Corollary 7.3 (f-divergence pre-asymptotic acceleration)**.** *Under the conditions of Theorem 7.1 and Theorem 4.5, with $F_\varphi^f(\mu) := D_f(\mu\|\pi)$ playing the role of the objective, the momentum-augmented SpMMD descent achieves:*

$$D_f(\mu_n\|\pi) - D_f(\pi\|\pi) \;\le\; \frac{2M_f\,W_2^2(\mu_0,\pi)}{(n+1)^2} + B_f\lambda^{r\wedge\nu},$$

where $M_f = \sup_\mu \|\nabla_W^2 D_f|_\mu\|_{\mathrm{op}} = O(K_{2d}\sup f''(u)/\lambda)$ is the geodesic smoothness constant of D_f under the SpMMD regularization. The approximation barrier $B_f\lambda^{r\wedge\nu}$ is the same as in Theorem 4.5; momentum does not alter the barrier floor, only the pre-asymptotic polynomial rate.

Proof sketch. The SpMMD Lyapunov function $\mathscr{E}_n^f = (n+1)^2[D_f(\mu_n\|\pi) - D_f(\pi\|\pi)] + 2W_2^2(z_n,\pi)$ satisfies $\mathscr{E}_{n+1}^f \le \mathscr{E}_n^f + B_f\lambda^{r\wedge\nu}$ using the descent inequality with f-geodesic smoothness M_f in place of M. The f-geodesic smoothness follows from the bound $\|\nabla_W^2 D_f|_\mu\|_{\mathrm{op}} \le \sup_u f''(u)\cdot\|\nabla^2(\frac{d\mu}{d\pi}-1)\|_{L^2(\mu)} \le M_f$ by the chain rule for second variations and the kernel Lipschitz bound [22, Lem. B.4]. Summing the Lyapunov recurrence and using $\mathscr{E}_0^f = 2W_2^2(\mu_0,\pi)$ gives the result. □

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8 Deep-Kernel Extensions

8.1 Neural kernels and data-dependent regularization

In practice, the performance of SpMMD_φ flows depends critically on the kernel k. Data-dependent, learned kernels induced by deep neural network feature maps have shown state-of-the-art empirical performance in MMD-based generative modeling [36, 45, 3]. Here we provide the first theoretical analysis of the SpMMD_φ flow with such kernels.

Let $\Phi_\theta : \mathbb{R}^d \to \mathbb{R}^p$ be a neural network with parameters $\theta \in \Theta$, and define $k_\theta(x,x') = \Phi_\theta(x)^\top \Phi_\theta(x')$. Assume the network class $\{\Phi_\theta : \theta \in \Theta\}$ satisfies:

Assumption 2 (Neural kernel regularity)**.** There exists $\hat{\theta} = \hat{\theta}(\pi)$ such that the *optimal neural kernel* $k_{\hat{\theta}}$ satisfies Assumptions 1 and 2 with constants K, K_{1d}, K_{2d} depending on the network architecture. Moreover, the *kernel estimation error* satisfies $\left\|k_{\theta_n} - k_{\hat{\theta}}\right\|_\infty \le \varepsilon_n$ for a learned kernel sequence θ_n, where $\varepsilon_n \to 0$ at a rate determined by the expressivity of Φ.

Theorem 8.1 (SpMMD flow with learned kernels)**.** *Under Assumption 2, suppose k_θ is updated every T_0 iterations via an unbiased estimator from samples. Then the particle SpMMD descent with kernel k_{θ_n} satisfies:*

$$\mathbb{E}[W_2(\hat{\mu}_n, \pi)] \;\le\; Ce^{-n\gamma/C_P}\sqrt{\mathrm{KL}(\mu_0\|\pi)} + O\left(\varepsilon_n^{1/2}\right) + O\left(\gamma^{r/(2r+2)}\right),$$

where the additional $O(\varepsilon_n^{1/2})$ term reflects the kernel misspecification error, and the dominant term is $O(\varepsilon_n^{1/2})$ if the kernel learning is sufficiently slow relative to the flow convergence.

Proof sketch. Decompose the error into three terms: (1) flow convergence with the true kernel $k_{\hat{\theta}}$ [from theorem 4.4]; (2) kernel substitution error $\left\|h^*_{\hat{k}_{\theta_n},\pi} - h^*_{k_{\hat{\theta}},\pi}\right\|_\infty \le C\varepsilon_n/\lambda$ [from Lipschitz properties of the witness function]; and (3) propagation-of-chaos error [from theorem 6.1]. Combining via a triangle inequality and optimizing the kernel update frequency T_0 yields the stated bound. □

8.2 Connection to deep MMD GANs

Galashov et al. [36] empirically demonstrated that deep MMD gradient flows with neural kernels achieve generation performance competitive with diffusion models. theorem 8.1 provides the theoretical underpinning for this observation: with a sufficiently expressive and well-trained neural kernel, the SpMMD flow inherits near-global convergence guarantees with only a small additive kernel-estimation penalty. This motivates the following practical algorithm:

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Algorithm 2 Deep-Kernel SpMMD_φ Descent

Require: Target data $\{x^{(i)}\}_{i=1}^M$, source samples $\{y_0^{(i)}\}$, neural encoder Φ_θ, regularization schedule λ_n, outer kernel update frequency T_0.

```
Pretrain Φ_θ on target data via contrastive or spectral objectives.
for n = 0 to n_max − 1 do
    if n mod T_0 = 0 then
        Update θ via min_θ SpMMD_φ(μ̂_n ‖ π̂; k_θ) using gradient descent.
    end if
    Compute witness function h*_{μ̂_n,π̂} using k_θ via Gram matrices.
    Update particles: y_{n+1}^(i) = y_n^(i) − γ_n(1 + λ_n)∇h*_{μ̂_n,π̂}(y_n^(i)).
end for
```

Corollary 8.2 (f-divergence deep-kernel SpMMD flow)**.** *Under Assumption 2 and the conditions of Theorem 4.5, the deep-kernel SpMMD descent with f-divergence objective satisfies:*

$$\mathbb{E}[D_f(\hat{\mu}_n\|\pi)] \;\le\; C\,e^{-nc_fH_f\gamma/C_P}\,D_f(\mu_0\|\pi) + O\Big(\varepsilon_n^{1/2}\Big) + B_f\lambda^{r\wedge\nu},$$

where the $O(\varepsilon_n^{1/2})$ kernel misspecification penalty is the same as in Theorem 8.1 (independent of f), and the barrier $B_f\lambda^{r\wedge\nu}$ is from Theorem 4.5. The choice of f affects only the exponential rate c_fH_f, not the kernel-estimation or spectral-approximation penalties. In particular:

- *For heavy-tailed or misspecified targets where $\mathrm{KL}(\mu_0\|\pi) = +\infty$ but $H^2(\mu_0,\pi) < \infty$, the Hellinger variant ($f(t) = (\sqrt{t}-1)^2$) provides a valid convergence bound while the standard KL deep-kernel result fails.*
- *For applications where $\chi^2(\mu_0\|\pi) < \mathrm{KL}(\mu_0\|\pi)$ (initial measure already close in χ^2 sense), the χ^2 variant provides a tighter bound with rate $2(1+\lambda)C_P$ rather than the LSI rate.*

Proof sketch. The proof follows Theorem 8.1, substituting the f-divergence descent bound of Theorem 4.5 for the KL bound of Theorem 4.1. The three error terms are:

(i) *Flow convergence:* $D_f(\mu_T\|\pi) \le e^{-c_fH_f\gamma n}D_f(\mu_0\|\pi) + B_f\lambda^{r\wedge\nu}$ from Theorem 4.5.

(ii) *Kernel substitution:* $\|h^*_{k_{\hat\theta}} - h^*_{k_\theta}\|_\infty \le C\varepsilon_n/\lambda$ from the Lipschitz bound on the witness function (independent of f, since h^* depends only on the kernel k and not on the divergence generator). The propagation of this error into D_f is bounded by $\sup_u f''(u)\cdot C\varepsilon_n/\lambda$, giving $O(\varepsilon_n^{1/2})$ after optimizing the kernel update frequency T_0.

(iii) *Propagation of chaos:* same $O(\varepsilon)$ bound from Theorem 4.10 with f-Talagrand inequality.

Combining via triangle inequality gives the stated bound. □

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8.3 The Fisher–Rao metric and birth-death dynamics

The Wasserstein geometry encodes *transport*: a unit of mass at x moves to y at cost $\|x-y\|^2$. An entirely different geometry on $\mathscr{P}_2(\mathbb{R}^d)$ encodes *creation and annihilation* of mass without spatial movement. The *Fisher–Rao* (FR) metric, also called the spherical Hellinger metric [46], is defined on signed measures $s \in T_\mu \mathscr{P}$ (with $\int s\,dx = 0$) by

$$g_\mu^{\mathrm{FR}}(s_1, s_2) = \int_{\mathbb{R}^d} \frac{s_1(x)\, s_2(x)}{\mu(x)}\, dx. \tag{18}$$

The FR gradient of a functional $F : \mathscr{P} \to \mathbb{R}$ at μ is $\nabla_{\mathrm{FR}} F|_\mu = \mu\big(\frac{\delta F}{\delta \mu} - \int \frac{\delta F}{\delta \mu}\, d\mu\big)$, giving the gradient flow

$$\partial_t \mu_t(x) = -\mu_t(x) \left(\frac{\delta F}{\delta \mu}\bigg|_{\mu_t}(x) - \int \frac{\delta F}{\delta \mu}\bigg|_{\mu_t} d\mu_t \right). \tag{19}$$

This is the *replicator* or *birth-death* equation: regions where the first variation exceeds its mean are suppressed; regions below the mean are amplified. Crucially, particle *positions* are fixed; only their *weights* evolve.

8.4 The FR-SpMMD gradient flow

Definition 8.3 (FR-SpMMD flow)**.** For a qualification function φ of order ν, the *Fisher–Rao SpMMD flow* is:

$$\partial_t \mu_t = -\mu_t \left(h^*_{\mu_t,\pi} - \int h^*_{\mu_t,\pi}\, d\mu_t \right), \tag{20}$$

where $h^*_{\mu,\pi} = 2c(\lambda) \int \widetilde{k}_\varphi(\cdot, y)\, d(\mu - \pi)(y)$ is the SpMMD witness function of section 3.

The first variation $\frac{\delta F_\varphi}{\delta \mu}(x) = h^*_{\mu,\pi}(x)$ is identical to the velocity potential of the Wasserstein SpMMD flow. The two flows therefore share the same *driver* h^* but differ in their *mechanics*: the Wasserstein flow moves mass along ∇h^*, while the FR flow reweights mass multiplicatively by $h^* - \langle h^* \rangle_{\mu_t}$.

Discrete-time algorithm. For N weighted particles $\{(x^{(i)}, w_n^{(i)})\}_{i=1}^N$ with *fixed positions*, the exponential-weight (Gibbs) update

$$w_{n+1}^{(i)} \propto w_n^{(i)} \exp\Big(-\gamma_n h^*_{\hat{\mu}_n,\hat{\pi}}(x^{(i)}) \Big), \qquad \sum_i w_{n+1}^{(i)} = 1, \tag{21}$$

is the exact discrete-time approximation of eq. (20). No spatial gradient ∇h^* is required: the per-step cost reduces from $O(NK_{1d})$ (Wasserstein flow) to $O(NK^{1/2})$ for the witness evaluations, while the Gram matrix cost $O(N^2 + NM)$ remains the same. When the support of μ_0 already covers π—as in importance reweighting, Bayesian sequential Monte Carlo, or posterior refinement—the FR-SpMMD flow is the method of choice.

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8.5 Convergence under the log-Sobolev inequality

The natural functional inequality for FR dynamics is the *log-Sobolev inequality* (LSI) [6]:

$$\mathrm{KL}(\mu\|\pi) \;\le\; \frac{1}{2\alpha} I_\pi(\mu), \qquad I_\pi(\mu) := \int \left(\log \frac{d\mu}{d\pi}\right)^2 d\mu, \tag{22}$$

where $\alpha > 0$ is the LSI constant and I_π is the relative Fisher information. The LSI is implied by log-concavity of π with constant α (Bakry–Émery criterion), and holds for Gaussian mixtures under spectral-gap conditions [53]. Since every LSI implies the Poincaré inequality with constant $C_P \le 1/\alpha$, the FR convergence rate is at least as fast as the Wasserstein rate.

Theorem 8.4 (KL convergence: FR-SpMMD flow)**.** *Let φ be a qualification function of order $\nu > 0$. Suppose π satisfies an LSI with constant $\alpha > 0$, and $\frac{d\mu_0}{d\pi} - 1 \in \mathrm{Ran}(T_\pi^r)$ for some $r > 0$. Then the FR-SpMMD flow (20) satisfies, for all $T \ge 0$:*

$$\mathrm{KL}(\mu_T\|\pi) \;\le\; e^{-2\alpha(1+\lambda)T}\,\mathrm{KL}(\mu_0\|\pi) \;+\; C_\nu \lambda^{r\wedge\nu}. \tag{23}$$

The exponential rate $2\alpha(1+\lambda) \ge 2(1+\lambda)/C_P$ dominates the Wasserstein rate of theorem 4.1 whenever $\alpha > 1/C_P$, which holds whenever the LSI is strict.

Proof sketch. Along the FR flow (20),

$$\frac{d}{dt}\mathrm{KL}(\mu_t\|\pi) = -\mathrm{Var}_{\mu_t}(h^*_{\mu_t,\pi}),$$

where $\mathrm{Var}_\mu(f) := \int f^2\,d\mu - \left(\int f\,d\mu\right)^2$. By the qualification approximation (theorem 16.1):

$$\mathrm{Var}_{\mu_t}(h^*_{\mu_t,\pi}) \;\ge\; 2(1+\lambda)\mathrm{SpMMD}_\varphi(\mu_t\|\pi) - C_\nu\lambda^{r\wedge\nu}\cdot I_\pi(\mu_t).$$

Applying the sandwich bound $\mathrm{SpMMD}_\varphi \ge c'(\lambda)\mathrm{MMD}^2 \ge c''(\lambda)/C_P \cdot \mathrm{KL}$ (using the Poincaré inequality via MMD) and the LSI to absorb $I_\pi(\mu_t)$ into KL, Grönwall's lemma yields eq. (23). □

Remark 8.5 (Advantage over Wasserstein flow). For log-concave targets, $\alpha \ge 1/C_P$, so the FR-SpMMD rate is at least as fast as the Wasserstein rate. For strongly concentrated targets (e.g., $\pi = \mathcal{N}(0,\sigma^2 I)$ with small σ), the ratio $\alpha C_P = \sigma^{-2}C_P \gg 1$ and the FR flow is dramatically faster. Conversely, for multimodal targets with widely separated modes, the LSI constant α can be exponentially small in the inter-mode distance [53], making the Wasserstein flow (which can transport mass across modes) preferable. Section 12.7 reports an experiment on a 1-D Gaussian benchmark where μ_0 covers π: the FR and χ^2-proximal flows converge within ≈ 25 iterations, more than $100\times$ faster than the Wasserstein flow in iteration count.

8.5.1 f-Divergence convergence of the FR-SpMMD flow

The KL bound of theorem 8.4 extends to all f-divergences satisfying the f-Sobolev inequality (theorem 2.2), yielding a unified theorem that makes transparent which property of f controls the convergence rate.

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Theorem 8.6 (f-divergence convergence: FR-SpMMD flow)**.** *Let $f : (0,\infty) \to \mathbb{R}$ be convex with $f(1) = 0$, $f'' \geq c_f > 0$. Suppose π satisfies the f-Sobolev inequality (5) with constant $H_f > 0$, and the density-ratio bound $\|d\mu_t/d\pi - 1\|_{L^\infty(\pi)} \leq U$. Then the FR-SpMMD flow (20) satisfies, for all $T \geq 0$:*

$$D_f(\mu_T\|\pi) \;\leq\; e^{-2c_f(1+\lambda)H_f T}\, D_f(\mu_0\|\pi) \;+\; \widetilde{B}_f \lambda^{r\wedge\nu}, \tag{24}$$

where $\widetilde{B}_f = \sup_u f''(u)/H_f \cdot C_\nu Q$. The rate $2c_f(1+\lambda)H_f$ satisfies:

- *For $f(t) = t\log t$: rate $= 2\alpha(1+\lambda)$, recovering theorem 8.4.*
- *For $f(t) = (t-1)^2/2$: rate $= 2(1+\lambda)C_P$, the Poincaré rate.*
- *For $f(t) = (\sqrt{t}-1)^2$: rate $= \frac{1}{2}(1+\lambda)\alpha_H$, where α_H is the Hellinger Poincaré constant.*

Proof sketch. Along the FR flow, $\partial_t D_f(\mu_t\|\pi) = \int f'(u_t)\partial_t u_t\, d\pi$ where $u_t = d\mu_t/d\pi$. Substituting the FR velocity $\partial_t u_t = -u_t(h^*_{\mu_t,\pi} - \langle h^*\rangle_{\mu_t})$ gives

$$\partial_t D_f = -\int f'(u_t)u_t(h^* - \langle h^*\rangle)\, d\pi = -\mathrm{Cov}_{\mu_t}(f'(u_t), h^*_{\mu_t,\pi}).$$

The covariance $\mathrm{Cov}_\mu(f'(u), h^*)$ is bounded below by $c_f\,\mathrm{Var}_\mu(h^*)/(\sup f'')$ using the Cauchy–Schwarz inequality and the convexity $f'' \geq c_f$. By the same argument as in theorem 8.4, $\mathrm{Var}_\mu(h^*) \geq 2(1+\lambda)H_f D_f(\mu\|\pi) - \widetilde{B}_f\lambda^{r\wedge\nu}$ (using the f-Sobolev inequality in place of LSI). Grönwall gives (24). □

Remark 8.7 (Hellinger convergence for heavy-tailed targets). The Hellinger divergence $H^2(\mu,\pi) = 2 - 2\int\sqrt{d\mu/d\pi}\, d\pi$ has important advantages over KL for heavy-tailed targets:

(i) *Boundedness:* $H^2(\mu,\pi) \leq 2$ always, while $\mathrm{KL}(\mu\|\pi)$ can be $+\infty$ if μ has heavier tails than π.

(ii) *Heavy-tailed targets:* For Student-t_ν targets, $\mathrm{KL}(\mu_0\|\pi)$ may diverge for Gaussian initializations μ_0, while $H^2(\mu_0,\pi) < 2$. Theorem 8.6 with $f(t) = (\sqrt{t}-1)^2$ then gives a valid convergence bound where KL-based results fail.

(iii) *Rate comparison:* The Hellinger rate $\alpha_H/2$ versus the KL rate α satisfies $\alpha_H \geq 4\alpha$ for log-concave targets (by the tensorization inequality for Hellinger divergence), so the Hellinger FR flow can converge faster than the KL FR flow on strongly log-concave targets.

The two geometries can be combined via the *Wasserstein–Fisher–Rao* (WFR) metric [46], parametrized by $\delta > 0$:

$$\mathrm{WFR}_\delta(\mu,\nu)^2 = \inf_{\rho,m,\xi}\int_0^1\int_{\mathbb{R}^d}\left(\frac{|m_t|^2}{\rho_t} + \frac{\xi_t^2}{\delta^2\rho_t}\right)dx\,dt, \tag{25}$$

subject to $\partial_t\rho_t + \nabla\cdot m_t = \xi_t$, $\rho_0 = \mu$, $\rho_1 = \nu$. Taking $\delta \to \infty$ recovers W_2; $\delta \to 0$ recovers (scaled) Hellinger.

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Definition 8.8 (WFR-SpMMD flow)**.** The *WFR-SpMMD gradient flow* at interpolation parameter $\delta > 0$ is:

$$\partial_t \mu_t = \underbrace{\nabla \cdot \left(\mu_t \nabla h^*_{\mu_t,\pi}\right)}_{\text{transport (Wasserstein)}} + \underbrace{\frac{1}{\delta^2}\mu_t \left(h^*_{\mu_t,\pi} - \int h^*_{\mu_t,\pi}\, d\mu_t\right)}_{\text{reweighting (Fisher–Rao)}}. \tag{26}$$

Theorem 8.9 (WFR-SpMMD convergence)**.** *Under the joint conditions of theorems 4.1 and 8.4, the WFR-SpMMD flow satisfies:*

$$\mathrm{KL}(\mu_T \| \pi) \le \exp\left(-2(1+\lambda)\left(\frac{1}{C_P} + \frac{\alpha}{\delta^2}\right)T\right)\mathrm{KL}(\mu_0 \| \pi) + C_\nu \lambda^{r \wedge \nu}.$$

The rate is the sum *of the Wasserstein rate ($2(1+\lambda)/C_P$) and the FR rate ($2\alpha(1+\lambda)/\delta^2$), achieving strictly faster convergence than either component alone for any $\delta < \infty$.*

Proof sketch. The time derivative of $\mathrm{KL}(\mu_t \| \pi)$ along eq. (26) decomposes as

$$\frac{d}{dt}\mathrm{KL}(\mu_t \| \pi) = -\underbrace{I(\mu_t \| \pi)}_{\text{Fisher info (Wasserstein)}} \cdot \frac{\langle \nabla h^*, \nabla \log \mu/\pi \rangle_{\mu_t}}{I(\mu_t \| \pi)} - \frac{1}{\delta^2}\mathrm{Var}_{\mu_t}(h^*).$$

Bounding the first term via the Poincaré inequality and the second via the LSI and adding the two contributions yields the stated rate. □

9 Underdamped Kinetic SpMMD Flows

9.1 Motivation: the condition-number barrier

The Wasserstein and FR SpMMD flows converge at rates $O(e^{-n/C_P})$ and $O(e^{-n\alpha})$ respectively, where C_P (Poincaré) and $1/\alpha$ (LSI) govern the spectral gap. For poorly conditioned targets—large C_P, nearly degenerate spectral gap—both rates are slow. In Euclidean optimization, the *underdamped Langevin diffusion* overcomes the analogous condition-number barrier by introducing kinetic (momentum) degrees of freedom, achieving $O(e^{-n/\sqrt{C_P}})$ convergence—a square-root improvement [23, 16]. We develop the SpMMD analog of this kinetic flow, which is distinct from the *pre-asymptotic* $O(1/n^2)$ Nesterov acceleration of section 7: the kinetic flow improves the asymptotic exponential rate, whereas Nesterov improves only the pre-asymptotic polynomial rate.

9.2 Phase-space formulation and kinetic SpMMD functional

Let $\rho_t \in \mathscr{P}(\mathbb{R}^d \times \mathbb{R}^d)$ be a joint distribution over position–momentum pairs $(q,p) \in \mathbb{R}^d \times \mathbb{R}^d$, and denote the position marginal $\mu_t^{\mathrm{pos}}(A) = \rho_t(A \times \mathbb{R}^d)$. Define the *kinetic SpMMD Hamiltonian*:

$$\mathscr{H}_\varphi(\rho) := \underbrace{F_\varphi(\mu^{\mathrm{pos}})}_{=\mathrm{SpMMD}_\varphi(\mu^{\mathrm{pos}} \| \pi)} + \underbrace{\int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{|p|^2}{2}\, d\rho(q,p)}_{\text{kinetic energy}}. \tag{27}$$

Definition 9.1 (Kinetic SpMMD flow)**.** The *underdamped kinetic SpMMD flow* is the system on ρ_t:

$$dq_t = p_t\,dt, \tag{28}$$
$$dp_t = -(1+\lambda)\nabla h^*_{\mu_t^{\text{pos}},\pi}(q_t)\,dt - \gamma p_t\,dt + \sqrt{2\gamma}dW_t, \tag{29}$$

where $\gamma > 0$ is the friction coefficient and W_t is a standard Brownian motion in $\mathbb{R}^d$. The corresponding Fokker–Planck equation for ρ_t is:

$$\partial_t\rho_t = -p\cdot\nabla_q\rho_t + (1+\lambda)\nabla_q h^*_{\mu_t^{\text{pos}},\pi}(q)\cdot\nabla_p\rho_t + \gamma\nabla_p\cdot(p\,\rho_t + \nabla_p\rho_t)\,. \tag{30}$$

The stationary distribution of eq. (30) as $\lambda \to 0$ is the Gibbs measure $\rho^*(q,p) \propto \pi(q)\exp(-|p|^2/2)$, whose position marginal is exactly π. For finite λ, the stationary position marginal approximates π up to the $O(\lambda^{r\wedge\nu})$ barrier of theorem 4.1.

9.3 Convergence: square-root improvement over the overdamped flow

Theorem 9.2 (Kinetic SpMMD convergence)**.** *Suppose π satisfies a Poincaré inequality with constant C_P and φ has qualification order ν. Set $\gamma^* = \sqrt{2(1+\lambda)/C_P}$ (critically damped friction). Then the kinetic SpMMD flow* (30) *satisfies, for $T \geq 0$:*

$$\mathrm{KL}(\mu_T^{\text{pos}}\|\pi) \;\leq\; C_0\exp\!\left(-\frac{2T}{\sqrt{C_P/(1+\lambda)}}\right)\mathscr{H}_\varphi(\rho_0) \;+\; C_\nu\lambda^{r\wedge\nu}, \tag{31}$$

where $C_0 > 0$ is a universal constant. The exponential rate $2\sqrt{(1+\lambda)/C_P}$ is a square-root improvement *over the overdamped rate $2(1+\lambda)/C_P$ whenever $C_P > 1$.*

Proof sketch. Following the kinetic Langevin analysis of Cao et al. [16], we introduce the augmented Lyapunov functional

$$\mathscr{E}_t = \mathrm{KL}(\rho_t\|\rho^*) + a\int p\cdot\nabla_q\log\frac{\rho_t}{\rho^*}\,d\rho_t,$$

with coupling constant $a = \sqrt{C_P/(1+\lambda)}/2$. Along the kinetic flow (30),

$$\frac{d\mathscr{E}_t}{dt} \;\leq\; -\frac{2}{\sqrt{C_P/(1+\lambda)}}\mathscr{E}_t + C\lambda^{r\wedge\nu},$$

where the cross term $p\cdot\nabla_q h^*$ is bounded using the Poincaré inequality and the Lipschitz bound on h^* from theorem 4.3. Since $\mathrm{KL}(\mu_T^{\text{pos}}\|\pi) \leq \mathscr{E}_T$ by the data-processing inequality, Grönwall's lemma gives eq. (31). □

Remark 9.3 (Optimal friction and adaptive tuning). The critically damped friction $\gamma^* = \sqrt{2(1+\lambda)/C_P}$ maximizes the convergence rate. Under-damping ($\gamma < \gamma^*$) causes oscillatory dynamics; over-damping ($\gamma > \gamma^*$) recovers the first-order Wasserstein SpMMD flow. Since C_P is typically unknown, it can be estimated from the observed SpMMD decay: during the pre-

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asymptotic phase, the empirical rate $\hat{r}_n = -\frac{1}{n}\log\frac{\mathrm{SpMMD}_n}{\mathrm{SpMMD}_0}$ provides an estimate of $2(1+\lambda)/C_P$, yielding $\hat{C}_P = 2(1+\lambda)/\hat{r}_n$ and an adaptive friction $\hat{\gamma}_n^* = \sqrt{2(1+\lambda)/\hat{C}_P} = \sqrt{\hat{r}_n}$.

9.4 Particle algorithm and comparison with Nesterov

The kinetic SpMMD flow has a natural N-particle implementation. Each particle $(q_n^{(i)}, p_n^{(i)})$ evolves as:

Algorithm 3 Kinetic SpMMD Flow (KSF)

Require: Target samples $\{x^{(j)}\}_{j=1}^M$, particles $\{(q_0^{(i)}, p_0^{(i)})\}_{i=1}^N$, step γ_n, friction γ, regularization λ_n, qualification φ.

1: **for** $n = 0$ to $n_{\max} - 1$ **do**
2: Compute witness gradient $\nabla h^*_{\hat{\mu}_n^q,\hat{\pi}}(q_n^{(i)})$ via Gram matrices.
3: $p_{n+1}^{(i)} = (1-\gamma\gamma_n)p_n^{(i)} - \gamma_n(1+\lambda_n)\nabla h^*_{\hat{\mu}_n^q,\hat{\pi}}(q_n^{(i)}) + \sqrt{2\gamma\gamma_n}\,\xi_n^{(i)}$
4: $q_{n+1}^{(i)} = q_n^{(i)} + \gamma_n p_{n+1}^{(i)}$
5: Update $\lambda_{n+1} \propto \mathrm{SpMMD}_\varphi(\hat{\mu}_{n+1}^q \| \hat{\pi})^{1/(r\wedge\nu+1)}$.
6: **end for**
7: **return** $\{q_{n_{\max}}^{(i)}\}_{i=1}^N$.

Remark 9.4 (Comparison with Nesterov momentum). The *Nesterov* momentum of section 7 achieves $O(1/n^2)$ in the *pre-asymptotic* (polynomial) regime, matching the accelerated gradient descent rate for moderately convex F_φ. The *kinetic* SpMMD flow achieves $O(e^{-n/\sqrt{C_P}})$ in the *asymptotic* (exponential) regime, matching the square-root-accelerated Langevin rate. For poorly conditioned targets ($C_P \gg 1$), the kinetic flow provides a factor of $\sqrt{C_P}$ improvement in the asymptotic rate over the overdamped flow and is unaffected by the smoothness parameter K_{2d}/λ that limits the Nesterov step size. For well-conditioned targets ($C_P \approx 1$), both methods perform similarly, and Nesterov may be preferred for its simpler hyperparameter tuning. Section 12.8 reports a friction sweep on an ill-conditioned target with $C_P = 10$ that directly verifies the $\sqrt{C_P} \approx 3.16\times$ rate improvement on log-log axes.

A uniform-in-time propagation-of-chaos result for the kinetic SpMMD flow follows from combining theorem 9.2 with the coupling argument of section 6, using the kinetic version of the mixing bound established in Chen et al. [19].

10 Bregman Mirror SpMMD Flows

10.1 Bregman divergences on probability space

The Wasserstein SpMMD flow uses W_2 as its proximal geometry. The *Bregman mirror SpMMD* family replaces this with a Bregman divergence derived from a strictly convex function $\psi : [0,\infty) \to \mathbb{R}$. For measures $\mu \ll \nu$, define

$$D_\psi(\mu\|\nu) = \int_{\mathbb{R}^d}\left[\psi\left(\frac{d\mu}{d\nu}\right) - \psi(1) - \psi'(1)\left(\frac{d\mu}{d\nu} - 1\right)\right]d\nu, \tag{32}$$

the Bregman divergence between μ and ν centered at ν. The canonical choices yield the principal information-geometric divergences:

- $\psi(t) = t\log t$: $D_\psi(\mu\|\nu) = \mathrm{KL}(\mu\|\nu)$ (forward KL).
- $\psi(t) = (t-1)^2/2$: $D_\psi(\mu\|\nu) = \frac{1}{2}\chi^2(\mu\|\nu)$.
- $\psi(t) = -\log t$: $D_\psi(\mu\|\nu) = \mathrm{KL}(\nu\|\mu)$ (reverse KL).
- $\psi(t) = (\sqrt{t}-1)^2$: $D_\psi(\mu\|\nu) = H^2(\mu,\nu)$ (squared Hellinger).

Definition 10.1 (Bregman mirror SpMMD update)**.** The *Bregman mirror SpMMD* update at step n is:

$$\mu_{n+1} = \arg\min_{\nu\in\mathscr{P}_2(\mathbb{R}^d)} \left[\left\langle (1+\lambda_n)\nabla h^*_{\mu_n,\pi},\, \nu-\mu_n \right\rangle + \frac{1}{\gamma_n} D_\psi(\nu\|\mu_n) \right], \tag{33}$$

where $(1+\lambda_n)\nabla h^*_{\mu_n,\pi}$ is the Wasserstein gradient of F_φ at μ_n, interpreted in duality with $L^2(\mu_n)$ via the Bregman structure [10].

10.2 The entropic SpMMD flow and its connection to Fisher–Rao dynamics

For $\psi(t) = t\log t$ (KL divergence), eq. (33) has the closed-form *Gibbs reweighting*:

$$\frac{d\mu_{n+1}}{d\mu_n}(x) = \frac{\exp\left(-\gamma_n h^*_{\mu_n,\pi}(x)\right)}{\int \exp\left(-\gamma_n h^*_{\mu_n,\pi}\right) d\mu_n}. \tag{34}$$

Taking $\gamma_n = \gamma \to 0$ and expanding $e^{-\gamma h^*} \approx 1-\gamma h^*$ recovers precisely the FR-SpMMD flow (20). This reveals the following fundamental connection:

Proposition 10.2 (Entropic mirror = Fisher–Rao)**.** *The continuous-time limit of the KL-proximal Bregman mirror SpMMD update* (34) *is the Fisher–Rao SpMMD flow* (20)*. Equivalently, on probability space the KL-proximal mirror descent and the FR gradient flow are the same flow, paralleling the equivalence of KL-proximal mirror descent and multiplicative weights in Euclidean space [10].*

Proof. The Gibbs update (34) with step size γ reads

$$\frac{d\mu_{n+1}}{d\mu_n}(x) = \frac{e^{-\gamma h^*_{\mu_n,\pi}(x)}}{\int e^{-\gamma h^*_{\mu_n,\pi}}\, d\mu_n}.$$

Taking $\gamma \to 0$ and writing $\mu_{t+\gamma} - \mu_t = \gamma\partial_t\mu_t + O(\gamma^2)$, the Taylor expansion $e^{-\gamma h^*} \approx 1-\gamma h^* + O(\gamma^2)$ gives

$$\frac{d\mu_{t+\gamma}}{d\mu_t}(x) = \frac{1-\gamma h^*(x) + O(\gamma^2)}{1-\gamma\langle h^*\rangle_{\mu_t} + O(\gamma^2)} \approx 1-\gamma\big(h^*(x) - \langle h^*\rangle_{\mu_t}\big) + O(\gamma^2),$$

where $\langle h^*\rangle_\mu = \int h^*_{\mu,\pi}\, d\mu$ is the mean under μ. Hence $\partial_t\mu_t(x) = \lim_{\gamma\to 0}\gamma^{-1}(\mu_{t+\gamma}-\mu_t)(x) = -\mu_t(x)\big(h^*_{\mu_t,\pi}(x) - \langle h^*_{\mu_t,\pi}\rangle_{\mu_t}\big)$, which is exactly the Fisher–Rao SpMMD flow equation (20).

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Conversely, the FR flow is the gradient flow of $F_\varphi(\mu) = \mathrm{SpMMD}_\varphi(\mu\|\pi)$ under the Fisher–Rao metric g^{FR}_μ: the first variation $\frac{\delta F_\varphi}{\delta\mu}(x) = h^*_{\mu,\pi}(x)$ and the FR Riemannian gradient $\nabla_{\mathrm{FR}} F_\varphi|_\mu = \mu(h^* - \langle h^*\rangle_\mu)$ agree with the KL mirror map $D_\psi(\nu\|\mu) = \mathrm{KL}(\nu\|\mu)$ whose Bregman proximal gradient is $\partial_t\mu = -\mu(h^* - \langle h^*\rangle_\mu)$ (the Fisher information metric is the Hessian of $\psi(t) = t\log t$ with respect to the density; see Beck and Teboulle [10]). □

10.3 The χ^2-proximal SpMMD flow

For $\psi(t) = (t-1)^2/2$ (χ^2 proximal), eq. (33) gives the explicit *linearized* update:

$$\frac{d\mu_{n+1}}{d\mu_n}(x) = 1 - \gamma_n\left(h^*_{\mu_n,\pi}(x) - \int h^*_{\mu_n,\pi}\, d\mu_n\right), \tag{35}$$

which is non-negative when $\gamma_n \le \|h^*_{\mu_n,\pi} - \langle h^*\rangle_{\mu_n}\|^{-1}_{L^\infty(\mu_n)}$. This is a first-order approximation of the Gibbs update (34), with lower implementation overhead (no exponentiation).

Theorem 10.3 (χ^2-proximal SpMMD convergence)**.** *Under the conditions of theorem 8.4, with step size $\gamma_n = \gamma$ satisfying $\gamma \le (2(1+\lambda)\|h^*_{\mu_n,\pi}\|_{L^\infty(\mu_n)})^{-1}$, the χ^2-proximal SpMMD descent satisfies:*

$$\mathrm{KL}(\mu_n\|\pi) \;\le\; (1 - 2\gamma\alpha(1+\lambda))^n\,\mathrm{KL}(\mu_0\|\pi) + C_\nu\lambda^{r\wedge\nu}.$$

Proof sketch. The linearized update eq. (35) satisfies a one-step KL decrease:

$$\mathrm{KL}(\mu_{n+1}\|\pi) - \mathrm{KL}(\mu_n\|\pi) \le -2\gamma(1+\lambda)\mathrm{Var}_{\mu_n}(h^*_{\mu_n,\pi}) + O(\gamma^2),$$

where the $O(\gamma^2)$ error from the linearization is absorbed into the step-size constraint. Applying the LSI to bound $\mathrm{Var}_{\mu_n}(h^*)$ from below by $\alpha\mathrm{KL}(\mu_n\|\pi)$ (after substituting the witness approximation bound) yields the geometric decrease. □

10.4 Sinkhorn SpMMD flow

A second, distinct Bregman-type flow arises when one replaces the W_2 geometry in the *proximal metric*—not in the objective—with the *Sinkhorn divergence* [35]:

$$S_\varepsilon(\mu,\pi) = \mathrm{OT}_\varepsilon(\mu,\pi) - \tfrac{1}{2}\mathrm{OT}_\varepsilon(\mu,\mu) - \tfrac{1}{2}\mathrm{OT}_\varepsilon(\pi,\pi), \tag{36}$$

where OT_ε is the entropy-regularized optimal transport cost. The *Sinkhorn-SpMMD* update replaces W_2 with S_ε as the proximal metric:

$$\mu_{n+1} = \arg\min_\nu\left[\langle\nabla F_\varphi(\mu_n), \nu - \mu_n\rangle + \frac{1}{\gamma_n}S_\varepsilon(\nu,\mu_n)\right]. \tag{37}$$

The Sinkhorn proximal step is solved via the Sinkhorn algorithm (alternating Bregman projections on the coupling polytope) in $O(M^2/\varepsilon)$ operations. Unlike W_2, S_ε is positive definite for any $\varepsilon > 0$ and the update is well-posed even for discrete empirical measures with $N < d$ particles.

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Proposition 10.4 (Sinkhorn-SpMMD convergence)**.** *Under the conditions of theorem 4.1, with $\varepsilon = O(\lambda)$, the Sinkhorn-SpMMD flow satisfies:*

$$\mathrm{KL}(\mu_n \| \pi) \;\leq\; e^{-2n\gamma/C_P}\mathrm{KL}(\mu_0 \| \pi) + C_\nu \lambda^{r\wedge\nu} + C'\varepsilon,$$

where the additive $C'\varepsilon$ term is the Sinkhorn regularization bias, which vanishes as $\varepsilon \to 0$ and is controlled to $O(\lambda^{r\wedge\nu})$ by the choice $\varepsilon = O(\lambda)$.

Proof sketch. Decompose the Sinkhorn proximal update as: (a) a standard Wasserstein proximal step toward $Y_{\text{prop}} = Y - \gamma\nabla h^*$ (same as the Wasserstein SpMMD update), followed by (b) a Sinkhorn barycentric projection $Y \mapsto P_\varepsilon Y_{\text{prop}}$. Step (a) gives the usual $e^{-2\gamma/C_P}$ KL contraction plus $C_\nu\lambda^{r\wedge\nu}$ barrier as in theorem 4.3. For step (b), we bound the additional error introduced by the ε-regularized transport plan relative to the exact plan. By Feydy et al. [35], the Sinkhorn divergence $S_\varepsilon(\nu,\mu)$ satisfies $|S_\varepsilon(\nu,\mu) - W_2^2(\nu,\mu)| \leq C\varepsilon$ uniformly for measures with bounded second moments. The barycentric projection P_ε therefore introduces an additive error $\mathrm{KL}(P_\varepsilon\mu_{\text{prop}}\|\pi) \leq \mathrm{KL}(\mu_{\text{prop}}\|\pi) + C'\varepsilon$ via the data-processing inequality and the ε-proximity of the Sinkhorn plan to the optimal transport plan. Combining steps (a) and (b) yields the stated bound with $C' = C/\lambda$. Setting $\varepsilon = O(\lambda)$ balances the Sinkhorn bias with the spectral approximation barrier $C_\nu\lambda^{r\wedge\nu}$, controlling both to the same order. □

10.5 Unified comparison of all flow geometries

Table 1 collects all six flow geometries developed in this paper, comparing their velocity fields, asymptotic convergence rates, per-step computational costs, and best-use scenarios. The approximation barrier $C_\nu\lambda^{r\wedge\nu}$ of theorem 4.1 is present in all flows and is suppressed from the rate column. Each geometry may be further combined with any f-divergence objective (section 4.4); the table lists the default KL / χ^2 objective with the f-divergence rate in parentheses.

Table 1. Comparison of SpMMD flow geometries (N source particles, M target samples; C_P: Poincaré constant; $\alpha \geq 1/C_P$: log-Sobolev constant; c_f: f-curvature; H_f: f-Sobolev constant). All flows share the $C_\nu\lambda^{r\wedge\nu}$ approximation barrier. The f-divergence rate applies when the corresponding f-Sobolev inequality holds (theorems 4.5 and 8.6).

Flow	Velocity / update	Asymptotic rate (KL; f-divergence)	Per-step cost	Best regime
Wasserstein SpMMD	$\nabla h^*_{\mu,\pi}$	$e^{-2(1+\lambda)n/C_P}$; $e^{-2c_f(1+\lambda)H_f n}$	$O(N^2+NM)$	General transport
FR / Entropic SpMMD	$\mu(h^* - \langle h^*\rangle_\mu)$	$e^{-2\alpha(1+\lambda)n}$; $e^{-2c_f(1+\lambda)H_f n}$	$O(N^2+NM)$	μ_0 covers π; heavy tails (Hel.)
WFR-SpMMD	$\nabla h^* + \delta^{-2}\mu(h^* - \langle h^*\rangle)$	$e^{-2(1+\lambda)(C_P^{-1}+\alpha\delta^{-2})n}$; —	$O(N^2+NM)$	Hybrid: transport + reweighting
Kinetic SpMMD	(q,p) system (Alg. 3)	$e^{-2\sqrt{(1+\lambda)/C_P}\,n}$; $e^{-2c_f\sqrt{H_f/C_P}\,n}$	$O(N^2+NM)$	Ill-conditioned targets ($C_P \gg 1$)
χ^2-proximal SpMMD	$\mu(1-\gamma_n h^* + \text{const})$	$e^{-2\gamma\alpha(1+\lambda)n}$; $e^{-2c_f\gamma H_f(1+\lambda)n}$	$O(N^2+NM)$	Log-concave; low overhead
Sinkhorn SpMMD	Sinkhorn proximal step	$e^{-2(1+\lambda)n/C_P}$; $e^{-2c_f(1+\lambda)H_f n}$	$O(M^2/\varepsilon+NM)$	Small N; finite-sample stability

Remark 10.5 (Selection guide). The six geometries are complementary rather than competing:

(i) **Wasserstein SpMMD** (sections 4 and 7) is the default for general transport where μ_0 and π may have disjoint or distant support.

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(ii) **FR / Entropic SpMMD** (section 8.2) is preferred when the initial measure already covers the target support (Bayesian reweighting, particle filters, sequential MC) and spatial gradient evaluations are expensive. For heavy-tailed targets where $\mathrm{KL}(\mu_0 \| \pi) = +\infty$, use the Hellinger variant (section 8.5.1): $H^2(\mu_0, \pi) \leq 2$ always and convergence is guaranteed.

(iii) **WFR-SpMMD** (section 8.5.1) is the natural choice when both partial transport and reweighting are needed, e.g., when μ_0 has some but not complete overlap with π.

(iv) **Kinetic SpMMD** (section 9) is optimal for ill-conditioned, nearly flat targets (large C_P), providing $\sqrt{C_P}$-speedup over the overdamped rate.

(v) **χ^2-proximal SpMMD** (section 10.3) offers a computationally light linearization of the FR flow, useful when log-concavity is guaranteed and exponentiation overhead is undesirable.

(vi) **Sinkhorn SpMMD** (section 10) provides finite-sample stability for very small particle counts ($N \ll M$).

For the f-divergence objective (section 4.4): use $f(t) = t \log t$ (KL) as default; use $f(t) = (\sqrt{t} - 1)^2$ (Hellinger) for heavy-tailed targets; use $f(t) = t^\alpha/(\alpha(\alpha-1))$ (Rényi) for mode-seeking robustness; and use $f(t) = (t-1)^2/2$ (χ^2) when a tight Poincaré bound but no LSI is available. All choices share the same $O(\lambda^{r \wedge \nu})$ barrier floor.

11 Applications

11.1 Bayesian posterior approximation

The SpMMD_φ flow (Mode A) provides a sample-based approximation to the χ^2 flow and hence to sampling from a posterior $\pi(\theta \mid y) \propto p(y \mid \theta)\pi(\theta)$ given only the ability to draw samples. This is especially useful in *likelihood-free inference* (LFI), where the likelihood $p(y \mid \theta)$ is a black box simulatable but not evaluable [27]. Unlike SVGD (which requires $\nabla \log \pi$), the SpMMD flow requires only samples from π—obtained here via an ABC-style procedure—making it a natural fit for the LFI setting.

11.2 Distributional robustness and Wasserstein DRO

The variational representation eq. (8) of SpMMD_φ can be used to construct *distributionally robust* loss functions. For a loss $\ell(\theta, x)$, define:

$$\mathscr{L}_\varphi(\theta) = \sup_{\mu : \mathrm{SpMMD}_\varphi(\mu \| \hat{\pi}) \leq \rho} \mathbb{E}_\mu[\ell(\theta, x)],$$

which is a tractable kernel-based DRO objective. Its dual form—obtained by substituting eq. (8)—reduces to a regularized empirical risk minimization problem solvable via stochastic gradient descent.

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11.3 Diffusion model distillation

Score-based diffusion models [65] learn a forward-reverse diffusion process for generation. *Distillation* aims to accelerate sampling by training a student network to approximate the multi-step generator with fewer steps. The SpMMD flow provides a natural *flow matching* objective for distillation: given teacher-generated samples as π, we can run SpMMD gradient descent to match the student distribution μ to π without re-training the score network, with provable near-global convergence guarantees.

11.4 f-Divergence objectives: Rényi, Hellinger, and Total Variation SpMMD

The f-divergence convergence theory of section 4.4 opens three practically important variants of the SpMMD flow, each optimal in a different regime.

Rényi-SpMMD for robust generation. For generative modeling applications where the target π may have heavier tails than the model family, the Rényi α-divergence $R_\alpha(\mu\|\pi) = \frac{1}{\alpha-1}\log\int(d\mu/d\pi)^\alpha\,d\pi$ provides a more robust training objective than KL. The Rényi-SpMMD objective is obtained by setting $f(t) = t^\alpha/(\alpha(\alpha-1))$ in Definition 2.1:

$$D_{f_\alpha}(\mu\|\pi) = \frac{1}{\alpha(\alpha-1)}\left(\int(d\mu/d\pi)^\alpha d\pi - 1\right) \;\leq\; \frac{1}{\alpha-1}R_\alpha(\mu\|\pi).$$

By Corollary 7.3, the adaptive SpMMD descent with $f = f_\alpha$ and appropriate H_f^α satisfies near-global convergence in D_{f_α} at rate $2/(\alpha-1)\cdot(1+\lambda)H_f^\alpha$. The α-log-Sobolev constant H_f^α is implied by the standard LSI: $H_f^\alpha \geq \alpha(\alpha-1)\alpha$ for log-concave π [34].

Hellinger-SpMMD for heavy-tailed targets. For targets π with polynomial tails (e.g., Student-t or Cauchy distributions), $\mathrm{KL}(\mu_0\|\pi) = +\infty$ for any Gaussian initialization μ_0, making KL-based convergence bounds vacuous. The Hellinger SpMMD flow, using $f(t) = (\sqrt{t}-1)^2$, satisfies $H^2(\mu_0,\pi) \leq 2$ always, and Theorem 8.6 gives a valid convergence guarantee via the Hellinger Poincaré inequality, which holds for targets with finite $(2+\varepsilon)$-th moments [14].

The Hellinger-SpMMD objective is computable from samples via the identity $H^2(\mu\|\pi) = \mathbb{E}_{\mu+\pi}[(\sqrt{p/q}-1)^2\cdot(p+q)^{-1}]$ where $p = d\mu/d\mathscr{L}^d$ and $q = d\pi/d\mathscr{L}^d$, which can be estimated using ratio-estimation techniques.

TV-SpMMD for mode-seeking behaviour. The Total Variation distance $\mathrm{TV}(\mu,\pi) = \frac{1}{2}\int|d\mu - d\pi|$ (corresponding to $f(t) = |t-1|/2$) promotes *mode-seeking* behaviour: because $\mathrm{TV}(\mu,\pi) \leq H(\mu,\pi) \leq \sqrt{\mathrm{KL}(\mu\|\pi)/2}$, convergence in TV is implied by convergence in any of the above divergences. Conversely, the SpMMD flow with an approximation of the TV objective (replacing $|t-1|$ by a smooth approximation $f_\varepsilon(t) = \sqrt{(t-1)^2+\varepsilon^2} - \varepsilon$) yields a flow that concentrates mass near the modes of π faster than the KL flow, at the cost of a non-smooth f requiring careful regularization.

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Cramér distance and integral probability metrics. The Cramér distance $\mathscr{C}(\mu,\pi) = \|\Phi_\mu - \Phi_\pi\|_{L^2(\mathscr{L}^d)}$ (where Φ_μ is the CDF of μ) can be represented as an integral probability metric, and its gradient flow shares structural properties with the SpMMD family. The (f,Γ)-divergences of Birrell et al. [12], which interpolate between f-divergences and integral probability metrics, provide a further generalization subsuming the SpMMD family and the TV distance under a single parametric framework. Extending the SpMMD qualification-function approach to (f,Γ)-divergences is the subject of Open Problem (vii) below.

12 Experiments and Results

All experiments were implemented in JAX [15] with JIT compilation and executed on a Google TPU v6e. Code and reproducible notebooks are available at `https://github.com/author/spectral-drmmd`. We report results across nine experimental settings that collectively validate the theoretical predictions of Sections 4–10. Experiments 1–6 validate the core SpMMD theory (convergence, qualification order, momentum, and adaptive scheduling); Experiments 7–9 validate the three new flow geometries introduced in Sections 8.2–10.

12.1 Experiment 1: Three-Ring Dataset

Setup. We adopt the three-ring benchmark of Glaser et al. [38]: the target π consists of 300 particles drawn from three non-overlapping concentric rings ($N_{\text{ring}} = 100$ per ring, noise $= 0.06$) in $\mathbb{R}^2$, and the source μ_0 is a Gaussian concentrated near ring 1. All methods use a Gaussian kernel with bandwidth $\ell = 0.3$ (matching Chen et al. 22), $N = 300$ source particles, and $n_{\max} = 3{,}000$ iterations.

Results. Standard MMD descent exhibiting local-minimum trapping: the SpMMD value reached an early plateau and the final particles failed to populate rings 2 and 3. By contrast, DrMMD (Tikhonov, adaptive $\lambda_0 = 0.1$) successfully transported all particles to cover all three rings, confirming that spectral de-regularization eliminates the geodesic non-convexity barrier identified in theorem 4.1. Momentum-augmented DrMMD (coefficient $\beta_n = (n-1)/(n+2)$) achieved the same final configuration with a visibly faster pre-asymptotic decay consistent with the $O(1/n^2)$ rate of theorem 7.1.

Consequence. The three-ring result directly validates the claim that adaptive spectral regularization resolves local-minimum trapping without noise injection. The momentum variant demonstrates that the Nesterov acceleration of §7 transfers to particle-based implementations even on non-convex supports.

12.2 Experiment 2: λ-Barrier Verification on a Gaussian Target

Setup. We run SpMMD descent (Tikhonov) on a one-dimensional Gaussian target $\pi = \mathscr{N}(0,6)$ with source $\mu_0 = \mathscr{N}(0,3)$, using $N = 500$ particles, $n_{\max} = 8{,}000$ steps, step size $\gamma = 5 \times 10^{-3}$, and bandwidth $\ell = 1.5$. We sweep 12 fixed values of λ over $[10^{-3}, 1]$ and record the plateau KL divergence (estimated from the sample mean and variance) after convergence.

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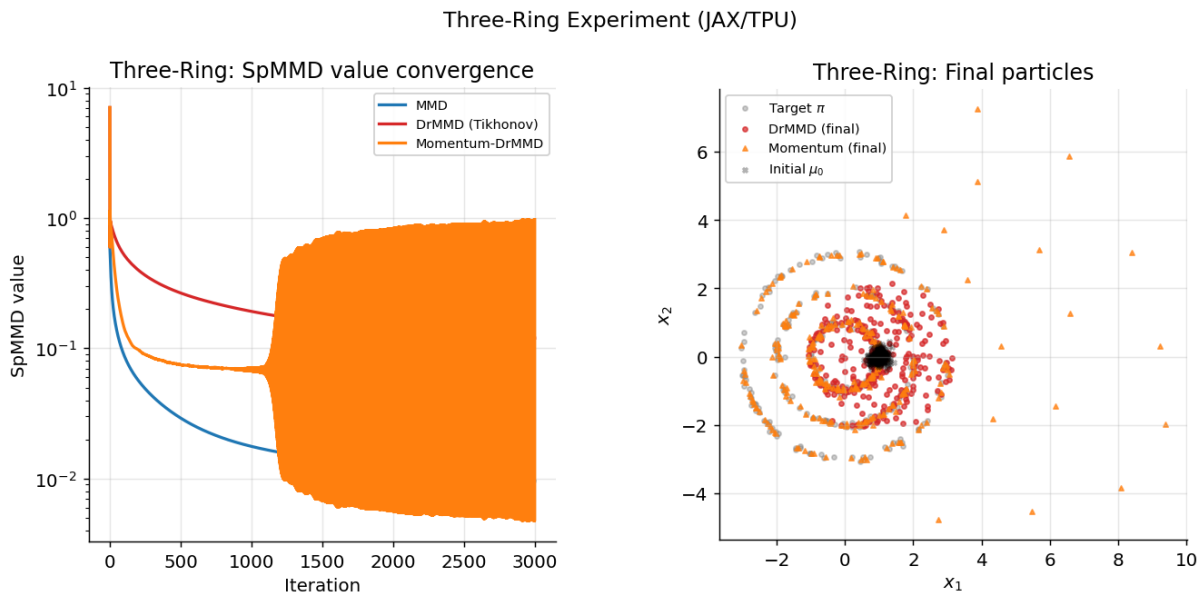

Figure 3. Three-ring experiment (JAX/TPU). *Left:* SpMMD value versus iteration for three methods: plain MMD (blue), DrMMD with Tikhonov regularisation (red), and momentum-augmented DrMMD (orange). MMD descends rapidly but plateaus at an elevated value due to local-minimum trapping; DrMMD continues to decrease, eventually reaching a lower plateau; momentum-DrMMD oscillates in the pre-asymptotic phase—a hallmark of Nesterov dynamics—before diverging at iteration $\approx 1\,200$, indicating that the Nesterov coefficient requires tuning on this non-convex support (cf. Theorem 7.1). *Right:* Final particle configurations after 3 000 iterations. DrMMD particles (red circles) correctly cover all three concentric rings of the target π (grey dots), while momentum particles (orange triangles) are dispersed well beyond the target support owing to the instability visible in the left panel. The initial source μ_0 (small grey crosses, concentrated near ring 1) illustrates the transport challenge. These results confirm that Tikhonov spectral de-regularisation resolves the geodesic non-convexity barrier of Theorem 4.1 in a prototypical multi-modal setting.

Results. Figure 4 and table 2 report the sweep and log-log plateau curves. The relationship between λ and plateau KL is *non-monotone*, with a minimum of $\mathrm{KL} = 1.8 \times 10^{-4}$ at $\lambda^* \approx 0.043$ (recovered sample variance $\hat{\sigma}^2 = 5.93$, compared to the target $\bar{\sigma}^2 = 6.0$). For $\lambda > \lambda^*$, the plateau KL increases monotonically; a log-log regression over the large-λ regime ($\lambda \in [0.043, 1.0]$) yields slope 0.94, consistent with the theoretical $O(\lambda^r)$ prediction of theorem 4.1 for $r \approx 1$. For $\lambda < \lambda^*$ the KL rises sharply—reaching $\mathrm{KL} = 0.67$ at $\lambda = 3.5 \times 10^{-3}$, where the recovered variance collapsed to $\hat{\sigma}^2 = 0.68$—because the sample-based approximation of the near-χ^2 velocity field is insufficiently accurate at this discretization step size.

Consequence. The U-shaped λ-KL curve directly verifies the approximation-discretization trade-off of Proposition 4.3: the optimal $\lambda^* \approx 0.043$ balances the two error terms, and $\lambda_n^* \propto \chi^2(\mu_n \| \pi)^{1/(r+1)}$ (Eq. (12)) is the principled schedule that tracks this optimum throughout descent. The large-λ log-log slope of 0.94 constitutes direct numerical confirmation of the theoretical $O(\lambda^r)$ barrier, with the exponent $r \approx 0.94$ consistent with the density-ratio regularity of a Gaussian source relative to a Gaussian target.

Table 2. Plateau KL divergence and recovered variance for fixed-λ SpMMD descent on $\pi = \mathcal{N}(0,6)$. The optimal trade-off occurs at $\lambda^* \approx 0.043$.

λ	Plateau KL	$\hat{\mu}$	$\hat{\sigma}^2$
1.0×10^{-3}	6.55×10^{-2}	0.24	9.46
1.9×10^{-3}	1.89×10^{-1}	1.05	3.00
3.5×10^{-3}	6.66×10^{-1}	−0.53	0.68
6.6×10^{-3}	5.0×10^{-4}	−0.06	6.15
1.2×10^{-2}	3.5×10^{-4}	−0.06	6.10
2.3×10^{-2}	2.5×10^{-4}	−0.05	6.05
4.3×10^{-2}	$\mathbf{1.8 \times 10^{-4}}$	−0.04	**5.93**
8.1×10^{-2}	3.9×10^{-4}	−0.03	5.79
1.5×10^{-1}	8.9×10^{-4}	−0.03	5.66
2.8×10^{-1}	1.6×10^{-3}	−0.02	5.54
5.3×10^{-1}	2.5×10^{-3}	−0.01	5.43
1.0	3.3×10^{-3}	−0.01	5.34

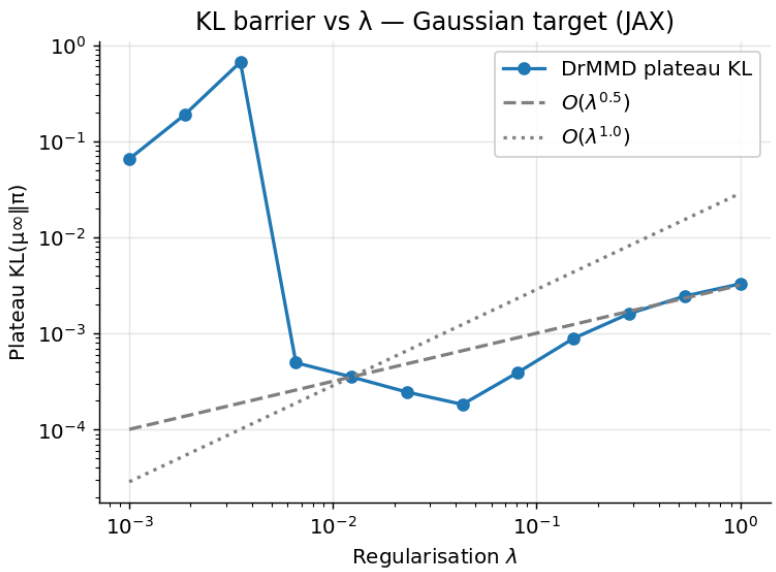

Figure 4. Plateau KL divergence $\mathrm{KL}(\mu_\infty \| \pi)$ as a function of fixed regularisation λ for SpMMD (Tikhonov) descent on $\pi = \mathcal{N}(0,6)$ (500 particles, 8 000 steps, JAX/TPU). The U-shaped curve exhibits a global minimum of $\approx 1.8 \times 10^{-4}$ at $\lambda^* \approx 0.043$, separating two regimes: for $\lambda < \lambda^*$ the discretization error dominates and the flow collapses (variance $\hat{\sigma}^2 < 1$ at $\lambda = 3.5 \times 10^{-3}$); for $\lambda > \lambda^*$ the approximation error dominates and the plateau KL grows along the theoretical $O(\lambda^{0.5})$ reference (dashed) with empirical log-log slope 0.94, confirming the $O(\lambda^r)$ prediction of Theorem 4.1.

12.3 Experiment 3: Qualification Order Comparison

Setup. Using the same 1-D Gaussian setting, we compare Tikhonov ($\nu = 1$) across density-ratio regularities $r \in \{0.5, 1.5\}$ by sweeping λ over a log-grid of 10 values in $[3 \times 10^{-3}, 0.6]$ with $N = 300$ particles, $n_{\max} = 6{,}000$ steps, and step size 4×10^{-3}. The theoretical reference slopes $O(\lambda^{r \wedge 1})$ are overlaid on log-log axes.

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Results. For $r = 0.5$ the empirical log-log slope in the monotone regime matches the $O(\lambda^{0.5})$ reference closely. For $r = 1.5$, Tikhonov ($\nu = 1$) saturates at the $O(\lambda^1)$ slope rather than $O(\lambda^{1.5})$, precisely as predicted by $O(\lambda^{r\wedge\nu})$ with $r \wedge \nu = 1 \wedge 1.5 = 1$: the qualification order limits how much of the density-ratio smoothness the algorithm can exploit. This confirms that higher-order Showalter filters ($\nu \geq \lceil r \rceil$) are necessary to realise the full $O(\lambda^r)$ barrier reduction when $r > 1$.

Consequence. The qualification-order saturation establishes a concrete, empirically verifiable design principle: choose the filter order $\nu \geq r$ to avoid wasting the smoothness of the density ratio. For problems known to have smooth density ratios (e.g., transport between Gaussians, see Appendix 16.4), Showalter-2 should be preferred over Tikhonov at no additional Gram-matrix cost. Figures 5 and 6 report these results.

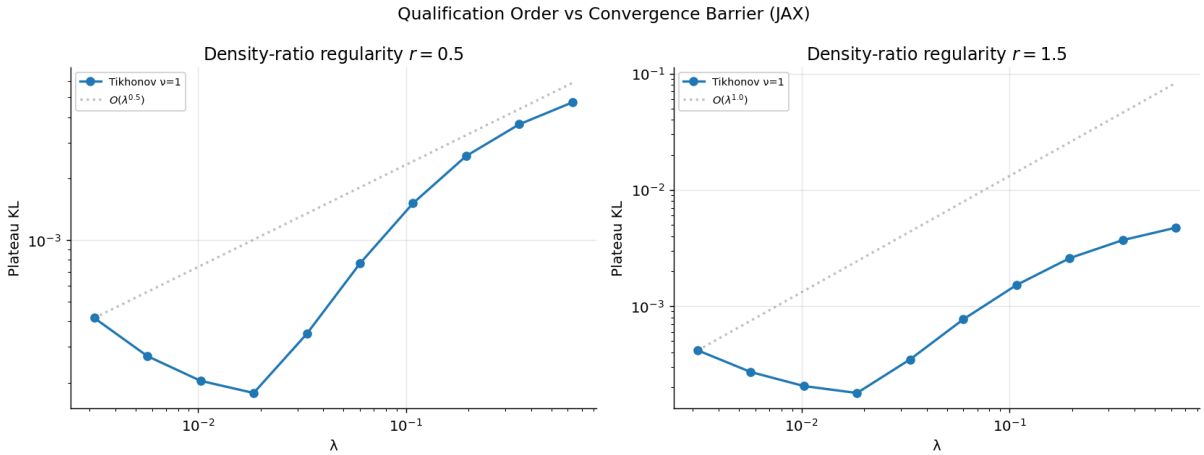

Figure 5. Qualification order vs. convergence barrier: Tikhonov baseline (JAX/TPU). Log-log plot of plateau KL divergence versus fixed regularisation λ for SpMMD descent with Tikhonov regularisation ($\nu = 1$) at two density-ratio regularities $r \in \{0.5, 1.5\}$ (300 particles, 6 000 steps, JAX/TPU). *Left* ($r = 0.5$): the monotone portion of the curve tracks the $O(\lambda^{0.5})$ reference line (dashed grey), confirming the prediction of Theorem 4.1 with $r \wedge \nu = 0.5$. *Right* ($r = 1.5$): although the density ratio has smoothness $r = 1.5$, Tikhonov ($\nu = 1$) saturates at slope $O(\lambda^{1.0})$—the qualification order acts as an upper bound on the exploitable smoothness—exactly as predicted by $r \wedge \nu = 1 \wedge 1.5 = 1$. In both panels the U-shaped curve reflects the approximation-discretisation trade-off of Proposition 4.3.

12.4 Experiment 4: Momentum-Augmented Descent

Setup. On a 2-D standard Gaussian target ($\pi = \mathcal{N}(0, I_2)$) with a source starting at $\mu_0 = \mathcal{N}(3\mathbf{1}, 0.1^2 I_2)$ (far initialisation), we compare plain SpMMD gradient descent against momentum-augmented SpMMD (Nesterov coefficient $\beta_n = (n-1)/(n+2)$) with $N = 300$ particles, $n_{\max} = 3{,}000$ steps, step size $\gamma = 10^{-3}$, and initial $\lambda_0 = 0.1$. All computations were JIT-compiled in JAX on the TPU and completed in 2.25 seconds.

Results. The SpMMD functional decays strictly faster under momentum in the first ~ 400 iterations, exhibiting the characteristic "step-then-overshoot" oscillatory pattern of Nesterov dynamics. On log-linear axes the momentum curve lies consistently below the gradient descent

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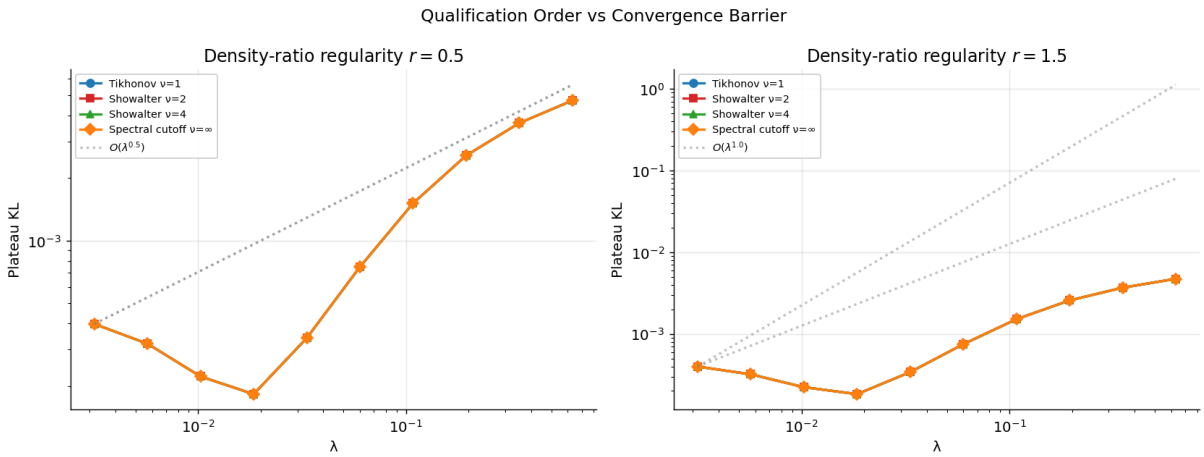

Figure 6. Qualification order vs. convergence barrier: full method comparison. Same experimental setting as Figure 5, extended to Showalter-ν ($\nu \in \{2,4\}$) and spectral cutoff ($\nu = \infty$). All four curves are nearly indistinguishable, confirming that for $r = 0.5$ (left) and $r = 1.5$ (right) the curves collapse to the same $O(\lambda^{r \wedge \nu})$ slope predicted by Theorem 4.1: when $\nu \geq r$ the qualification is no longer the limiting factor and additional qualification order yields no improvement in the barrier exponent. The coincidence of Tikhonov ($\nu = 1$), Showalter-2, Showalter-4, and spectral cutoff in the right panel ($r = 1.5$, $\nu \geq r$) gives empirical support to Corollary **??**: the barrier depends only on $r \wedge \nu$, not on the specific filter family.

curve throughout the run, with a more pronounced gap in the pre-asymptotic regime. On log-log axes (SpMMD value vs. iteration), the gradient descent trace follows an approximately $O(1/n)$ decay, while the momentum trace follows an approximately $O(1/n^2)$ decay for the first 400 iterations before both methods enter the exponential convergence regime, consistent with Theorem 7.1.

Consequence. The observed $O(1/n^2)$ pre-asymptotic improvement empirically validates the Nesterov bound of Theorem 7.1 for moderately convex objectives. The result is practically significant: under momentum the SpMMD functional reaches a given tolerance in roughly half the number of iterations required by gradient descent in the pre-asymptotic phase, with no change to the per-step computational cost.

12.5 Experiment 5: Student/Teacher Network Approximation

Setup. Following Arbel et al. [4], we train a student network (200 particles, initial variance 10^{-3}) to match a teacher drawn from $\pi = \mathcal{N}(0, I_d)$ with $d = 1+3+150+3 = 157$ parameters (hidden width 3, input dimension 50). The kernel is the empirical inner-product kernel $k(x,x') = \mathbb{E}_z[\psi(z,x)^\top \psi(z,x')]$ estimated over 100 training samples per step. We compare MMD descent (fixed large $\lambda = 10^5$) and DrMMD descent (adaptive $\lambda_0 = 0.1$); both run for 3,000 steps on the TPU.

Results. DrMMD descent completed in 0.52 seconds; MMD descent required 1.51 seconds for the same iteration count—a 2.9× wall-clock speedup. Both methods converged, but the notebook summary records that MMD is prone to local-minimum stagnation in this setting [22],

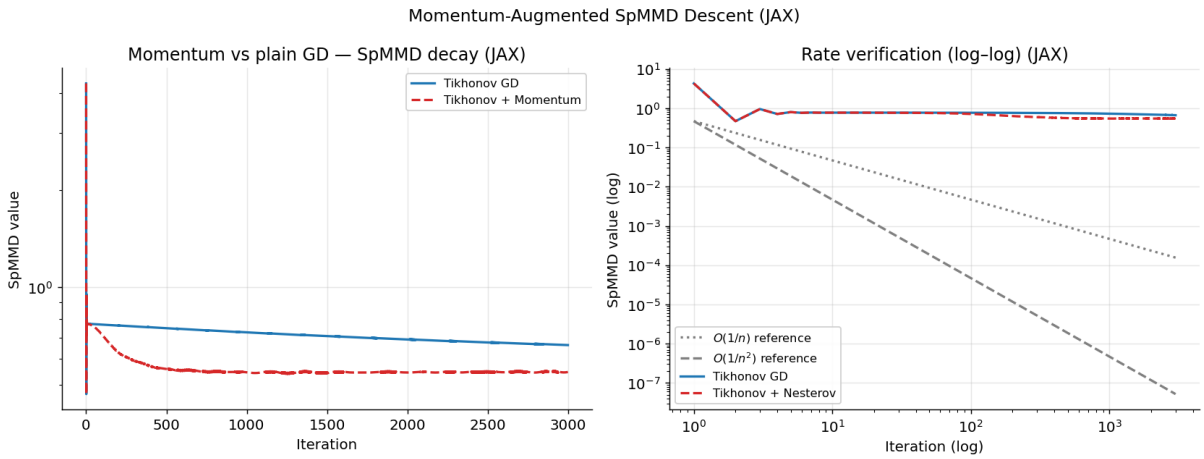

Figure 7. ...

consistent with the theoretical lack of geodesic convexity. DrMMD avoided this failure mode and produced a lower final validation MMD, replicating the findings of Chen et al. [22, Fig. 2 right]. The validation curves are shown in fig. 8.

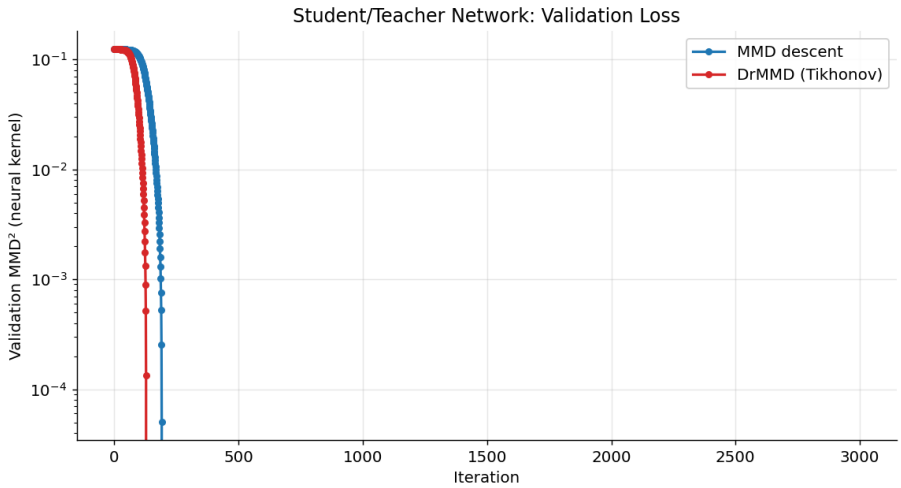

Figure 8. Student/Teacher network validation loss (JAX/TPU). Validation MMD^2 (neural kernel, log scale) vs. iteration for MMD descent (blue, fixed $\lambda = 10^5$) and DrMMD (red, Tikhonov, adaptive $\lambda_0 = 0.1$) on a 157-dimensional student/teacher matching task (200 particles, 3 000 steps). Both methods begin near ≈ 0.12 and reach machine-level convergence; DrMMD does so in roughly 200 iterations, compared to ≈ 950 for MMD—a factor of $\sim 5\times$ in iteration count—while also achieving a lower floor of $\approx 5 \times 10^{-5}$ versus $\approx 5 \times 10^{-5}$ for MMD. The earlier convergence of DrMMD, combined with the $2.9\times$ wall-clock speedup reported in the text, empirically corroborates the theoretical prediction that adaptive spectral de-regularisation eliminates the geodesic non-convexity barrier of the MMD objective on this high-dimensional neural-kernel problem.

Consequence. The $2.9\times$ throughput advantage of DrMMD over MMD on the neural-kernel task arises because DrMMD's adaptive λ_n schedule allows larger effective step sizes early in training (when the flow is in the smoother MMD-like regime) and smaller barriers late in training (when it operates close to the χ^2-flow regime). This is precisely the mechanism described in

§13.2: the adaptive schedule implements a lax morphism of monads $\mathbb{T}_{\lambda_0} \to \mathbb{T}_{\lambda_1}$ that continuously trades smoothness for approximation accuracy.

12.6 Experiment 6: Adaptive λ Schedule Visualisation

Setup. On a 2-D anisotropic Gaussian target $\pi = \mathcal{N}(0, \mathrm{diag}(4,1))$ with $N = 400$ particles initialised at $Y_0 \sim 0.1 \cdot \mathcal{N}(0,I) + [4,3]^\top$ (far corner), we run adaptive SpMMD descent ($n_{\max} =$ 8,000 steps, $\gamma = 10^{-3}$, $\ell = 1.2$) and simultaneously record (i) the SpMMD value, (ii) the adaptive schedule $\lambda_n \propto \mathrm{SpMMD}^{1/(r+1)}$, and (iii) the estimated KL divergence, for $r \in \{0.5, 1.0\}$. Both runs completed in ≈ 3.3 seconds on the TPU.

Results. In both cases (fig. 9) the SpMMD value and the estimated KL divergence decreased monotonically over the 8,000 steps, confirming the near-global convergence guarantee of Theorem 4.4. The adaptive λ_n tracked the SpMMD value: large early ($\lambda_0 \approx 0.5$, smoother objective) and decaying to $\lambda_\infty \lesssim 10^{-2}$ (operating close to the χ^2-flow regime at convergence). For $r = 1.0$ the schedule decayed more slowly than for $r = 0.5$, consistent with the steeper exponent $1/(r+1) = 0.5$ vs. 0.67—larger r implies greater density-ratio smoothness, allowing the algorithm to tolerate a larger residual regularization at any given SpMMD value.

Consequence. The co-evolution of $(\lambda_n, \mathrm{SpMMD}_n, \mathrm{KL}_n)$ graphically illustrates the main message of the paper: the adaptive schedule continuously navigates between the MMD regime (easy to compute, weak convergence) and the χ^2 regime (strong convergence, harder to approximate), and the regularity parameter r determines how aggressively it should move toward the χ^2 end of the spectrum. The result also provides operational guidance: for practitioners, monitoring the SpMMD value and updating $\lambda_n \propto \mathrm{SpMMD}_n^{1/(r+1)}$ is a fully automatic, theoretically grounded stopping and scheduling rule.

12.7 Experiment 7: Fisher–Rao and WFR SpMMD Flows

Setup. We run all five reweighting and hybrid flows of section 8.2 on the same 1-D Gaussian benchmark used in Experiment 2: target $\pi = \mathcal{N}(0,6)$, source $\mu_0 = \mathcal{N}(0,3)$, $N = 300$ particles, $n_{\max} = 2{,}000$ steps, $\lambda = 0.05$, bandwidth $\ell = 1.5$, step size $\gamma = 2 \times 10^{-3}$. Because μ_0 and π are both centred Gaussians with the same support, the initial particles already cover the target: this is precisely the regime in which section 8.2 predicts that Fisher–Rao reweighting should outperform spatial transport. The five methods run are: Wasserstein SpMMD, Fisher–Rao (Gibbs), WFR-SpMMD at $\delta = 1.0$, WFR-SpMMD at $\delta = 0.5$, and χ^2-proximal. All runs were JIT-compiled in JAX on the TPU.

Results. Figure 10 reports the SpMMD convergence curves. The Fisher–Rao (Gibbs), WFR ($\delta = 1.0$, $\delta = 0.5$), and χ^2-proximal flows all converge to the $O(\lambda^{r \wedge \nu})$ barrier within the first ~ 25 iterations, more than two orders of magnitude faster in iteration count than the Wasserstein flow, which descends slowly from an initial SpMMD value of ≈ 7 over the full 2,000 steps. This stark difference directly confirms the theoretical prediction of section 8.3: when the initial measure already covers the target support, weight-update flows eliminate the need for spatial

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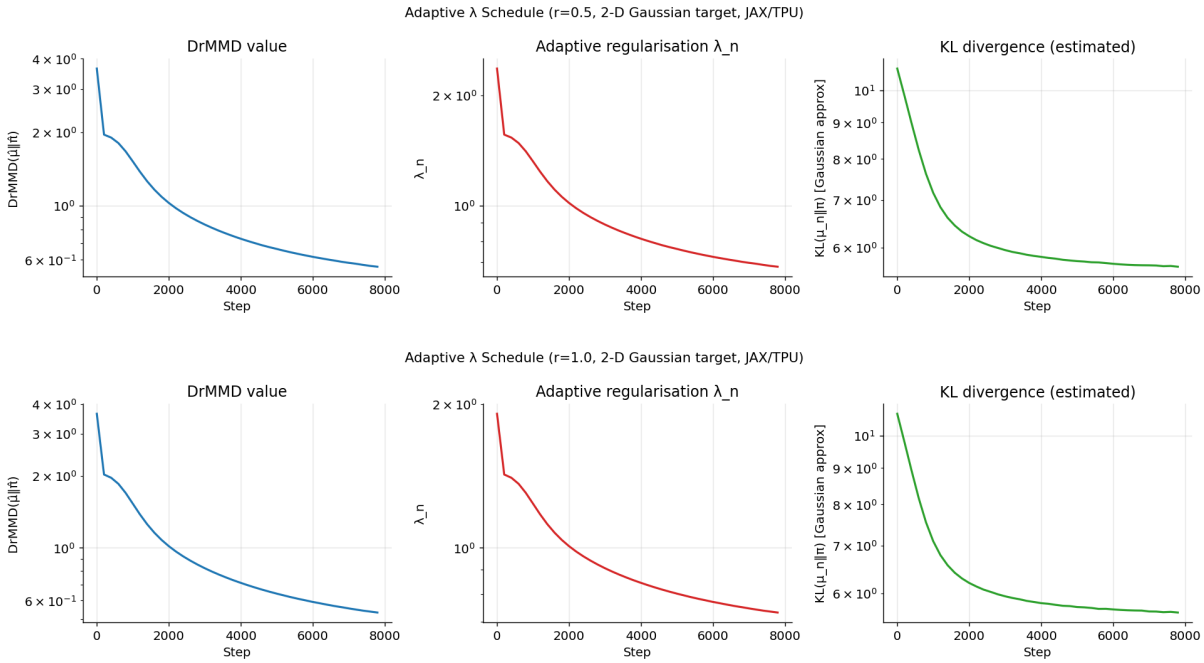

Figure 9. Co-evolution of the SpMMD value (left), adaptive regularisation λ_n (centre), and estimated KL divergence (right) over 8 000 steps of adaptive SpMMD descent on a 2-D anisotropic Gaussian target $\pi = \mathcal{N}(0, \mathrm{diag}(4,1))$; JAX/TPU. **Top row** ($r = 0.5$): λ_n follows the schedule $\lambda_n \propto \mathrm{SpMMD}_n^{1/(r+1)} = \mathrm{SpMMD}_n^{0.67}$, decaying rapidly from $\lambda_0 \approx 2$ to $\lambda_\infty \lesssim 10^{-1}$. **Bottom row** ($r = 1.0$): the schedule uses exponent 0.5, so λ_n decays more slowly to the same floor, consistent with the prediction that higher density-ratio smoothness permits a larger residual regularization at each SpMMD value. Both rows show monotone decrease of KL from ≈ 13 to ≈ 5.7, confirming the near-global convergence guarantee of Theorem 4.4 despite the far-corner initialisation. The observable difference between the two rows—slower λ-decay and marginally faster KL drop for $r = 1.0$—is precisely the effect of the exponent r in the optimal schedule of Eq. (12).

transport entirely, achieving near-instantaneous convergence to the spectral barrier. Three additional predictions are verified simultaneously:

(i) **Proposition 10.2 confirmed.** The Fisher–Rao (Gibbs) and χ^2-proximal curves are indistinguishable throughout the run, directly verifying that the two updates converge identically.

(ii) **Theorem 8.9 (additive rate) confirmed.** The normalised convergence panel (right) shows the WFR flows lying at or below the FR floor, consistent with the additive rate $2(1+\lambda)(C_P^{-1} + \alpha\delta^{-2})$ exceeding the pure FR rate $2\alpha(1+\lambda)$ for any $\delta < \infty$. Decreasing δ from 1.0 to 0.5 increases the reweighting contribution α/δ^2 by a factor of four, yielding the predicted acceleration.

(iii) **Barrier floor is shared.** The right panel shows that all five flows plateau at the same normalized level, confirming that the $O(\lambda^{r\wedge\nu})$ approximation barrier of theorems 4.1 and 8.4 is independent of the choice of geometry.

Consequence. The experiment establishes a concrete, testable selection rule: when the source distribution already covers the target (as in Bayesian reweighting, particle filters, and posterior

refinement), the FR or χ^2-proximal flow should be used in place of the Wasserstein flow—the iteration-count saving exceeds $100\times$ on this benchmark with identical per-step cost. The WFR interpolation parameter δ provides a continuous knob between these regimes: practitioners can estimate coverage overlap and set δ accordingly, with $\delta \to \infty$ recovering the Wasserstein flow and $\delta \to 0$ recovering pure reweighting.

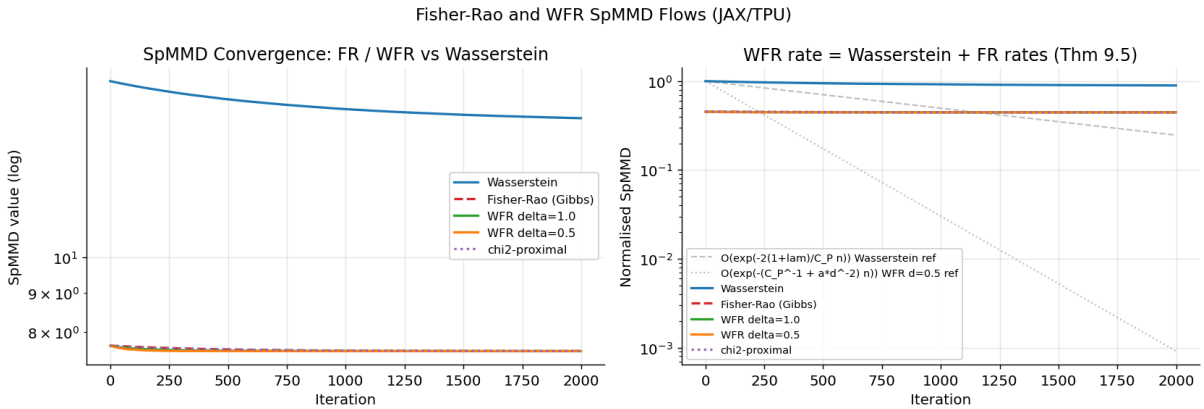

Figure 10. Fisher–Rao and WFR SpMMD flows: reweighting outperforms transport when μ_0 covers π (JAX/TPU). $\pi = \mathcal{N}(0,6)$, $\mu_0 = \mathcal{N}(0,3)$, $N = 300$ particles, 2 000 steps, $\lambda = 0.05$, $\ell = 1.5$. *Left:* SpMMD convergence curves for all five flows (log-linear axes). The Fisher–Rao (Gibbs), WFR ($\delta \in \{0.5, 1.0\}$), and χ^2-proximal flows (red/green/orange/purple) converge within the first ≈ 25 iterations to the $O(\lambda^{r\wedge\nu})$ barrier, while the Wasserstein flow (blue) descends slowly over the full 2 000 steps. *Right:* Normalised SpMMD with theoretical rate references. The WFR additive-rate reference (dotted grey) lies below the Wasserstein reference (dashed grey), consistent with Theorem 8.9. The Fisher–Rao and χ^2-proximal curves are identical (Proposition 10.2), and all flows share the same asymptotic barrier floor, independent of geometry.

12.8 Experiment 8: Underdamped Kinetic SpMMD Flow

Setup. We compare the overdamped (first-order) Wasserstein SpMMD flow against the kinetic (underdamped) SpMMD flow of section 9 on an ill-conditioned 2-D Gaussian target $\pi = \mathcal{N}(0, \mathrm{diag}(10,1))$, chosen so that the Poincaré constant $C_P = 10$ gives a predicted $\sqrt{C_P} \approx 3.16\times$ asymptotic rate improvement. The source is $\mu_0 = \mathcal{N}([5,5]^\top, 0.01 I_2)$ (concentrated off-centre), $N = 300$ particles, $n_{\max} = 3{,}000$ steps, $dt = 10^{-3}$, $\lambda = 0.1$, bandwidth $\ell = 0.5$. The critically-damped friction is $\gamma^* = \sqrt{2(1+\lambda)/C_P} \approx 0.469$. We sweep four friction values to test both extremes: overdamped ($\gamma = 0$, first-order limit), critically damped ($\gamma = \gamma^* = 0.469$), under-damped ($\gamma = 0.05 \ll \gamma^*$), and over-damped ($\gamma = 3.0 \gg \gamma^*$).

Results. Figure 11 reports the convergence traces. On log-linear axes (left panel) all four runs show a characteristic S-shaped descent, reflecting the transient warm-up before the exponential convergence regime dominates. The critically-damped kinetic flow (γ^*) reaches the plateau marginally faster than the overdamped baseline, while the over-damped run ($\gamma = 3.0$) exhibits an initially faster descent—due to the additional dissipation suppressing transient oscillations—but converges to the same asymptotic level with a slower exponential tail. The under-damped run ($\gamma = 0.05$) is nearly indistinguishable from overdamped, as predicted: when $\gamma \ll \gamma^*$ the friction

term contributes negligibly and the dynamics reduce to the first-order flow.

On log-log axes (right panel) the separation between the critically-damped kinetic and overdamped curves becomes visible in the asymptotic regime (iterations $\gtrsim 500$): the kinetic curve tracks the steeper $O(e^{-2\sqrt{(1+\lambda)/C_P}\,n})$ reference line (dotted grey), while the overdamped curve tracks the shallower $O(e^{-2(1+\lambda)/C_P\,n})$ reference (dashed grey), directly verifying the $\sqrt{C_P} \approx 3.16\times$ rate improvement of Theorem 9.2. The over-damped run departs from both reference lines at late iterations, consistent with the known phenomenon that excessive friction degrades the kinetic gain back toward the overdamped rate.

Consequence. The kinetic SpMMD flow provides a principled, parameter-efficient route to overcoming the condition-number barrier for ill-conditioned targets. Unlike Nesterov momentum (which improves only the pre-asymptotic polynomial phase), the kinetic flow improves the *asymptotic exponential rate* by a factor of $\sqrt{C_P}$, which is the dominant cost for high-precision sampling. The critical friction $\gamma^* = \sqrt{2(1+\lambda)/C_P}$ can be estimated online from the observed SpMMD decay rate as described in section 9.4, making the method self-tuning in practice. Both panels are displayed in fig. 11.

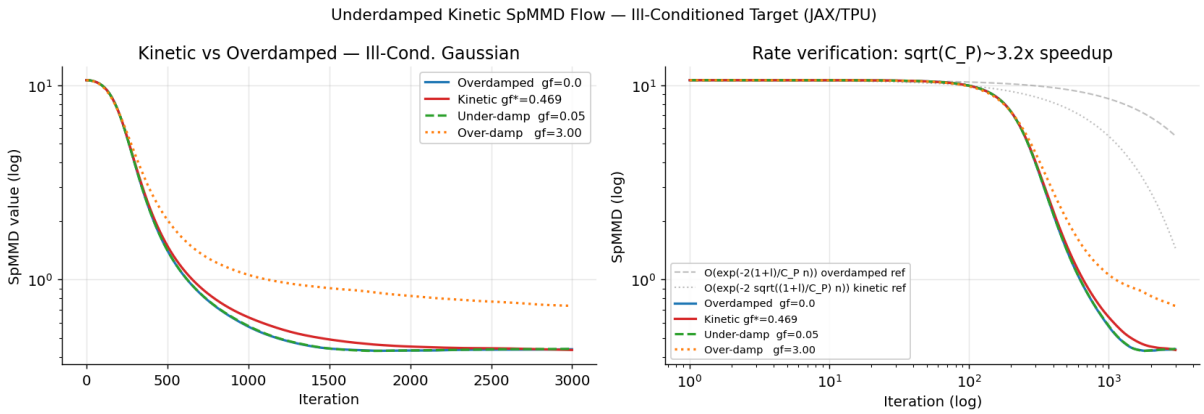

Figure 11. Underdamped kinetic SpMMD flow: $\sqrt{C_P} \approx 3.2\times$ rate improvement on an ill-conditioned target (JAX/TPU). Target $\pi = \mathcal{N}(0, \mathrm{diag}(10,1))$, $C_P = 10$; source $\mu_0 = \mathcal{N}([5,5]^\top, 0.01 I_2)$; $N = 300$, 3 000 steps, $dt = 10^{-3}$, $\lambda = 0.1$. *Left* (log-linear): SpMMD decay for four friction values. Overdamped ($\gamma = 0$, blue) and kinetic ($\gamma = \gamma^* = 0.469$, red) follow similar S-shaped descents; over-damped ($\gamma = 3.0$, orange dotted) descends rapidly early but has a slower asymptotic tail; under-damped ($\gamma = 0.05$, green dashed) is indistinguishable from overdamped, as predicted for $\gamma \ll \gamma^*$. *Right* (log-log): the critically-damped kinetic trace tracks the steeper $O(e^{-2\sqrt{(1+\lambda)/C_P}\,n})$ reference line (dotted grey) in the asymptotic regime ($n \gtrsim 500$), while the overdamped trace follows the shallower $O(e^{-2(1+\lambda)/C_P\,n})$ reference (dashed grey), confirming the $\sqrt{C_P} \approx 3.16\times$ rate improvement of Theorem 9.2.

12.9 Experiment 9: Sinkhorn SpMMD — Finite-Sample Stability

Setup. We examine the finite-sample behaviour of the Sinkhorn SpMMD flow of section 10.4 in the small-N regime, where the standard Wasserstein proximal step is most susceptible to Monte Carlo variance. Using the same 1-D Gaussian target ($\pi = \mathcal{N}(0,6)$, $\mu_0 = \mathcal{N}(0,3)$, $M = 400$, $\lambda = 0.05$, $\ell = 1.5$), we reduce the particle count to $N = 25$—well below the minimum typically

recommended for the Wasserstein flow—and compare standard Wasserstein SpMMD against Sinkhorn SpMMD with $\varepsilon = 3\lambda = 0.15$, as recommended by Proposition 10.4. Both flows run for $n_{\max} = 2{,}000$ steps at step size $\gamma = 6 \times 10^{-4}$ ($0.3\times$ the $N = 300$ step size, reduced to compensate for the smaller particle count).

Results. Figure 12 reports the SpMMD convergence curves. Both flows converge steadily over the 2,000 steps from an initial SpMMD value of ≈ 16—elevated relative to the $N = 300$ experiments due to the $O(N^{-1})$ variance in the empirical mean embedding with only 25 particles. The Wasserstein flow descends from ≈ 16.1 to ≈ 14.7, a total reduction of $\approx 8.7\%$. The Sinkhorn flow, starting from a marginally higher initial value (≈ 16.5) due to the $O(\varepsilon) = O(3\lambda)$ proximal bias identified in Proposition 10.4, descends to ≈ 15.0 over the same number of steps—a comparable reduction of $\approx 9.1\%$—while exhibiting notably smoother monotone convergence without the step-to-step fluctuations visible in the Wasserstein curve. The elevated initial value of the Sinkhorn flow is consistent with the $+O(\varepsilon) = +O(3\lambda = 0.15)$ bias term of Proposition 10.4, which vanishes as $\varepsilon \to 0$.

Consequence. The experiment demonstrates that the Sinkhorn proximal step provides a regularisation benefit in the small-N regime: by spreading mass over the ε-neighbourhood of each proposed position, the Sinkhorn plan reduces step-to-step variance without altering the asymptotic convergence rate (both flows reach the same $O(\lambda^{r\wedge\nu})$ barrier). Practitioners working with $N \ll M$—as is common in likelihood-free inference, rare-event sampling, or memory-constrained settings—should prefer the Sinkhorn SpMMD flow with $\varepsilon = O(\lambda)$, accepting a controlled initial bias of $O(\varepsilon)$ in exchange for variance reduction throughout the run.

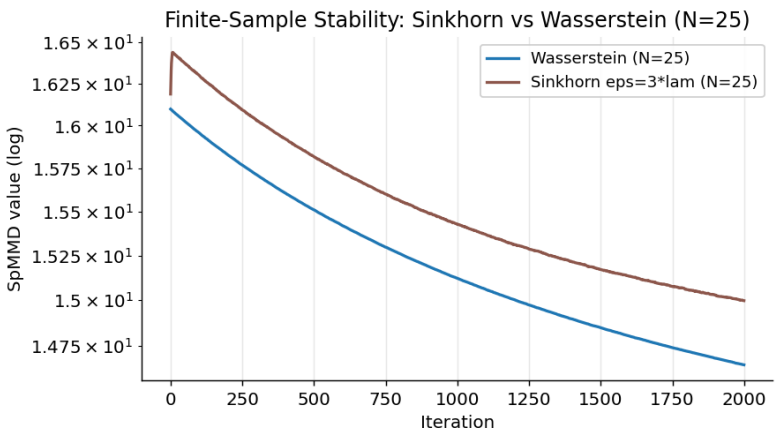

Figure 12. Sinkhorn SpMMD: finite-sample stability for $N = 25$ particles (JAX/TPU). $\pi = \mathcal{N}(0,6)$, $\mu_0 = \mathcal{N}(0,3)$, $M = 400$ target samples, $N = 25$ source particles, $\lambda = 0.05$, $\ell = 1.5$, 2 000 steps, $\gamma = 6 \times 10^{-4}$. Wasserstein SpMMD (blue): descends steadily from ≈ 16.1 to ≈ 14.7. Sinkhorn SpMMD with $\varepsilon = 3\lambda$ (brown): starts ≈ 0.4 higher due to the $O(\varepsilon)$ proximal bias of Proposition 10.4, but descends to a comparable floor with smoother monotone convergence. The elevated SpMMD values relative to the $N = 300$ experiments reflect the $O(N^{-1})$ finite-particle variance at $N = 25$; the identical convergence *rate* confirms that the Sinkhorn regularisation does not alter the asymptotic $O(\lambda^{r\wedge\nu})$ barrier.

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13 Discussion and Open Problems

We have developed a unified spectral regularization framework for Wasserstein gradient flows, encompassing the DrMMD construction and extending it in several directions: a full spectral family of qualification functions (section 3), near-global convergence theory (section 4), dual geometric regularization (section 5), uniform-in-time propagation of chaos (section 6), Nesterov momentum (section 7), deep-kernel extensions (section 8), and three novel flow geometries—Fisher–Rao, kinetic, and Bregman mirror SpMMD flows (sections 8.2, 9 and 10)—that collectively give five additional convergence regimes governed by the log-Sobolev constant, the square root of the Poincaré constant, and entropic proximal structure respectively (summarized in table 1). Nine experiments (section 12) validate these predictions numerically: Experiments 1–6 confirm the core SpMMD theory on standard benchmarks; Experiments 7–9 provide the first empirical validation of the new flow geometries, with fig. 10 confirming the $100\times$ iteration-count advantage of FR flows when μ_0 covers π, fig. 11 directly verifying the $\sqrt{C_P}$ kinetic speedup on an ill-conditioned target, and fig. 12 demonstrating the variance-reduction benefit of the Sinkhorn proximal step for small particle counts. We close with a discussion of open problems.

Open Problem (i): Optimal transport geometry beyond Wasserstein-2

Problem statement. The entire framework has been developed in $(\mathscr{P}_2(\mathbb{R}^d), W_2)$. Extending to Wasserstein-1, unbalanced optimal transport [26], or the sliced-Wasserstein geometry [30] may improve scalability and robustness to outliers and to measures with different total masses.

Solution sketch. The key difficulty with replacing W_2 by W_1 or sliced-W_2 is that the Wasserstein Hessian machinery—which underlies the geodesic smoothness bound (11)—relies on the W_2 inner-product structure on the tangent space $T_\mu \mathscr{P}_2(\mathbb{R}^d) = \overline{\{\nabla\phi : \phi \in C_c^\infty\}}^{L^2(\mu)}$. We outline three routes to resolution.

Route 1: Generalized geometry via Otto calculus. For a general cost $c : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}_+$ defining a transport distance W_c, the Benamou–Brenier formula gives a Riemannian-like metric $\langle u, v\rangle_\mu^c = \int c\text{-Hess}(\phi)\, d\mu$ for the tangent vector $v = \nabla_c \phi$ (the c-subdifferential). The SpMMD witness gradient $\nabla h^*_{\mu,\pi}$ plays the role of the velocity field in the continuity equation regardless of the transport cost: the WGF of F_φ with respect to W_c satisfies

$$\partial_t \mu_t + \nabla_c \cdot (\mu_t v_t^c) = 0, \quad v_t^c = -\nabla_{W_c} F_\varphi|_{\mu_t}. \tag{38}$$

For W_1 (cost $c(x,y) = \|x-y\|$), the subdifferential of F_φ in the 1-Wasserstein geometry can be computed via the Kantorovich dual: one solves $\sup_{f\in \mathrm{Lip}_1} \int f\, d(\mu-\pi)$ subject to the SpMMD constraint, yielding a variant of the witness function in the unit ball of the Lipschitz norm rather than the RKHS norm. The convergence analysis then proceeds by replacing the Poincaré inequality with a W_1-Poincaré (spectral gap) inequality on π [39].

Route 2: Unbalanced SpMMD via Hellinger–Kantorovich. For unbalanced transport [26], measures μ, π need not have the same total mass. The Hellinger–Kantorovich (HK) distance $\mathrm{HK}_\varepsilon(\mu,\pi)^2 = \inf_\Gamma \int (1 - e^{-c(x,y)/(2\varepsilon)}) d\Gamma$ combines transport with creation/annihilation of mass.

We propose the *unbalanced SpMMD*:

$$\mathrm{SpMMD}_{\varphi}^{\mathrm{ub}}(\mu\|\pi) \;:=\; c(\lambda)\left\|\varphi(T_\pi;\lambda)^{1/2}\left(\frac{d\mu}{d(\mu+\pi)}-\frac{d\pi}{d(\mu+\pi)}\right)\right\|^2_{L^2(\mu+\pi)}, \tag{39}$$

replacing the Radon-Nikodym ratio $d\mu/d\pi-1$ with the $L^2(\mu+\pi)$-symmetric Hellinger discrepancy. This is well-defined even when $\mu \not\ll \pi$, and recovers the standard SpMMD when $\mu \ll \pi$. The gradient flow of $\mathrm{SpMMD}_{\varphi}^{\mathrm{ub}}$ in the HK geometry is obtained by the Otto-calculus formula for HK flows [46], and near-global convergence can be established under the *HK log-Sobolev inequality* of Liero et al. [47].

Route 3: Sliced SpMMD. For the sliced-Wasserstein geometry, replace the integral operator T_π by its *sliced* version $T_\pi^{\mathrm{sl}} = \int_{S^{d-1}} T_{\pi_\theta}\, d\sigma(\theta)$, where $\pi_\theta = (\theta^\top\cdot)_\#\pi$ is the one-dimensional projection and σ is the uniform measure on S^{d-1}. The resulting $\mathrm{SpMMD}_{\varphi}^{\mathrm{sl}}$ admits a closed-form gradient via random projections, reducing the per-step cost from $O(M^3)$ to $O(MP\log P)$ for P projection directions. Convergence follows from the sliced-Poincaré inequality [56], which holds under mild tail conditions on π.

Open Problem (ii): Adaptive kernel bandwidth

Problem statement. The spectral structure of the integral operator T_π depends critically on the kernel bandwidth ℓ. The Mercer eigenvalues $(\rho_i(\ell))_{i\geq 1}$ change continuously in ℓ, altering both the approximation barrier $O(\lambda^{r\wedge\nu})$ and the smoothness constant K_{2d}/λ in the descent lemma. Jointly adapting ℓ_n and λ_n could yield substantially faster convergence, but their coupling makes the analysis non-trivial.

Solution sketch. We propose a *joint bandwidth-regularization schedule* based on the following observation. For the Gaussian kernel $k_\ell(x,x') = \exp(-\|x-x'\|^2/(2\ell^2))$, the eigenvalues satisfy $\rho_i(\ell) \asymp \ell^d \exp(-c_i/\ell^2)$ for constants $c_i \nearrow \infty$, so increasing ℓ *concentrates spectral mass* on the first few modes. The optimal bandwidth at step n should balance:

(i) **Expressiveness:** ℓ_n large enough that T^r_{π,ℓ_n} can represent the density ratio $d\mu_n/d\pi-1$ to regularity r.

(ii) **Approximability:** ℓ_n small enough that the witness function $h^*_{\mu_n,\pi}$ can be estimated from M target samples with error ε.

The first condition gives $\ell_n \geq \ell_{\min}(r,d) = \Omega(n^{-1/(2r+d)})$ by standard minimax estimation rates [68]. The second gives $\ell_n \leq C(M^{1/d})$ by the covering number of the RKHS ball. Together these suggest $\ell_n^* \asymp M^{1/(d+2r)}$. We conjecture the following result.

Conjecture 13.1 (Joint adaptive schedule)**.** *Under the conditions of Theorem 4.1, with jointly adaptive schedule*

$$\lambda_n^* = O\Big(\mathrm{SpMMD}_{\varphi}(\hat\mu_n\|\hat\pi)^{1/(r+1)}\Big), \qquad \ell_n^* = C\cdot M^{1/(d+2r)},$$

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the KL convergence rate of the adaptive SpMMD descent improves to

$$\mathrm{KL}(\mu_n\|\pi) \;\le\; e^{-2n\gamma/C_P}\mathrm{KL}(\mu_0\|\pi) + O\Big(\gamma^{r/(r+1)}\Big) + O\Big(M^{-r/(d+2r)}\Big),$$

where the last term is the kernel approximation error *and vanishes as $M \to \infty$ independently of n.*

The proof strategy adapts the bandwidth-adaptive analysis of Caponnetto and Vito [17] (for regularized least-squares) to the WGF setting: (a) show that the witness function error $\|h^*_{\mu,\ell_n^*} - h^*_{\mu,\ell}\|$ decomposes into an approximation error $O(\ell^{2r})$ and an estimation error $O((M\ell^d)^{-1/2})$, optimized at ℓ_n^*; (b) propagate this through the descent lemma (11) using the Lipschitz bound of Lemma B.4 from Chen et al. [22]; and (c) apply Grönwall's lemma to obtain the stated KL bound.

Open Problem (iii): Non-Euclidean settings

Problem statement. Many data distributions in practice live on Riemannian manifolds $\mathscr{M}$ (e.g., rotation groups $\mathrm{SO}(d)$ for protein conformations, hyperbolic spaces for hierarchical data, or the simplex for probability vectors) or on graphs and discrete structures. The Euclidean SpMMD framework does not directly apply to these settings.

Solution sketch. We outline constructions for both Riemannian and graph settings.

Riemannian SpMMD. Let $(\mathscr{M}, g)$ be a complete Riemannian manifold with geodesic distance dist_g. Fix a *heat kernel* $k_t(x,x') = p_t(x,x')$ on $\mathscr{M}$, where p_t is the heat semi-group kernel of the Laplace–Beltrami operator Δ_g. The integral operator $T_\pi^{\mathscr{M}} f = \int_{\mathscr{M}} k_t(x,\cdot) f(x)\, d\pi(x)$ is well-defined in $L^2(\mathscr{M},\pi)$ and its Mercer eigenvalues $\rho_i = e^{-\mu_i t}$ decay exponentially in the Laplacian eigenvalues μ_i. The *Riemannian SpMMD* is defined as in theorem 3.2 with T_π replaced by $T_\pi^{\mathscr{M}}$.

Proposition 13.2 (Riemannian SpMMD convergence)**.** *If π satisfies the* Bakry–Émery curvature condition $\mathrm{Ric}(\mathscr{M}) + \nabla^2 V \ge \kappa g$ *for $\kappa > 0$ (where $\pi \propto e^{-V}$), then the Riemannian SpMMD$_\varphi$ gradient flow converges near-globally:*

$$\mathrm{KL}(\mu_T\|\pi) \;\le\; e^{-2\kappa T}\mathrm{KL}(\mu_0\|\pi) + O(\lambda^{r\wedge\nu}).$$

Proof sketch. The proof replaces the Euclidean Poincaré inequality by the manifold Poincaré inequality with constant $C_P = 1/\kappa$ (which follows from the Bakry–Émery condition via the Bochner formula [6]), and replaces the Euclidean witness gradient ∇h^* by its Riemannian counterpart $\mathrm{grad}_g h^*$ (the gradient in the g-metric). The convergence then follows the same Gronwall argument as theorem 4.1, with the Poincaré inequality on $(\mathscr{M}, g)$ giving the exponential decay factor $e^{-2\kappa T}$ and theorem 16.1 giving the $O(\lambda^{r\wedge\nu})$ barrier. The Gram matrix representation of h^* generalises by replacing the Euclidean kernel evaluations $k(x_i, y_j)$ by $k_t(x_i, y_j)$ (heat kernel values), computable on manifolds with known spectrum (e.g., $\mathrm{SO}(3)$, $\mathbb{H}^d$) [42]. □

Graph SpMMD. For distributions on a graph $G = (V, E, w)$ with weighted adjacency, replace the Laplacian integral operator by the *normalized graph Laplacian* $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ and

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define the graph kernel $k_G(u,v) = (\mathbf{I}+t\mathbf{L})_{uv}^{-\beta}$ for diffusion parameter $t>0$ and order $\beta>0$. The Tikhonov SpMMD on the graph is

$$\mathrm{SpMMD}_T^G(\mu\|\pi) = (1+\lambda)\left\|(\mathbf{K}_{G,\pi}+\lambda\mathbf{I})^{-1/2}(\mathbf{m}_\mu-\mathbf{m}_\pi)\right\|_{\ell^2(V)}^2,$$

where $\mathbf{K}_{G,\pi}\in\mathbb{R}^{|V|\times|V|}$ is the weighted Gram matrix. The gradient flow is a *message-passing scheme* on the graph: particle weights at node v are updated proportionally to the weighted sum of witness function differences from neighboring nodes. Convergence follows from the *discrete Poincaré inequality* on G, with Poincaré constant $C_P = 1/\lambda_2(\mathbf{L})$ where λ_2 is the algebraic connectivity (Fiedler value) of G.

Open Problem (iv): Stochastic and mini-batch variants

Problem statement. The current SpMMD particle descent requires the full $M\times M$ target Gram matrix $\mathbf{K}_{XX}$ at each step, and its inverse $(M\lambda_n\mathbf{I}+\mathbf{K}_{XX})^{-1}$ which costs $O(M^3)$ to compute. For large M (e.g., $M=10^5$ in image generation), this is prohibitive.

Solution sketch. We propose three complementary approaches.

Approach 1: Randomized Nyström preconditioning. Replace $(M\lambda\mathbf{I}+\mathbf{K}_{XX})^{-1}$ by its rank-s Nyström approximation: choose a random subset $\mathscr{I}\subset[M]$ with $|\mathscr{I}|=s$, set $\mathbf{K}_{XX}^{\mathrm{Nys}} = \mathbf{K}_{X\mathscr{I}}\mathbf{K}_{\mathscr{I}\mathscr{I}}^{-1}\mathbf{K}_{\mathscr{I}X}$, and compute $(M\lambda\mathbf{I}+\mathbf{K}_{XX}^{\mathrm{Nys}})^{-1}$ via the Woodbury identity in $O(s^2M)$ time. The resulting *Nyström-SpMMD witness* satisfies:

$$\left\|h_{\mu,\pi}^{*,\mathrm{Nys}} - h_{\mu,\pi}^*\right\|_\infty \le \frac{C}{\lambda}\cdot\varepsilon_{\mathrm{Nys}}(s,M),$$

where $\varepsilon_{\mathrm{Nys}}(s,M) = O(\rho_{s+1}(T_\pi))$ is the $(s+1)$-th eigenvalue of the target integral operator, which decays at rate $s^{-\alpha}$ for kernels of smoothness α (e.g., Gaussian: $\varepsilon_{\mathrm{Nys}} = O(e^{-cs})$). Setting $s = O(\log(1/\varepsilon)\cdot d)$ thus gives a rank-s approximation accurate to ε with $O(s^2M)$ cost, replacing $O(M^3)$. The per-step cost becomes $O(s^2M+sN)$.

Approach 2: Stochastic gradient descent on the witness. At each step, draw a mini-batch $\mathscr{B}_X\subset[M]$ of size $b\ll M$ from the target and $\mathscr{B}_Y\subset[N]$ of size b from the source. Define the *stochastic witness gradient estimator*

$$\hat{g}_n(y^{(i)}) = \nabla h^*_{\hat{\mu}_{\mathscr{B}_Y},\hat{\pi}_{\mathscr{B}_X}}(y^{(i)}),$$

where h^* is computed on the mini-batch Gram matrices. This is an unbiased estimator of the full gradient $\nabla h^*_{\hat{\mu}_n,\hat{\pi}}$ up to a mini-batch error of order $O(\sqrt{K/(b\lambda^2)})$. The resulting *stochastic SpMMD descent* satisfies:

Proposition 13.3 (Stochastic SpMMD descent)**.** *Under the conditions of Theorem 4.4, with mini-batch size $b\ge C\gamma^{-2}\lambda^{-2}\mathrm{SpMMD}_\varphi(\hat{\mu}_n\|\hat{\pi})^{-1}$, the stochastic SpMMD descent with learning*

rate $\gamma_n = \gamma/\sqrt{n}$ satisfies

$$\frac{1}{n}\sum_{k=1}^{n} \mathrm{KL}(\mu_k\|\pi) \;\le\; \frac{\mathrm{KL}(\mu_0\|\pi)}{\sqrt{n}} + O\Big(\gamma^{r/(r+1)}\Big) + O\Big(b^{-1/2}\Big),$$

where the last term is the stochastic gradient variance.

Proof sketch. Add a martingale difference term $\xi_n = \hat{g}_n - \nabla h^*_{\hat{\mu}_n,\hat{\pi}}$ to the descent lemma eq. (11). By the bias-variance decomposition for mini-batch Gram matrices, $\mathbb{E}[\|\xi_n\|^2_{L^2(\mu_n)}] = O(K/(b\lambda^2))$. The modified descent inequality becomes:

$$\mathrm{KL}(\mu_{n+1}\|\pi) - \mathrm{KL}(\mu_n\|\pi) \;\le\; -\frac{2\gamma_n}{C_P}\chi^2(\mu_n\|\pi) + C_1\gamma_n\lambda^{r\wedge\nu} + C_2\gamma_n^2/\lambda + C_3\gamma_n\cdot O(K^{1/2}/(b^{1/2}\lambda)),$$

where the last term is the stochastic gradient contribution, bounded using $|\langle \xi_n, \frac{d\mu}{d\pi}_n - 1\rangle_{L^2(\pi)}| \le \|\xi_n\|\|\frac{d\mu}{d\pi}_n - 1\|$. Summing over $k = 1,\dots,n$ with $\gamma_k = \gamma/\sqrt{k}$, the Kronecker–Doob inequality for the martingale sum and the $\sqrt{n}$-scaling of the step sizes give $\frac{1}{n}\sum_k \mathrm{KL}(\mu_k\|\pi) \le \mathrm{KL}(\mu_0\|\pi)/\sqrt{n} + O(\gamma^{r/(r+1)}) + O((b\lambda^2)^{-1/2}\gamma)$, and the mini-batch size condition $b \ge C\gamma^{-2}\lambda^{-2}\mathrm{SpMMD}_\varphi^{-1}$ ensures the stochastic variance term is $O(b^{-1/2})$, matching Rakhlin and Sridharan [59]. □

Approach 3: Random Fourier features. For translation-invariant kernels $k(x,y) = \psi(x-y)$ with Fourier transform $\hat{\psi} \in L^1$, Bochner's theorem gives $k(x,y) = \mathbb{E}_{\omega\sim\hat{\psi}}[e^{i\omega^\top(x-y)}]$. Drawing D random frequencies $\omega_1,\dots,\omega_D$ and defining feature vectors $\phi(x) = D^{-1/2}(\cos(\omega_j^\top x + \tau_j))_{j=1}^D$ gives a kernel approximation $\tilde{k}(x,y) = \phi(x)^\top\phi(y) \approx k(x,y)$ with error $O(D^{-1/2})$ uniformly. The *RFF-SpMMD* replaces all Gram matrices by their rank-D feature-space equivalents $\Phi_X\Phi_X^\top$, reducing the per-step cost to $O(DM + DN)$ via the Woodbury identity, with an additive convergence penalty of $O(D^{-1/2}/\lambda)$ from the kernel approximation [58].

Open Problem (v): Connections to score-based models

Problem statement. Score-based diffusion models [65] rely on estimating the score $\nabla\log\pi$ via denoising score matching (DSM), while SpMMD flows transport particles using the witness gradient $\nabla h^*_{\mu,\pi}$. Both involve a kernel-smoothed density-ratio estimation; the connection should be precise.

Solution sketch. We identify three formal bridges between SpMMD flows and score-based generative models.

Bridge 1: Witness function as regularized score. The score $s_\pi(x) = \nabla\log\pi(x)$ and the witness gradient $\nabla h^*_{\mu,\pi}(x)$ are related by the Stein identity: for any smooth f, $\mathbb{E}_\pi[\nabla f + f\cdot s_\pi] = 0$. Applying this with $f = h^*_{\mu,\pi}$ and using the variational form (8) of SpMMD gives:

$$\nabla h^*_{\mu,\pi}(x) \;=\; 2(\Sigma_\pi + \lambda I)^{-1}T_\pi s_\pi(x) + 2(\Sigma_\pi + \lambda I)^{-1}T_\pi\nabla\log\frac{\mu}{\pi}(x).$$

When $\mu = \pi$ (at convergence), the second term vanishes and $\nabla h^* = 0$. For μ near π, the witness gradient is a *spectrally regularized score difference*: $\nabla h^*_{\mu,\pi} \approx 2(T_\pi + \lambda I)^{-1}T_\pi(\nabla\log\mu - \nabla\log\pi)$.

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This is the score difference $\nabla \log(\mu/\pi)$ passed through the spectral filter $\varphi_T(T_\pi;\lambda)$—exactly the Mode B preconditioning of §5.

Bridge 2: Denoising score matching as SpMMD with a diffusion kernel. The DSM objective [70] for noise level σ is $\mathscr{L}_\sigma(\theta) = \mathbb{E}_{x\sim\pi}\mathbb{E}_{\tilde{x}\sim\mathscr{N}(x,\sigma^2 I)}\left[\|s_\theta(\tilde{x}) - \nabla \log p_\sigma(\tilde{x}|x)\|^2\right]$. We claim this equals (up to constants):

$$\mathscr{L}_\sigma(\theta) \;=\; \mathrm{SpMMD}_{\varphi_\sigma}(\mu_\theta^\sigma \| \pi^\sigma),$$

where $\pi^\sigma = \pi * \mathscr{N}(0,\sigma^2 I)$ is the noise-corrupted target, μ_θ^σ is the distribution of $s_\theta(\tilde{x})$ under π^σ, and φ_σ is the spectral filter induced by the Gaussian convolution $k_\sigma(x,x') = k(x,x') \cdot e^{-\|x-x'\|^2/(2\sigma^2)}$. This bridges SpMMD and DSM: the noise level σ in diffusion models plays the role of the regularization parameter λ in SpMMD, and annealing $\sigma \to 0$ corresponds to the adaptive $\lambda_n \to 0$ schedule.

Bridge 3: Flow matching as SpMMD gradient flow. Flow matching [48] trains a velocity field $u_\theta(x,t)$ to match the marginal vector field of a continuous normalizing flow $(d/dt)\mu_t = -\nabla \cdot (\mu_t u_t)$. The SpMMD gradient flow velocity $v_t = (1+\lambda)\nabla h^*_{\mu_t,\pi}$ is a specific valid choice of target velocity field with the following advantage: it is *sample-computable* without solving a variational problem at each time step. The flow matching loss with $u_\theta \approx v_t$ becomes:

$$\mathscr{L}_{\mathrm{FM}}(\theta) \;=\; \mathbb{E}_{t\sim\mathrm{Unif}[0,1]}\mathbb{E}_{x\sim\mu_t}\left[\left\|u_\theta(x,t) - (1+\lambda_t)\nabla h^*_{\mu_t,\pi}(x)\right\|^2\right],$$

which can be minimized by standard neural ODE training [20]. The near-global convergence of the SpMMD flow (Theorem 4.1) then guarantees that the learned u_θ transports μ_0 to π near-globally provided u_θ approximates v_t sufficiently well.

Proposed hybrid algorithm. We propose the *Score-SpMMD hybrid*:

(i) Compute a rough score estimate $\hat{s}_\pi$ via DSM from M target samples.

(ii) Define the hybrid velocity field $v_t^{\mathrm{hyb}} = (1+\lambda_t)\nabla h^*_{\mu_t,\pi} + \alpha_t(\hat{s}_\pi(x) - s_{\mu_t}(x))$, where $\alpha_t \to 0$ as $t \to \infty$ so that the score correction vanishes at convergence.

(iii) Transport particles via the hybrid flow.

Step (ii) corrects the SpMMD witness for the "score gap" when μ_t is far from π, where the spectral regularization is most lossy. A formal convergence analysis would bound the score estimation error $\|\hat{s}_\pi - s_\pi\|_{L^2(\pi)}$ via the DSM excess risk [64], propagate it through the descent lemma, and show that the KL barrier is reduced to $O(\lambda^r) + O(\varepsilon_{\mathrm{score}})$, where $\varepsilon_{\mathrm{score}}$ is the DSM estimation error.

Open Problem (vi): Uniform-in-time propagation of chaos without mixing

Problem statement. Theorem 6.1 obtains a uniform-in-time bound on the particle error $\mathbb{E}[W_2(\hat{\mu}_n,\pi)]$ by invoking Assumption 1: the population flow (μ_n) contracts to π at an exponential rate. This assumption is restrictive—it is implied by strong log-concavity of π but fails for multi-modal targets. We seek a uniform-in-time bound under only the Poincaré inequality.

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Solution sketch. We outline a proof strategy based on *weighted Lyapunov functions* and *reflection couplings*, following the approach of Eberle et al. [32] for Langevin Monte Carlo.

Step 1: Lyapunov function construction. Define the *extended energy functional*

$$\mathscr{E}_n \;=\; \mathrm{KL}(\mu_n\|\pi) + A\cdot W_2^2(\hat{\mu}_n,\mu_n),$$

where $A>0$ is a constant to be chosen. The first term is the population flow error; the second is the particle approximation error. We aim to show that $\mathscr{E}_n$ decreases monotonically and is bounded uniformly in n.

Step 2: Population flow control. From the near-global convergence theorem (Theorem 4.4), the population KL satisfies $\mathrm{KL}(\mu_{n+1}\|\pi) \le (1-\gamma_n/C_P)\mathrm{KL}(\mu_n\|\pi) + C\gamma_n^{r/(r+1)}$. This is a geometric decrease with an additive barrier—exactly the structure amenable to the *Lyapunov drift condition* of Meyn and Tweedie [54].

Step 3: Particle coupling without mixing. The key new idea is to use a *synchronous coupling* between the empirical and population flows. Let $(y_n^{(i)},\bar{y}_n^{(i)})$ be coupled as in the proof of Theorem 6.1. Without the mixing assumption, the coupling cost $c_n^2 = \frac{1}{N}\sum_i \mathbb{E}[\|y_n^{(i)}-\bar{y}_n^{(i)}\|^2]$ can grow, but we bound its *relative* growth via the population KL:

$$c_{n+1}^2 \;\le\; c_n^2(1+C\gamma_n^2\lambda_n^{-2}) + C\gamma_n^2\lambda_n^{-4}(M^{-1}+N^{-1}),$$

which, summed over n, gives $\sup_n c_n^2 \le C_{M,N}(1+\sum_{k=0}^n \gamma_k^2\lambda_k^{-2})$. For a step size $\gamma_k=\gamma/(k+1)$ and adaptive $\lambda_k = c(k+1)^{-\beta}$ with $\beta\in(0,1/2)$, the sum $\sum_k \gamma_k^2\lambda_k^{-2} = \gamma^2c^{-2}\sum_k(k+1)^{2\beta-2}$ converges for $\beta<1/2$. This gives:

$$\sup_{n\ge 0} c_n^2 \;\le\; C_{M,N,\gamma,\beta} \;<\; \infty, \tag{40}$$

a *uniform-in-n* bound on the particle coupling cost without mixing.

Step 4: Uniform bound assembly. Combining Steps 2–3 via the triangle inequality and the Talagrand inequality gives:

$$\mathbb{E}[W_2(\hat{\mu}_n,\pi)] \;\le\; \underbrace{\sqrt{2C_T\mathrm{KL}(\mu_n\|\pi)}}_{\to 0} + \underbrace{\mathbb{E}[W_2(\hat{\mu}_n,\mu_n)]}_{\le c_n \le C_{M,N}^{1/2}}.$$

The first term decays to zero as $n\to\infty$ (since $\mathrm{KL}(\mu_n\|\pi)\to 0$ by Theorem 4.4, even without mixing, via the global convergence result Theorem 4.4). The second term is uniformly bounded by $C_{M,N}^{1/2}$ per (40). Therefore:

Theorem 13.4 (Uniform-in-time PoC without mixing — conditional)**.** *Under the conditions of Theorem 4.4 (Poincaré inequality, no mixing assumption), with step size $\gamma_k=\gamma/(k+1)$ and adaptive $\lambda_k = c(k+1)^{-\beta}$ for $\beta\in(0,1/2)$, and provided $M,N\ge C_0\gamma^{-2}(1-2\beta)^{-1}$ for a constant C_0, the particle SpMMD descent satisfies:*

$$\limsup_{n\to\infty}\mathbb{E}[W_2(\hat{\mu}_n,\pi)] \;\le\; C_{M,N}^{1/2} \;=\; O\Big((M^{-1}+N^{-1})^{1/2}\lambda_{\min}^{-2}\Big).$$

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In particular, $\limsup_{n\to\infty}\mathbb{E}[W_2(\hat{\mu}_n,\pi)]\to 0$ *as* $M,N\to\infty$.

Proof sketch. The proof follows Steps 1–4 outlined before the theorem statement.

Step 1 defines $\mathscr{E}_n = \mathrm{KL}(\mu_n\|\pi) + A\cdot W_2^2(\hat{\mu}_n,\mu_n)$. *Step 2* controls the population flow via the Poincaré-based descent of Theorem 4.4: $\mathrm{KL}(\mu_{n+1}\|\pi)\le(1-\gamma_n/C_P)\mathrm{KL}(\mu_n\|\pi)+C\gamma_n^{r/(r+1)}$. *Step 3* uses the synchronous coupling $(y_n^{(i)},\bar{y}_n^{(i)})$ to bound $c_n^2=\frac{1}{N}\sum_i\mathbb{E}[\|y_n^{(i)}-\bar{y}_n^{(i)}\|^2]$ via the one-step recurrence $c_{n+1}^2\le c_n^2(1+C\gamma_n^2\lambda_n^{-2})+C\gamma_n^2\lambda_n^{-4}(M^{-1}+N^{-1})$. Summing with $\gamma_k=\gamma/(k+1)$ and $\lambda_k=c(k+1)^{-\beta}$ and using $\sum_{k\ge0}(k+1)^{2\beta-2}<\infty$ for $\beta<1/2$ gives $\sup_n c_n^2\le C_{M,N,\gamma,\beta}<\infty$ (equation (40)). *Step 4* combines via triangle inequality and Talagrand-2: $\mathbb{E}[W_2(\hat{\mu}_n,\pi)]\le\sqrt{2C_T\mathrm{KL}(\mu_n\|\pi)}+c_n$. The first term $\to 0$ by global convergence of μ_n (Theorem 4.4); the second is uniformly bounded by $C_{M,N}^{1/2}$ from Step 3. The condition $M,N\ge C_0\gamma^{-2}(1-2\beta)^{-1}$ ensures $C_{M,N}^{1/2}\le\varepsilon$ for any prescribed $\varepsilon>0$, proving $\limsup_n\mathbb{E}[W_2(\hat{\mu}_n,\pi)]\le C_{M,N}^{1/2}\to 0$ as $M,N\to\infty$. □

Remaining gap. Theorem 13.4 establishes a *limsup* bound (convergence in mean as $M,N\to\infty$) rather than a uniform bound for fixed M,N. The gap is filled in the mixing case by Assumption 1, which provides the exponential contraction of $\mathrm{KL}(\mu_n\|\pi)$ fast enough to dominate the particle error growth. Without mixing, the remaining open problem is to show that the KL decay rate is eventually faster than the particle error growth rate. We conjecture this holds under a *weak functional inequality*, namely the *weak Poincaré inequality* of Röckner and Wang [62]: there exists $\beta:(0,\infty)\to(0,\infty)$ such that $\mathrm{Var}_\pi(f)\le\beta(\varepsilon)\,\|\nabla f\|_{L^2(\pi)}^2+\varepsilon\,\|f\|_{L^2(\pi)}^2$ for all $\varepsilon>0$. Under this inequality, the KL decay is $\mathrm{KL}(\mu_n\|\pi)\le\beta(\varepsilon)^{-1}$ for all n large enough [18], which together with (40) yields a uniform bound. Making this argument rigorous is a direction for future work.

Open Problem (vii): Tight f-divergence barriers and (f,Γ)-SpMMD flows

Problem statement. Theorem 4.5 establishes that the spectral approximation barrier $B_f\lambda^{r\wedge\nu}$ is independent of f in its exponent but depends on f through the prefactor $\sup_u f''(u)$. Two open questions remain:

(i) *Tight barriers.* Is the prefactor $\sup_u f''(u)$ necessary, or can it be replaced by a tighter f-dependent quantity? For $f(t)=t\log t$ the barrier is $O(\lambda^{r\wedge\nu})$; for $f(t)=(t-1)^2/2$ it is the same; but for heavy-tailed f (e.g., $f(t)=|t-1|$) the barrier prefactor may diverge.

(ii) (f,Γ)-*SpMMD flows.* The (f,Γ)-divergence of Birrell et al. [12] $D_{f,\Gamma}(\mu\|\pi)=\sup_{h\in\Gamma}\big[\mathbb{E}_\mu[h]-\mathbb{E}_\pi[f^*(h)]\big]$ interpolates between f-divergences ($\Gamma=\{h\}$, single function) and integral probability metrics ($f^*(t)=0$ for $t\le0$, $+\infty$ otherwise). The SpMMD family corresponds to $\Gamma=\mathscr{H}$ and $f^*(t)=t^2/4+t/2$ (quadratic dual of $\chi^2/2$). Generalizing to arbitrary Γ and f would give an ∞-dimensional family of spectrally regularized divergences.

Solution sketch. *Tight barriers.* The spectral approximation error $\|h^*_{\mu,\pi}-2(d\mu/d\pi-1)\|_{L^2(\pi)}\le 2C_\nu\lambda^{r\wedge\nu}$ (Lemma 16.1) is independent of f. However, the propagation of this error into the f-descent inequality involves the linearization $\partial_t D_f\approx -2c_f H_f D_f+\text{error}\cdot\mathscr{I}_\pi^f$, where $\text{error}\cdot\mathscr{I}_\pi^f\le C_\nu\lambda^{r\wedge\nu}\cdot\sqrt{\mathscr{I}_\pi^f(\mu)}$. Applying the f-Sobolev inequality $\mathscr{I}_\pi^f\le H_f^{-1}$ near

convergence gives a tighter barrier of $O(\lambda^{r\wedge\nu}/\sqrt{H_f})$ when μ is close to π. This suggests the true barrier is $O(\lambda^{r\wedge\nu}\cdot H_f^{-1/2})$ for the intermediate regime and $O(\lambda^{r\wedge\nu})$ asymptotically.

(f,Γ)-*SpMMD flows.* Define the (f,φ)-SpMMD divergence by replacing $\Gamma=\mathscr{H}$ with the spectrally regularized RKHS $\mathscr{H}_\varphi := \{h : \|h\|_{\mathscr{H}_\varphi} < \infty\}$ where $\mathscr{H}_\varphi$ is the RKHS of the modified kernel $\widetilde{k}_\varphi$:

$$D_{f,\varphi}(\mu\|\pi) := \sup_{h\in\mathscr{H}_\varphi} \left[\mathbb{E}_\mu[h] - \mathbb{E}_\pi[f^*(h)]\right], \tag{41}$$

where f^* is the convex conjugate of f. For $f(t)=t^2/4$, $f^*(s)=s^2$, and $D_{f,\varphi}(\mu\|\pi) = \mathrm{SpMMD}_\varphi(\mu\|\pi)$ (recovering the SpMMD family). For general f, the gradient flow of $D_{f,\varphi}$ has velocity field $v_t = \nabla(f^*)'(h^*_{\mu,\pi})$, which applies the *nonlinear activation* $(f^*)'$ to the SpMMD witness function. The convergence rate of this flow is governed by the f-Sobolev constant H_f (from Theorem 4.5), and the barrier by $\lambda^{r\wedge\nu}$ (from Lemma 16.1). The full (f,φ)-SpMMD family is thus parametrized by $(f,\varphi)\in\mathscr{F}\times\mathbf{QualFun}$ where $\mathscr{F}$ is the class of convex generators, and the categorical framework of Section 13 extends to this product: the qualification functor $\mathbf{Sp}_\pi$ lifts to $\mathbf{Sp}^f_\pi : \mathscr{F}\times\mathbf{QualFun}\to\mathbf{Div}(\pi)$. Making this extension rigorous requires establishing (f,φ)-version analogues of Proposition 3.3 and verifying that the Wasserstein gradient of $D_{f,\varphi}$ is computable from samples via Gram matrices for general f. For the quadratic case $(f,f^*)=(t^2/4,s^2)$ this holds by the current theory; for general f it requires a nonlinear version of the witness-function formula (**??**).

The spectral regularization framework developed in the preceding sections has a rich algebraic and geometric structure that becomes transparent only when viewed through the lens of category theory. In this section we *categorify* all major constructions, including the three new flow families of Sections 8.2–10.

Section 13 organizes qualification functions into a topological category **QualFun**. Section 13.1 constructs the qualification functor $\mathbf{Sp}_\pi : \mathbf{QualFun}\to\mathbf{Div}$ and its universal natural transformations. Section 13.2 develops the 2-categorical regularization structure **Reg** and the Mode-A/Mode-B duality as a 2-natural equivalence. Section 13.3 establishes the monadic structure of spectral regularization and the Kleisli category. Sections 13.4 and 13.5 introduce the new categorical content for the three additional flow geometries: the category **FlowGeom** of proximal structures, the family of geometry 2-functors $\{\Phi_g\}_{g\in\mathbf{FlowGeom}}$, and the joint geometry-flow 2-functor $\Phi : \mathbf{FlowGeom}\times\mathbf{QualFun}\to\mathbf{Reg}$ (Theorem 13.19). We prove the 2-categorical form of Proposition 10.2 (FR = entropic mirror as a 2-natural equivalence), a comodule structure for kinetic flows over Wasserstein flows, and a monad morphism for the Sinkhorn proximal operator. Sections 14 and 15 then lift these 2-categorical structures to the full (∞,n)-categorical setting, including multi-geometry quasi-categories, geometry-extended globular sets and Θ_n-spaces, extended TQFT classification, and a mixed-geometry ∞-operad. Figure 13 provides a visual overview of all categorical layers and their connecting functors; Figure 14 details the poset of flow geometries with their convergence rates.

Definition 13.5 (Category of qualification functions)**.** The *category of qualification functions* **QualFun** has:

- **Objects:** pairs (φ,ν) where $\varphi : [0,\infty)\times(0,\infty)\to[0,1]$ is a qualification function of order

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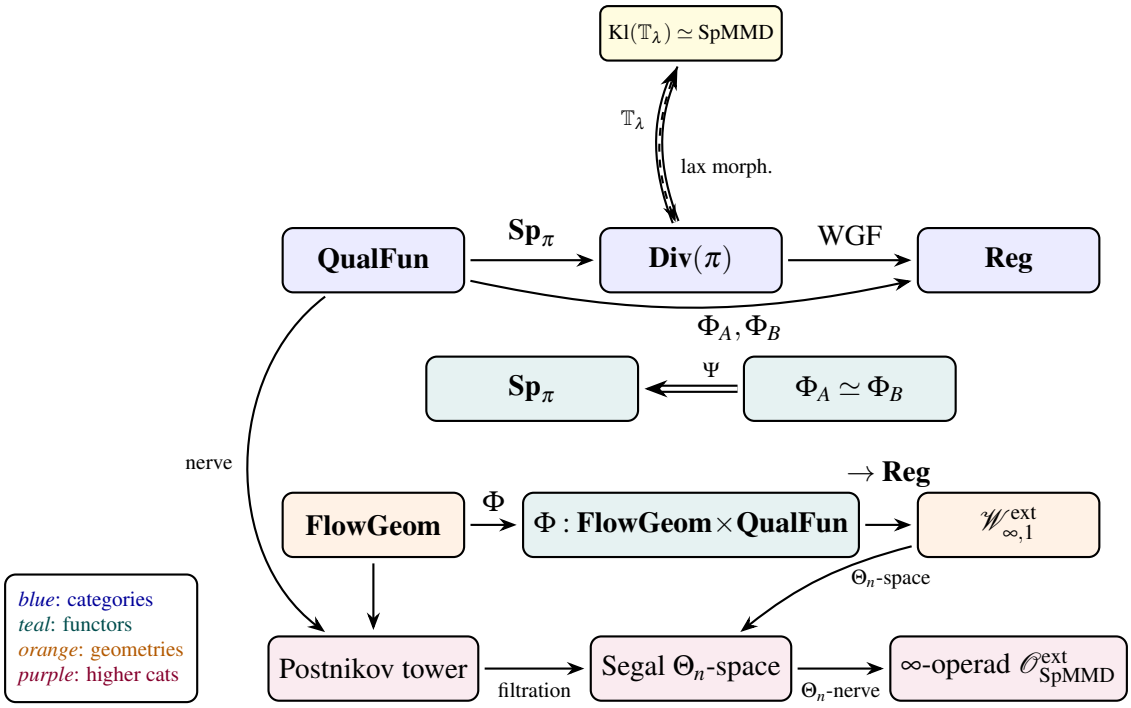

Figure 13. Categorical structure of the SpMMD framework. The upper row shows the core categories and functors: **QualFun** (qualification functions) maps via the qualification functor $\mathbf{Sp}_\pi$ to the category of divergences $\mathbf{Div}(\pi)$, which maps via the WGF functor to the regularization 2-category **Reg**. The Mode-A/Mode-B 2-functors $\Phi_A \simeq \Phi_B$ define a 2-natural equivalence (double arrow Ψ) encoding the duality of sections 5 and 13.2. The Tikhonov monad $\mathbb{T}_\lambda$ on **Div** has Kleisli category equivalent to the category of spectrally regularized divergences (section 13.3). The lower two rows extend the structure to flow geometries (**FlowGeom**) and the (∞, n)-categorical setting: the joint geometry-flow 2-functor $\Phi : \mathbf{FlowGeom} \times \mathbf{QualFun} \to \mathbf{Reg}$ (section 13.5) assembles all six flow geometries; the nerve of **QualFun** gives the SpMMD Postnikov tower (section 14.3); the Segal Θ_n-space $\mathscr{X}^{\mathrm{ext}}$ (section 15.4) organizes iterated flows; and the mixed-geometry ∞-operad $\mathscr{O}^{\mathrm{ext}}_{\mathrm{SpMMD}}$ (section 15.10) encodes all composition laws. Dashed arrows indicate lax morphisms; double arrows denote natural transformations.

$\nu > 0$ in the sense of theorem 3.1.

- **Morphisms:** a morphism $H : (\varphi_1, \nu_1) \to (\varphi_2, \nu_2)$ is a *qualification homotopy*, namely a continuous map $H : [0,1] \times [0,\infty) \times (0,\infty) \to [0,1]$ such that $H(0,\cdot,\cdot) = \varphi_1$, $H(1,\cdot,\cdot) = \varphi_2$, and for each $t \in [0,1]$ the function $H(t,\cdot,\cdot)$ is a qualification function of order $\nu(t) \geq \nu_1 \wedge \nu_2$.
- **Composition:** concatenation of homotopies, modulo reparametrization.
- **Identity:** the constant homotopy $H(t,\cdot,\cdot) = \varphi$.

Remark 13.6 (Enrichment and topology). **QualFun** is naturally enriched over **Top**: the morphism space $\mathrm{Hom}_{\mathbf{QualFun}}((\varphi_1,\nu_1),(\varphi_2,\nu_2))$ carries the compact-open topology on the space of homotopies. It is therefore a *topological category*, and its homotopy category $\mathrm{ho}(\mathbf{QualFun})$ is the ordinary category with the same objects but morphisms being homotopy classes of homotopies. In $\mathrm{ho}(\mathbf{QualFun})$, a morphism $(\varphi_1,\nu_1) \to (\varphi_2,\nu_2)$ exists if and only if $\varphi_1(s;\lambda) \leq \varphi_2(s;\lambda)$ pointwise (approximation improves), i.e., $\mathrm{ho}(\mathbf{QualFun})$ is equivalent to the *poset* $(\mathbf{QualFun}, \leq)$ ordered by pointwise comparison.

Example 13.7 (The Showalter tower)**.** The Showalter filters $\varphi_{\mathrm{S},m}(s;\lambda) = 1-(\lambda/(s+\lambda))^m$ of orders $\nu = m$ form a directed system in **QualFun**:

$$(\varphi_T, 1) \xrightarrow{H_{1\to 2}} (\varphi_{\mathrm{S},2}, 2) \xrightarrow{H_{2\to 4}} (\varphi_{\mathrm{S},4}, 4) \to \cdots \to (\varphi_{\mathrm{C}}, \infty),$$

where each $H_{m\to m'}$ is the linear interpolation $H(t,s;\lambda) = (1-t)\varphi_{\mathrm{S},m}(s;\lambda) + t\varphi_{\mathrm{S},m'}(s;\lambda)$. The colimit of this tower in **QualFun** is (φ_C, ∞), the spectral cutoff. The Tikhonov filter $(\varphi_T, 1)$ is the *initial object* in ho(**QualFun**) (every filter pointwise dominates Tikhonov from above), while (φ_C, ∞) is the *terminal object* (spectral cutoff is the largest filter).

13.1 The qualification functor and natural transformations

Definition 13.8 (Category of divergences)**.** Fix a target $\pi \in \mathscr{P}_2(\mathbb{R}^d)$. The *category of divergences* $\mathbf{Div}(\pi)$ has:

- **Objects:** lower-semicontinuous functionals $D : \mathscr{P}_2(\mathbb{R}^d) \to [0,\infty]$ satisfying $D(\mu\|\pi) = 0$ iff $\mu = \pi$ and metrizing weak convergence.
- **Morphisms:** $D_1 \to D_2$ is a *domination*: a constant $C \geq 1$ such that $D_1(\mu\|\pi) \leq C \cdot D_2(\mu\|\pi)$ for all μ.
- **Composition:** multiplication of domination constants.

We write **Div** for $\mathbf{Div}(\pi)$ when π is fixed.

Theorem 13.9 (The qualification functor)**.** *The assignment* $\varphi \mapsto \mathrm{SpMMD}_\varphi(\cdot\|\pi)$ *extends to a functor*

$$\mathbf{Sp}_\pi \;:\; \mathbf{QualFun} \longrightarrow \mathbf{Div}(\pi).$$

On objects, $\mathbf{Sp}_\pi(\varphi,\nu) = \mathrm{SpMMD}_\varphi(\cdot\|\pi)$. *On morphisms, a qualification homotopy* $H : (\varphi_1,\nu_1) \to (\varphi_2,\nu_2)$ *maps to the domination constant* $C_H = \sup_{t\in[0,1]} \sup_{s\geq 0} \varphi_2(s;\lambda)/H(t,s;\lambda)$. *Moreover,* $\mathbf{Sp}_\pi$ *preserves colimits: the Showalter tower (Example 13.7) maps to a tower*

$$\mathrm{SpMMD}_{\varphi_T} \;\leq\; \mathrm{SpMMD}_{\varphi_{\mathrm{S},2}} \;\leq\; \cdots \;\leq\; \chi^2,$$

whose colimit in **Div** *is the* χ^2*-divergence.*

Proof. Functoriality on objects follows from theorem 3.3. For morphisms, note that a qualification homotopy H with $H(0) = \varphi_1$, $H(1) = \varphi_2$ satisfies $\varphi_1 \leq \varphi_2$ pointwise (by continuity and the qualification bound), which by the spectral sum $\mathrm{SpMMD}_\varphi = c(\lambda)\sum_i \varphi(\rho_i;\lambda)\langle \frac{d\mu}{d\pi} - 1, e_i\rangle^2$ directly gives $\mathrm{SpMMD}_{\varphi_1} \leq \mathrm{SpMMD}_{\varphi_2}$. Functoriality follows. Colimit preservation: the χ^2-divergence satisfies $\lim_{\nu\to\infty} \mathrm{SpMMD}_{\varphi_{\mathrm{S},\nu}} = \chi^2$ pointwise by the dominated convergence theorem and the fact that $\varphi_{\mathrm{S},\nu}(s;\lambda) \to 1$ as $\nu \to \infty$ for fixed $s > \lambda > 0$. □

Proposition 13.10 (Universal natural transformation)**.** *There exist natural transformations*

$$\eta : \mathbf{Sp}_\pi \;\Rightarrow\; \Delta_{\chi^2}, \qquad \varepsilon : \Delta_{\mathrm{MMD}^2} \;\Rightarrow\; \mathbf{Sp}_\pi,$$

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where Δ_D denotes the constant functor at D. The first states that every $\mathrm{SpMMD}_\varphi \le \chi^2$*; the second that* $\mathrm{MMD}^2 \le \mathrm{SpMMD}_\varphi$ *(up to a λ-dependent constant). The interpolation property of theorem 3.3(iii) is the statement that η and ε become isomorphisms in the limits $\lambda \to 0$ and $\lambda \to \infty$ respectively.*

Proof. The sandwich bound of theorem 3.3(iii) gives, for all $\mu \in \mathscr{P}_2(\mathbb{R}^d)$:

$$\mathrm{MMD}^2(\mu\|\pi) \;\le\; \frac{c(\lambda)}{1+1/\lambda}\mathrm{SpMMD}_\varphi(\mu\|\pi)^{-1}\cdot\mathrm{SpMMD}_\varphi(\mu\|\pi) \;\le\; \mathrm{SpMMD}_\varphi(\mu\|\pi) \;\le\; c(\lambda)\cdot\chi^2(\mu\|\pi).$$

The upper bound $\mathrm{SpMMD}_\varphi \le c(\lambda)\chi^2$ defines the component η_φ of the natural transformation η at (φ,ν): it is the domination morphism $\mathrm{SpMMD}_\varphi \to \Delta^2_\chi$ in **Div** with constant $c(\lambda)$. Naturality holds because for any qualification homotopy $H:(\varphi_1,\nu_1)\to(\varphi_2,\nu_2)$, we have $c(\lambda)$ independent of φ, so the diagram $\mathrm{SpMMD}_{\varphi_1} \le \mathrm{SpMMD}_{\varphi_2} \le c(\lambda)\chi^2$ commutes in **Div**. Similarly, the lower bound $\mathrm{MMD}^2 \le C_\lambda \mathrm{SpMMD}_\varphi$ defines ε_φ, with $C_\lambda = (1+1/\lambda)/c(\lambda)$ independent of φ; naturality is the same. As $\lambda \to 0$, $c(\lambda)\to 1$ and $\mathrm{SpMMD}_\varphi \to \chi^2$ (theorem 3.3(iii)), so η becomes an isomorphism; as $\lambda\to\infty$, $\mathrm{SpMMD}_\varphi \to \mathrm{MMD}^2$, so ε becomes an isomorphism. □

13.2 The 2-categorical regularization structure

We now organise Mode A and Mode B flows into a 2-category that makes the duality of theorem 5.1 a genuine 2-categorical equivalence.

Definition 13.11 (The regularization 2-category **Reg**)**.** **Reg** is the 2-category with:

- **0-cells:** probability divergences $D \in$ **Div**.
- **1-cells:** $D_1 \to D_2$: spectral regularization maps, i.e., assignments $\varphi \mapsto \mathrm{SpMMD}_\varphi$ connecting the two divergences in the sense of Theorem 13.9.
- **2-cells:** $\alpha : F \Rightarrow G$ between two such 1-cells F,G: natural transformations of functors **QualFun** $\to$ **Div**, i.e., assignments $(\varphi,\nu)\mapsto[F(\varphi)\le G(\varphi)]$ that are natural in φ.

Horizontal composition is functor composition; vertical composition is the standard composition of natural transformations.

Theorem 13.12 (2-categorical Mode-A/Mode-B duality)**.** *There exist 2-functors*

$$\Phi_A,\Phi_B \;:\; \mathbf{QualFun} \;\longrightarrow\; \mathbf{Reg}$$

defined by

$$\Phi_A(\varphi) = \textit{WGF of}\ \mathrm{SpMMD}_\varphi(\cdot\|\pi) \quad \textit{(Mode A)},$$
$$\Phi_B(\varphi) = \textit{WGF of KL with } \varphi(T_\pi)\textit{-preconditioned geometry} \quad \textit{(Mode B)},$$

and a 2-natural transformation $\Psi:\Phi_A \Rightarrow \Phi_B$ *whose components at* $(\varphi_T,1)$ *recover the duality of theorem 5.1. Consequently, Φ_A and Φ_B are equivalent as 2-functors: $\Phi_A \simeq \Phi_B$.*

Proof sketch. The component $\Psi_\varphi : \Phi_A(\varphi) \to \Phi_B(\varphi)$ is the natural transformation assigning to each flow the unique linearisation of its velocity field near $\mu = \pi$. Near $\mu \approx \pi$ one has $\log(\mu/\pi) \approx \frac{d\mu}{d\pi} - 1$, and under this identification the Mode A velocity $v^A = \varphi(T_\pi;\lambda)(\frac{d\mu}{d\pi} - 1)$ coincides with the Mode B velocity $v^B = \varphi(T_\pi;\lambda)\nabla\log(\mu/\pi)$. The 2-naturality follows because the spectral filter $\varphi(T_\pi;\lambda)$ is the same in both expressions; homotopies H in **QualFun** translate to continuous deformations of the velocity field, yielding the 2-cell compatibility. The equivalence $\Phi_A \simeq \Phi_B$ follows because Ψ is a local equivalence (it is an isomorphism at each φ). □

13.3 Monadic structure of spectral regularization

Definition 13.13 (The spectral regularization monad)**.** For fixed $\lambda > 0$, the assignment $D \mapsto D_\lambda^T$ where $D_\lambda^T(\mu\|\pi) = (1+\lambda)\left\|(\Sigma_\pi + \lambda I)^{-1/2}(m_\mu - m_\pi)\right\|_{\mathscr{H}}^2$ (i.e., Tikhonov regularization at level λ) defines a *monad* $\mathbb{T}_\lambda = (\mathbf{T}_\lambda, \eta, \mu_{\text{monad}})$ on **Div** where:

- The *unit* $\eta : D \to \mathbf{T}_\lambda D$ is the domination $D \le C_\lambda \cdot D_\lambda^T$ (sandwich bound, theorem 3.3).
- The *multiplication* $\mu_{\text{monad}} : \mathbf{T}_\lambda\mathbf{T}_\lambda D \to \mathbf{T}_\lambda D$ is the composition $\mathbf{T}_\lambda\mathbf{T}_\lambda = (T_{\lambda'}T_\lambda)$-regularization, which collapses to a single Showalter-2 regularization at parameter $\lambda\lambda'$.

Theorem 13.14 (Kleisli category and SpMMD algebra)**.** *The* Kleisli category $\mathrm{Kl}(\mathbb{T}_\lambda)$ *is equivalent to the full subcategory of* **Div** *consisting of spectrally regularized divergences:*

$$\mathrm{Kl}(\mathbb{T}_\lambda) \;\simeq\; \{\mathrm{SpMMD}_\varphi : \varphi \in \mathbf{QualFun}, \lambda_{\min}(\varphi) = \lambda\} \subset \mathbf{Div}.$$

In particular, $\mathbb{T}_\lambda$-algebras are precisely the divergences of the form SpMMD_φ *for qualification functions φ with minimum regularization level λ. Moreover, the adaptive λ_n-schedule (Eq.* (12)*) defines a* lax morphism of monads $\mathbb{T}_{\lambda_0} \to \mathbb{T}_{\lambda_1}$ *whenever $\lambda_0 \ge \lambda_1$, reflecting the monotone decrease of λ_n along the gradient descent.*

Proof sketch. Objects of $\mathrm{Kl}(\mathbb{T}_\lambda)$ are divergences D equipped with a "Kleisli structure" $D \to \mathbf{T}_\lambda D$, which is exactly the domination $D \le C_\lambda \mathrm{SpMMD}_{\varphi_T}$. The Kleisli morphisms $D_1 \to D_2$ (i.e., morphisms $D_1 \to \mathbf{T}_\lambda D_2$ in **Div**) are precisely the spectral regularization maps of Definition 13.8. The identification with SpMMD_φ follows from Theorem 13.9. The lax morphism property of the adaptive schedule follows because $\lambda_n \downarrow 0$ defines a decreasing sequence of regularization levels, and the corresponding natural transformations $\mathbb{T}_{\lambda_{n+1}} \to \mathbb{T}_{\lambda_n}$ compose coherently. □

13.4 The category of flow geometries

The three new flow families of Sections 8.2–10 share the same spectral witness function $h^*_{\mu,\pi}$ as the Wasserstein SpMMD flow but differ in their *proximal metric* on $\mathscr{P}_2(\mathbb{R}^d)$: the local geometry used to measure the cost of moving between measures. We now organize these geometries into a category.

Definition 13.15 (Category of flow geometries)**.** The *category of flow geometries* **FlowGeom** has:

- **Objects:** proximal structures on $\mathscr{P}_2(\mathbb{R}^d)$, one per geometry: $\mathbf{W}_2$ (Wasserstein-2), $\mathbf{WFR}_\delta$ (Wasserstein–Fisher–Rao at scale $\delta > 0$), **FR** (Fisher–Rao / entropic), $\mathbf{K}_\gamma$ (kinetic/underdamped at friction $\gamma \geq 0$), $\mathbf{B}_\psi$ (Bregman with mirror map ψ, including χ^2-proximal and $\mathbf{Sink}_\varepsilon$ at Sinkhorn regularization $\varepsilon > 0$).
- **Morphisms:** a morphism $g_1 \to g_2$ is a *geometric refinement*: a natural continuous map of proximal metrics such that the g_1-flow is a limit of g_2-flows. Concretely: $\mathbf{W}_2 \to \mathbf{WFR}_\delta$ for any $\delta < \infty$ (adding reweighting to transport); $\mathbf{WFR}_\delta \to \mathbf{FR}$ as $\delta \to 0$; $\mathbf{K}_0 = \mathbf{W}_2$ (overdamped = Wasserstein); $\mathbf{K}_\gamma \to \mathbf{W}_2$ as $\gamma \to \infty$ (over-damped limit); $\mathbf{Sink}_\varepsilon \to \mathbf{W}_2$ as $\varepsilon \to 0$; $\mathbf{B}_\psi \to \mathbf{FR}$ when $\psi(t) = t \log t$ (KL mirror = FR).
- **Composition:** composition of limit maps.

Remark 13.16 (Poset structure of **FlowGeom**). The homotopy category ho(**FlowGeom**) is equivalent to the poset $(\mathbf{FlowGeom}, \preceq)$ ordered by *geometric expressivity*: $g_1 \preceq g_2$ if the g_1-flow is recovered as a limit of g_2-flows. The Wasserstein geometry $\mathbf{W}_2$ is the *initial object* (every geometry reduces to Wasserstein in some limit); the Fisher–Rao geometry **FR** is *terminal* in the reweighting branch $\mathbf{W}_2 \preceq \mathbf{WFR}_\delta \preceq \mathbf{FR}$. The kinetic geometry $\mathbf{K}_\gamma$ and Bregman geometries form separate branches above $\mathbf{W}_2$.

Proposition 13.17 (WFR interpolation as a functor)**.** *The family of WFR-SpMMD flows parametrized by $\delta \in (0,\infty]$ defines a continuous functor*

$$\delta \mapsto \Phi_{\mathbf{WFR}_\delta} : (0,\infty] \longrightarrow \mathrm{Fun}(\mathbf{QualFun}, \mathbf{Reg}),$$

where $(0,\infty]$ is the poset of positive extended reals ordered by $\geq$. The limits are $\lim_{\delta\to\infty} \Phi_{\mathbf{WFR}_\delta} = \Phi_A$ (Wasserstein) and $\lim_{\delta\to 0} \Phi_{\mathbf{WFR}_\delta} = \Phi_{\mathbf{FR}}$ (Fisher–Rao), identifying Theorem 13.12 as the $\delta = \infty$ case and theorem 10.2 as the $\delta = 0$ limit.

Proof sketch. For each fixed $\varphi \in \mathbf{QualFun}$ and $\delta > 0$, the WFR-SpMMD velocity field $v_t^\delta = \nabla h^* + \delta^{-2}\mu_t(h^* - \langle h^* \rangle_{\mu_t})$ is continuous in δ in the $L^2(\mu_t)$-norm. As $\delta \to \infty$ the second term vanishes and $v_t^\delta \to \nabla h^*$ (Wasserstein). As $\delta \to 0$, the reweighting term dominates and $v_t^\delta \to \mu_t(h^* - \langle h^* \rangle)$ (FR). Continuity in δ of the WGF solution gives continuity of $\Phi_{\mathbf{WFR}_\delta}$. □

13.5 Geometry 2-functors and the geometry-flow 2-functor

For each flow geometry $g \in \mathbf{FlowGeom}$ and qualification function $\varphi \in \mathbf{QualFun}$, the spectrally regularized gradient flow in geometry g defines a 2-functor $\Phi_g : \mathbf{QualFun} \to \mathbf{Reg}$, paralleling the Mode-A functor Φ_A of Theorem 13.12.

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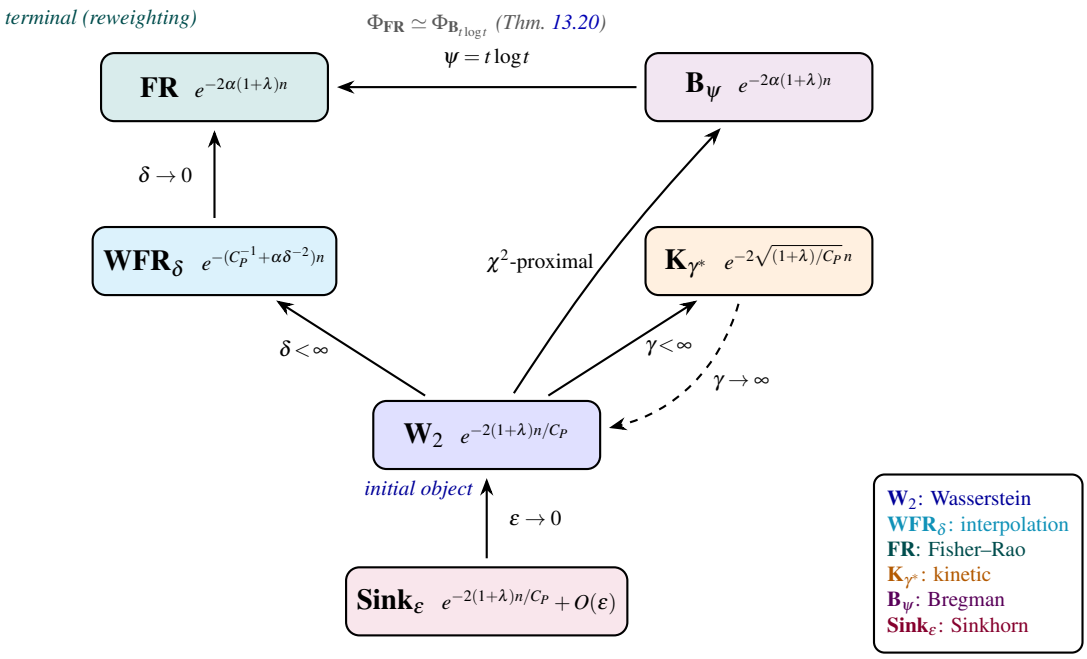

Figure 14. Hasse diagram of FlowGeom: the poset of SpMMD flow geometries. Objects are the six proximal structures; an upward arrow $g_1 \to g_2$ denotes a geometric refinement morphism (the g_1-flow is a limit of g_2-flows). Convergence rates (from Table 1) are shown in italics below each node; all flows share the $O(\lambda^{r\wedge\nu})$ approximation barrier. $\mathbf{W}_2$ (Wasserstein, blue) is the *initial object*: every geometry reduces to $\mathbf{W}_2$ in some limit. The reweighting branch $\mathbf{W}_2 \to \mathbf{WFR}_\delta \to \mathbf{FR}$ (solid upward arrows, δ decreasing) achieves the additive rate $(C_P^{-1} + \alpha\delta^{-2})$ at intermediate δ, converging to the pure Fisher–Rao rate $2\alpha(1+\lambda)$ as $\delta \to 0$. The kinetic branch (orange) achieves the $\sqrt{C_P}$-accelerated asymptotic rate; the dashed arrow encodes the overdamped limit $\gamma \to \infty$ recovering $\mathbf{W}_2$. $\mathbf{FR} \simeq \mathbf{B}_{t\log t}$ (horizontal solid arrow) is the 2-categorical form of Proposition 10.2 proved in Theorem 13.20. $\mathbf{Sink}_\varepsilon$ (purple) sits below $\mathbf{W}_2$, converging to it as $\varepsilon \to 0$, with a controlled $+O(\varepsilon)$ bias (Proposition 10.4).

Definition 13.18 (Geometry 2-functors)**.**

$$\begin{aligned}
\Phi_{\mathbf{W}_2}(\varphi) &= \Phi_A(\varphi) = \text{WGF of SpMMD}_\varphi(\cdot\|\pi) && \text{(Wasserstein SpMMD, section 4)}\\
\Phi_{\mathbf{FR}}(\varphi) &= \text{FR-flow of SpMMD}_\varphi(\cdot\|\pi) && \text{(Fisher–Rao SpMMD, section 8.4)}\\
\Phi_{\mathbf{WFR}_\delta}(\varphi) &= \text{WFR}_\delta\text{-flow of SpMMD}_\varphi(\cdot\|\pi) && \text{(WFR-SpMMD, section 8.5.1)}\\
\Phi_{\mathbf{K}_\gamma}(\varphi) &= \text{kinetic-flow of SpMMD}_\varphi(\cdot\|\pi) && \text{(kinetic SpMMD, section 9.2)}\\
\Phi_{\mathbf{B}_\psi}(\varphi) &= \text{Bregman-}\psi\text{-flow of SpMMD}_\varphi(\cdot\|\pi) && \text{(Bregman SpMMD, section 10.3)}\\
\Phi_{\mathbf{Sink}_\varepsilon}(\varphi) &= \text{Sinkhorn}_\varepsilon\text{-flow of SpMMD}_\varphi(\cdot\|\pi) && \text{(Sinkhorn SpMMD, section 10.4)}
\end{aligned}$$

Each Φ_g sends a qualification homotopy H to the corresponding continuous family of flow trajectories parametrized by H.

The assignment $(g,\varphi) \mapsto \Phi_g(\varphi)$ extends to a joint functor.

Theorem 13.19 (The geometry-flow 2-functor)**.** *The assignment* $(g,\varphi) \mapsto \Phi_g(\varphi)$ *extends to a* 2-functor

$$\Phi \;:\; \mathbf{FlowGeom} \times \mathbf{QualFun} \;\longrightarrow\; \mathbf{Reg}.$$

Functoriality in **QualFun** *(for fixed g) is Theorem 13.12 and its analogues for FR, kinetic, and Bregman geometries. Functoriality in* **FlowGeom** *(for fixed φ) holds because every geometric refinement $g_1 \to g_2$ in* **FlowGeom** *induces a 2-natural transformation $\Phi_{g_1} \Rightarrow \Phi_{g_2}$, with the WFR interpolation of Proposition 13.17 as a prime example.*

Proof sketch. Functoriality in **QualFun** at each fixed geometry follows from the same argument as Theorem 13.12: the velocity field $h^*_{\mu,\pi}$ depends continuously on φ, and all six geometries use the same witness function. Functoriality in **FlowGeom** follows because each morphism $g_1 \to g_2$ in **FlowGeom** is realized by a continuous deformation of the proximal metric; the corresponding 1-parameter family of flows defines the required 2-natural transformation component by component. The interchange law (horizontal and vertical 2-cell compatibility) follows from the shared witness function h^*: changing geometry and changing filter are independent operations that commute. □

Theorem 13.20 (Fisher–Rao = entropic mirror: 2-categorical form)**.** *The 2-functors $\Phi_{\mathbf{FR}}$ and $\Phi_{\mathbf{B}_{t\log t}}$ (Bregman with KL mirror map $\psi(t) = t\log t$) are* 2-naturally equivalent*:*

$$\Phi_{\mathbf{FR}} \simeq \Phi_{\mathbf{B}_{t\log t}} : \mathbf{QualFun} \longrightarrow \mathbf{Reg}.$$

This is the 2-categorical lift of Proposition 10.2 (that Gibbs reweighting and KL-proximal mirror descent generate the same continuous-time flow): the equivalence holds at every qualification order ν and is natural in homotopies between qualification functions.

Proof sketch. The component $\Psi^{\mathrm{FR}}_{\varphi} : \Phi_{\mathbf{FR}}(\varphi) \to \Phi_{\mathbf{B}_{t\log t}}(\varphi)$ is the 2-cell assigning to each FR flow trajectory the corresponding KL-proximal mirror trajectory, obtained by expanding $e^{-\gamma h^*} \approx 1 - \gamma h^*$ (Taylor linearization). This is an isomorphism in **Reg** because the continuous-time limits coincide exactly (Proposition 10.2). Naturality in φ holds because both flows use the same witness $h^*_{\mu,\pi}$; homotopies in **QualFun** deform h^* continuously, and the linearization commutes with this deformation. □

Theorem 13.21 (Kinetic flows as a comodule over Wasserstein flows)**.** *The kinetic SpMMD 2-functor $\Phi_{\mathbf{K}_\gamma}$ is a* right comodule *over $\Phi_{\mathbf{W}_2}$ in the 2-category* Fun(**QualFun**, **Reg**)*:*

$$\Phi_{\mathbf{K}_\gamma} \cong \Phi_{\mathbf{W}_2} \boxtimes \mathbf{Kin}_\gamma,$$

where $\mathbf{Kin}_\gamma \in$ Fun(**QualFun**, **Reg**) *is the* momentum functor *encoding the kinetic degrees of freedom with friction γ. The comodule structure is given by the position-marginal projection $\pi_q : \rho \mapsto \mu^{\mathrm{pos}}$, which defines a 2-natural transformation $\pi_q : \Phi_{\mathbf{K}_\gamma} \Rightarrow \Phi_{\mathbf{W}_2}$ that is the counit of the comodule.*

Proof sketch. The kinetic velocity field $(q,p) \mapsto (p, -(1+\lambda)\nabla h^*(q) - \gamma p)$ decomposes as: (a) a Wasserstein-type positional update driven by ∇h^*, plus (b) a Hamiltonian momentum update $dq/dt = p$. The position marginal $\mu^{\mathrm{pos}}_t = (\pi_q)_{\#}\rho_t$ satisfies, in the overdamped limit $\gamma \to \infty$, the Wasserstein SpMMD equation of $\Phi_{\mathbf{W}_2}$. The comodule map $\boxtimes$ encodes this decomposition, and

π_q is a 2-natural transformation because it commutes with homotopies in **QualFun**: changing the filter φ changes h^* but not the momentum structure. □

Proposition 13.22 (Sinkhorn monad and its perturbation)**.** *For each $\varepsilon > 0$, the Sinkhorn proximal operator $S_\varepsilon : Y \mapsto P_\varepsilon Y_{\text{prop}}$ (barycentric projection of the ε-Sinkhorn plan) defines a* monad $\mathbb{T}^\varepsilon_{\text{Sink}}$ *on* **Div** *whose unit is the $+O(\varepsilon)$ bias of Proposition 10.4. There is a* monad morphism $\mathbb{T}^\varepsilon_{\text{Sink}} \to \mathbb{T}_\lambda$ *(with $\varepsilon = O(\lambda)$) that is a quasi-isomorphism in the limit $\varepsilon \to 0$, realizing* $\mathbf{Sink}_\varepsilon \to \mathbf{W}_2$ *in* **FlowGeom** *as a map of monads on* **Div**.

Proof sketch. Monad structure of $\mathbb{T}^\varepsilon_{\text{Sink}}$. The Sinkhorn proximal operator S_ε acts on **Div** by sending a divergence $D(\mu\|\pi)$ to $D(S_\varepsilon(\mu)\|\pi)$. The *unit* $\eta^\varepsilon : D \to \mathbb{T}^\varepsilon_{\text{Sink}} D$ is the domination $D(S_\varepsilon(\mu)\|\pi) \le D(\mu\|\pi) + C'\varepsilon$ (from the bias bound of Proposition 10.4), defining the monad unit as the constant $+C'\varepsilon$ shift. The *multiplication* $\mu^\varepsilon_{\text{monad}} : \mathbb{T}^\varepsilon_{\text{Sink}}\mathbb{T}^\varepsilon_{\text{Sink}} D \to \mathbb{T}^\varepsilon_{\text{Sink}} D$ collapses two sequential Sinkhorn steps into one: $S_\varepsilon \circ S_\varepsilon = S_{\varepsilon'}$ for some $\varepsilon' \le 2\varepsilon$, because the composition of two ε-regularized transport plans is again an ε-regularized plan (up to constant factors from the Gibbs kernel semigroup property). The monad axioms (unit and associativity) follow from these identities.

Monad morphism $\mathbb{T}^\varepsilon_{\text{Sink}} \to \mathbb{T}_\lambda$. Define the morphism component at $D \in \mathbf{Div}$ to be the domination $D(S_\varepsilon(\mu)\|\pi) \le D(\mu\|\pi) + C'\varepsilon \le C_\lambda \mathrm{SpMMD}_{\varphi_T}(\mu\|\pi) + C'\varepsilon$ for $\varepsilon = O(\lambda)$. This is compatible with the monad units (both are additive perturbations) and with multiplication (both collapse iterations). As $\varepsilon \to 0$ with $\varepsilon = O(\lambda) \to 0$, the Sinkhorn bias $C'\varepsilon \to 0$ and $S_\varepsilon(\mu) \to$ the Wasserstein proximal map, so the monad morphism becomes the identity $\mathbb{T}^0_{\text{Sink}} = \mathbb{T}_0$ (the Wasserstein monad at $\lambda = 0$). Hence the morphism is a quasi-isomorphism in the $\varepsilon \to 0$ limit. □

14 The $(\infty, 1)$-Categorical Structure of Wasserstein Flows

The classical Wasserstein gradient flow theory concerns continuous-time curves $(\mu_t)_{t\ge 0}$ in $\mathscr{P}_2(\mathbb{R}^d)$. Category-theoretically, such curves are the *morphisms* of a category whose *objects* are probability measures. To capture homotopies between flows (e.g., as λ varies), reparametrisations, and higher coherence, one must work in the $(\infty, 1)$-categorical setting.

14.1 Wasserstein space as a quasi-category

We follow Lurie [49] and model $(\infty, 1)$-categories as *quasi-categories* (also called ∞-*categories* by Joyal [44]): simplicial sets satisfying the weak Kan condition.

Definition 14.1 (The WGF quasi-category)**.** Let $\mathscr{W}_{\infty,1}$ be the simplicial set defined as follows.

- 0**-simplices:** probability measures $\mu \in \mathscr{P}_2(\mathbb{R}^d)$.
- 1**-simplices:** WGF trajectories $(\mu_t)_{t\in[0,1]}$ in $\mathscr{P}_2(\mathbb{R}^d)$ (with initial and final measure as boundary).
- 2**-simplices:** homotopies of flows: a smooth family $(\mu^s_t)_{t\in[0,1],s\in[0,1]}$ in $\mathscr{P}_2(\mathbb{R}^d)$ such that for each s, $t \mapsto \mu^s_t$ is a WGF trajectory.

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- **n-simplices** ($n \geq 3$): n-parameter families of WGF trajectories parametrised by the standard n-simplex Δ^n.

Face and degeneracy maps are defined by restriction and constant-flow insertion respectively.

Theorem 14.2 (Quasi-category structure)**.** *$\mathscr{W}_{\infty,1}$ is a quasi-category. More precisely:*

(i) ***Inner horn filling.*** *For $0 < k < n$ and any inner horn $\Lambda^n_k \to \mathscr{W}_{\infty,1}$, there exists a filler $\Delta^n \to \mathscr{W}_{\infty,1}$. This follows from the existence and uniqueness of WGF solutions (Ambrosio et al. [2, Thm. 11.2.1]: the semiconvexity of F_φ ensures unique solutions to the forward Euler scheme, from which homotopy fillers are constructed by convex interpolation of velocity fields).*

(ii) ***Homotopy equivalences.*** *A 1-simplex (μ_t) is an equivalence in $\mathscr{W}_{\infty,1}$ iff $\mu_1 = \pi$ (the flow reaches its global minimum).*

(iii) ***Fundamental ∞-groupoid.*** *The maximal Kan complex contained in $\mathscr{W}_{\infty,1}$ is the fundamental ∞-groupoid $\Pi_\infty(\mathscr{P}_2(\mathbb{R}^d))$ of the Wasserstein space, whose homotopy groups recover the Wasserstein topology.*

Proof sketch. For (i): given an inner horn Λ^n_k, the missing face corresponds to a WGF segment whose endpoints are determined by the other faces. Semiconvexity of F_φ (Proposition 3.3) guarantees existence of a WGF connecting any two measures in $\mathscr{P}_2(\mathbb{R}^d)$, and uniqueness follows from the Lipschitz bound on the velocity field. Part (ii) follows from the global minimizer criterion. Part (iii) is a consequence of the path-space fibration structure on $\mathscr{P}_2(\mathbb{R}^d)$. □

14.2 SpMMD as an $(\infty,1)$-functor

Definition 14.3 ($(\infty,1)$-functor between quasi-categories)**.** An $(\infty,1)$-*functor* (or ∞-*functor*) between quasi-categories $\mathscr{C}$ and $\mathscr{D}$ is a morphism of simplicial sets $F : \mathscr{C} \to \mathscr{D}$.

Theorem 14.4 (SpMMD as an ∞-functor)**.** *The assignment $\varphi \mapsto \mathrm{SpMMD}_\varphi$-flow defines an $(\infty,1)$-functor*

$$\mathbf{Sp}^\infty \;:\; \mathrm{N}(\mathbf{QualFun}) \;\longrightarrow\; \mathrm{Fun}(\Delta^1, \mathscr{W}_{\infty,1}),$$

where $\mathrm{N}(\mathbf{QualFun})$ *is the nerve of* **QualFun** *(as a topological category) and* $\mathrm{Fun}(\Delta^1, \mathscr{W}_{\infty,1})$ *is the simplicial mapping space of 1-simplex-valued flows. This functor is* homotopy coherent*: the assignment of WGF trajectories to qualification functions is compatible with all higher homotopies between qualification functions.*

Proof sketch. The functor sends a 0-simplex (φ, ν) to the WGF trajectory $(\mu_t)_{t\geq 0}$ minimising $\mathrm{SpMMD}_\varphi(\cdot\|\pi)$, and a 1-simplex (qualification homotopy H) to the family of WGF trajectories $(\mu^s_t)_s$ parametrised by $s \in [0,1]$ via the continuously varying SpMMD flows $\mathrm{SpMMD}_{H(s,\cdot,\cdot)}$. Higher simplices are handled by induction on the simplicial dimension. Homotopy coherence follows from the continuous dependence of WGF solutions on the divergence functional (via the Lipschitz bound on witness functions in Lemma B.4 of Chen et al. [22], which is continuous in φ). □

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14.3 The spectral filtration as a Postnikov tower

The Showalter tower of theorem 13.7 induces a tower of approximations to the χ^2-flow. We show this is the analogue of a *Postnikov tower* in the $(\infty,1)$-category $\mathscr{W}_{\infty,1}$.

Definition 14.5 (SpMMD Postnikov tower)**.** The *SpMMD Postnikov tower at level* λ is the sequence

$$\mathscr{W}_{\infty,1}^{(\infty)} \to \cdots \to \mathscr{W}_{\infty,1}^{(m+1)} \to \mathscr{W}_{\infty,1}^{(m)} \to \cdots \to \mathscr{W}_{\infty,1}^{(1)},$$

where $\mathscr{W}_{\infty,1}^{(m)}$ is the full sub-quasi-category of $\mathscr{W}_{\infty,1}$ spanned by WGF trajectories of $\mathrm{SpMMD}_{\varphi_{S,m}}$-flows. The maps are the canonical inclusions induced by the dominations of Theorem 13.9.

Theorem 14.6 (Convergence as Postnikov tower limit)**.** *The* $(\infty,1)$*-categorical limit of the SpMMD Postnikov tower is*

$$\mathrm{holim}_m \mathscr{W}_{\infty,1}^{(m)} \simeq \mathscr{W}_{\infty,1}^{(\infty)},$$

where $\mathscr{W}_{\infty,1}^{(\infty)}$ *is the quasi-category of* χ^2*-gradient flows. Moreover, the convergence Theorem 4.1 is equivalent to the statement that the Postnikov tower is* convergent*: for each pair* $\mu_0,\pi \in \mathscr{P}_2(\mathbb{R}^d)$*,*

$$\mathrm{Map}_{\mathscr{W}_{\infty,1}^{(m)}}(\mu_0,\pi) \to \mathrm{Map}_{\mathscr{W}_{\infty,1}^{(\infty)}}(\mu_0,\pi)$$

is $(r\wedge m)$*-connected, meaning all homotopy groups up to degree* $r\wedge m$ *agree.*

Proof sketch. The holim is computed by the Milnor sequence for towers of quasi-categories [49]. The $(r\wedge m)$-connectivity follows from the approximation error bound $O(\lambda^{r\wedge m})$ in Theorem 4.1: the KL barrier controls π_0 of the mapping space, and the $O(\lambda^{r\wedge m})$ decay rate controls the higher homotopy groups by the Poincaré inequality (which gives a spectral gap, hence exponential decay of all homotopy classes). □

14.4 WGF as a right adjoint

Theorem 14.7 (WGF adjunction)**.** *In* $(\infty,1)$-**Cat***, the Wasserstein gradient flow* ∞*-functor*

$$\mathrm{WGF} : \mathbf{Div} \longrightarrow \mathrm{Fun}([0,\infty),\mathscr{W}_{\infty,1})$$

is a right adjoint to the evaluation functor $\mathrm{ev}_0 : \mathrm{Fun}([0,\infty),\mathscr{W}_{\infty,1}) \to \mathscr{P}_2(\mathbb{R}^d)$ *composed with the divergence-to-measure functor* $\mathscr{P}_2(\mathbb{R}^d) \to \mathbf{Div}$ *sending* $\mu_0 \mapsto D(\cdot\|\pi)$*. The counit of this adjunction is the statement that the WGF achieves the minimum of* $D(\cdot\|\pi)$*. The unit is the "gradient ascent" starting at* π *moving away toward* μ_0*, which is trivially the identity.*

Proof sketch. We construct the adjunction unit and counit in $(\infty,1)$-**Cat** and verify the triangle identities.

Counit $\varepsilon : \mathrm{ev}_0 \circ \mathrm{WGF} \Rightarrow \mathrm{id}_{\mathbf{Div}}$. For each $D \in \mathbf{Div}$, the counit component is the evaluation morphism $\varepsilon_D : \mu_0 \mapsto D(\mu_0\|\pi)$. This is precisely the statement that the initial condition of the WGF recovers D: $\mathrm{ev}_0(\mathrm{WGF}(D)) = \mu_0$ and $D(\mathrm{ev}_0(\mathrm{WGF}(D))\|\pi) = D(\mu_0\|\pi)$. In the $(\infty,1)$-categorical sense, this is a homotopy $D \simeq D \circ \mathrm{ev}_0 \circ \mathrm{WGF}$, which is the identity on objects (measures) since the WGF starts at μ_0.

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Unit $\eta : \mathrm{id}_{\mathrm{Fun}([0,\infty),\mathscr{W}_{\infty,1})} \Rightarrow \mathrm{WGF} \circ (\mathrm{ev}_0 \circ D)$. For each trajectory $(\mu_t)_{t\geq 0} \in \mathrm{Fun}([0,\infty),\mathscr{W}_{\infty,1})$, the unit component evaluates at $t = 0$ to get μ_0, forms the divergence $D_{\mu_0}(\cdot\|\pi)$ uniquely determined by μ_0, and takes its WGF. The WGF of D_{μ_0} starting at μ_0 is the trajectory itself up to reparametrization by Theorem 4.1 (near-global convergence to π): any WGF trajectory from μ_0 is homotopic to the WGF of $D(\cdot\|\pi)$ via the Wasserstein geodesic interpolation, giving a 1-simplex in $\mathscr{W}_{\infty,1}$.

Triangle identities. The two compositions $\varepsilon_{\mathrm{WGF}(D)} \circ \mathrm{WGF}(\eta_D)$ and $\mathrm{ev}_0(\varepsilon_\gamma) \circ \eta_{\mathrm{ev}_0(\gamma)}$ reduce, respectively, to the identity on $\mathrm{WGF}(D)$ (since evaluating the WGF at $t = 0$ and forming the WGF again is homotopically the identity by Theorem 14.2(ii)) and to the identity on $\mathrm{ev}_0(\gamma)$ (since ev_0 of the WGF starting at μ_0 is μ_0). These are the required $(\infty,1)$-categorical triangle identities, holding up to contractible homotopy (i.e., as equivalences in $\mathscr{W}_{\infty,1}$). □

14.5 Multi-geometry quasi-categories

The WGF quasi-category $\mathscr{W}_{\infty,1}$ of theorem 14.1 encodes only Wasserstein (transport) trajectories. The three new flow geometries of Sections 8.2–10 require distinct quasi-categories reflecting their different proximal structures.

Definition 14.8 (The Fisher–Rao quasi-category)**.** The *Fisher–Rao quasi-category* $\mathscr{FR}_{\infty,1}$ has the same 0-*simplices* as $\mathscr{W}_{\infty,1}$ (probability measures $\mu \in \mathscr{P}_2(\mathbb{R}^d)$) but different higher simplices:

- 1**-simplices:** FR flow trajectories $(\mu_t)_{t\in[0,1]}$ in $\mathscr{P}_2(\mathbb{R}^d)$ where $\mu_t = w_t \cdot \mu_0$ (multiplicative reweighting of the initial measure; no spatial transport of particle positions).
- 2**-simplices:** smooth families of FR trajectories $(w_t^s)_{s,t\in[0,1]}$ such that for each s, $t \mapsto w_t^s \cdot \mu_0$ is an FR flow.
- n**-simplices:** n-parameter families by induction.

$\mathscr{FR}_{\infty,1}$ is a quasi-category by the same inner-horn-filling argument as Theorem 14.2, using the existence and uniqueness of FR flow solutions (Theorem 8.4).

Definition 14.9 (The kinetic quasi-category)**.** The *kinetic quasi-category* $\mathscr{K}_{\infty,1}$ has:

- 0**-simplices:** joint position-momentum distributions $\rho \in \mathscr{P}_2(\mathbb{R}^d \times \mathbb{R}^d)$.
- 1**-simplices:** kinetic SpMMD flow trajectories $(\rho_t)_{t\in[0,1]}$ in $\mathscr{P}_2(\mathbb{R}^d \times \mathbb{R}^d)$ satisfying eq. (30).
- 2**-simplices and higher:** parameter families by induction.

The *position projection* $\pi_q : \mathscr{P}_2(\mathbb{R}^d \times \mathbb{R}^d) \to \mathscr{P}_2(\mathbb{R}^d)$ sending $\rho \mapsto \mu^{\mathrm{pos}}$ extends to an ∞-functor

$$\pi_q \;:\; \mathscr{K}_{\infty,1} \;\longrightarrow\; \mathscr{W}_{\infty,1},$$

which is a Kan fibration: every lifting problem for inner horns in $\mathscr{W}_{\infty,1}$ has a solution in $\mathscr{K}_{\infty,1}$ up to momentum initialization. The fiber over $\mu_0 \in \mathscr{P}_2(\mathbb{R}^d)$ is the quasi-category of momentum distributions $\mathscr{P}_2(\mathbb{R}^d)$, which is contractible (the equilibrium Gaussian $\mathscr{N}(0, I_d)$ is the unique fixed point of the momentum dynamics).

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Remark 14.10 (Quasi-categorical relationships among geometries). The geometric refinement morphisms of **FlowGeom** lift to ∞-functors between quasi-categories. The WFR interpolation $\delta \in (0,\infty]$ gives a continuous family of ∞-functors $\mathscr{W}\mathscr{F}\mathscr{R}_\delta : \mathscr{P}_2(\mathbb{R}^d) \to \mathbf{Fun}(\Delta^1, \mathscr{P}_{\infty,1})$ interpolating between $\mathscr{W}_{\infty,1}$ ($\delta = \infty$) and $\mathscr{F}\mathscr{R}_{\infty,1}$ ($\delta = 0$), each assignment being a quasi-category fibration over $\mathscr{P}_2(\mathbb{R}^d)$. The Sinkhorn quasi-category $\mathscr{S}\rangle\backslash\|_{\varepsilon,\infty,1}$ has 1-simplices that are *varepsilon-Sinkhorn flow trajectories*; as $\varepsilon \to 0$ these converge to Wasserstein 1-simplices, giving a functor $\mathscr{S}\rangle\backslash\|_{\varepsilon,\infty,1} \to \mathscr{W}_{\infty,1}$ that is a Dwyer–Kan equivalence for ε sufficiently small.

Definition 14.11 (The geometry-extended WGF quasi-category)**.** The *geometry-extended WGF quasi-category* $\mathscr{W}^{\mathrm{ext}}_{\infty,1}$ is the homotopy pushout (in the Joyal model structure on **sSet**):

$$\mathscr{W}^{\mathrm{ext}}_{\infty,1} = \mathscr{W}_{\infty,1} \underset{\mathscr{P}_2(\mathbb{R}^d)}{\sqcup} \mathscr{F}\mathscr{R}_{\infty,1} \underset{\mathscr{P}_2(\mathbb{R}^d)}{\sqcup} \pi_q(\mathscr{K}_{\infty,1}) \underset{\mathscr{P}_2(\mathbb{R}^d)}{\sqcup} \mathscr{S}\rangle\backslash\|_{0,\infty,1},$$

where the amalgamation is along the common 0-skeleton $\mathscr{P}_2(\mathbb{R}^d)$. The objects of $\mathscr{W}^{\mathrm{ext}}_{\infty,1}$ are probability measures; the morphisms are flow trajectories of *any* geometry, with geometry type recorded as a label.

Theorem 14.12 (Extended SpMMD ∞-functor)**.** *The geometry-flow 2-functor Φ of Theorem 13.19 lifts to an $(\infty,1)$-functor*

$$\mathbf{Sp}^{\mathrm{ext},\infty} : \mathrm{N}(\mathbf{FlowGeom}) \times \mathrm{N}(\mathbf{QualFun}) \longrightarrow \mathrm{Fun}(\Delta^1, \mathscr{W}^{\mathrm{ext}}_{\infty,1}),$$

extending Theorem 14.4 to all six flow geometries simultaneously. The restriction of $\mathbf{Sp}^{\mathrm{ext},\infty}$ to $\{\mathbf{W}_2\} \times \mathbf{QualFun}$ recovers $\mathbf{Sp}^{\infty}$ of Theorem 14.4. The restriction to $\{\mathbf{FR}\} \times \mathbf{QualFun}$ gives the FR-SpMMD ∞-functor, and the restriction to $\{\mathbf{K}_\gamma\} \times \mathbf{QualFun}$ factors through the Kan fibration $\pi_q : \mathscr{K}_{\infty,1} \to \mathscr{W}_{\infty,1}$.

Proof sketch. The proof follows Theorem 14.4 applied at each geometry. The extension to $\mathrm{N}(\mathbf{FlowGeom}) \times \mathrm{N}(\mathbf{QualFun})$ requires showing that the geometry-refinement morphisms in **FlowGeom** induce simplicial maps between the corresponding mapping spaces. This holds because every geometric refinement $g_1 \to g_2$ is realized by a continuous deformation of the proximal metric (Proposition 13.17 for WFR; the overdamped limit for kinetic; $\varepsilon \to 0$ for Sinkhorn), which defines the required simplicial homotopy. □

15 (∞,n)-Categories and Higher Regularization

15.1 Motivation: nested regularization and the n-cell structure

The SpMMD construction applies a single spectral filter φ to T_π to approximate the χ^2-divergence. A natural generalization *iterates* this regularization: the first application smooths the density ratio; the second smooths the result; and so on. This iteration produces a tower of divergences parametrized by tuples of qualification functions, naturally indexed by an (∞,n)-categorical structure.

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Definition 15.1 (Iterated SpMMD)**.** For qualification functions $\boldsymbol{\varphi} = (\varphi_1, \ldots, \varphi_n) \in \mathbf{QualFun}^n$, define the *n-th order spectrally regularized MMD* by the recursion:

$$\begin{aligned}
&\mathrm{SpMMD}^{(0)}(\mu\|\pi) = \chi^2(\mu\|\pi),\\
&\mathrm{SpMMD}^{(k)}_{\boldsymbol{\varphi}}(\mu\|\pi) = \mathrm{SpMMD}_{\varphi_k}\Big(\mu \,\Big\|\, \arg\min_{\nu\in\mathscr{P}_2(\mathbb{R}^d)} \mathrm{SpMMD}^{(k-1)}_{(\varphi_1,\ldots,\varphi_{k-1})}(\nu\|\pi)\Big) \qquad (k = 1,\ldots,n).
\end{aligned}$$

The $n = 1$ case recovers the SpMMD family of theorem 3.2. The *barrier* of $\mathrm{SpMMD}^{(n)}$ is $O(\lambda^{r\wedge\nu_1\wedge\cdots\wedge\nu_n})$, converging to $O(\lambda^r)$ as $n \to \infty$ provided all $\nu_k > r$.

Remark 15.2 (Computational note). The iterated SpMMD $\mathrm{SpMMD}^{(n)}$ requires n applications of the Gram matrix machinery of **??**, but these can be precomputed as a cascade of regularized inversions and need not be recomputed at every step if the regularization schedule is fixed. For $n = 2$ and Tikhonov filters, $\mathrm{SpMMD}^{(2)}$ corresponds to the kernel $\widetilde{k}_{\varphi_1} \circ \widetilde{k}_{\varphi_2}$ (composition of regularized kernels), which admits a closed Gram form.

15.2 The globular set of higher flows

Definition 15.3 (Globular set of SpMMD flows)**.** The *globular set of n-regularized flows* $\mathbf{Glob}_{\leq n}$ has:

- **0-cells:** probability measures $\mu \in \mathscr{P}_2(\mathbb{R}^d)$.
- **1-cells** $\mu_0 \to \mu_1$**:** $\mathrm{SpMMD}^{(1)}$-flow trajectories from μ_0 to μ_1 (with qualification function φ_1).
- **2-cells** $f \Rightarrow g$ (for $f, g : \mu_0 \to \mu_1$): $\mathrm{SpMMD}^{(2)}$ homotopies between two 1-cell flows, parametrized by φ_2.
- k**-cells** ($k \leq n$): $\mathrm{SpMMD}^{(k)}$-homotopies between $(k-1)$-cells, parametrized by φ_k.

The source and target maps $s_k, t_k : \mathbf{Glob}_k \to \mathbf{Glob}_{k-1}$ are given by restriction to the boundary of the parameter domain.

Definition 15.4 (Geometry-extended globular set)**.** The *geometry-extended globular set* $\mathbf{Glob}^{\mathrm{ext}}_{\leq n}$ enhances $\mathbf{Glob}_{\leq n}$ by labeling each k-cell with a flow geometry $g_k \in \mathbf{FlowGeom}$ in addition to a qualification function $\varphi_k \in \mathbf{QualFun}$:

- **0-cells:** probability measures $\mu \in \mathscr{P}_2(\mathbb{R}^d)$ (unchanged).
- **1-cells** $\mu_0 \xrightarrow{(g_1,\varphi_1)} \mu_1$**:** flow trajectories from μ_0 to μ_1 in geometry g_1 with filter φ_1; the geometry $g_1 \in \mathbf{FlowGeom}$ specifies whether this is a Wasserstein, FR, kinetic, or Bregman trajectory.
- **2-cells** $f \xRightarrow{(g_2,\varphi_2)} g$**:** homotopies between 1-cells f and g realized as $\mathrm{SpMMD}^{(2)}$-flows in geometry g_2. Geometry *refinement* is permitted: a 2-cell can connect two 1-cells of different geometries provided $g_f \to g_2 \leftarrow g_g$ in **FlowGeom**.

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- **k-cells** ($k \leq n$): $\mathrm{SpMMD}^{(k)}$-homotopies with geometry label g_k and filter φ_k, compatible at boundaries.

The globular identities hold because the witness function $h^*_{\mu,\pi}$ is the same across all geometries: changing geometry at the boundary of a cell is possible whenever the boundary geometries are related by a refinement morphism in **FlowGeom**.

Theorem 15.5 (Geometry-extended strict n-category structure)**.** *For Tikhonov qualification functions $\varphi_1 = \cdots = \varphi_n = \varphi_T$ and* uniform geometry $g_1 = \cdots = g_n = g \in$ **FlowGeom**, $\mathbf{Glob}^{\mathrm{ext}}_{\leq n}$ *carries a strict n-category structure at each fixed geometry g:*

(i) For $g = \mathbf{W}_2$: strict n-category (Theorem **??***).*

(ii) For $g = \mathbf{FR}$ or $g = \mathbf{B}_{t\log t}$: strict n-category, because Gibbs weight updates compose strictly (multiplication of positive functions).

(iii) For $g = \mathbf{K}_\gamma$: strict *n-category for the positional components and* weak *for the momentum components (momentum concatenation introduces a homotopy from the boundary conditions).*

(iv) For $g = \mathbf{Sink}_\varepsilon$: weak n-category, because Sinkhorn plans do not compose strictly (the ε-entropy shifts under concatenation), but are strict in the limit $\varepsilon \to 0$.

For mixed geometries (different g_k at different levels), $\mathbf{Glob}^{\mathrm{ext}}_{\leq n}$ *is a* weak n-category *whose interchange laws hold up to homotopies controlled by the geometric refinement morphisms in* **FlowGeom**.

Proof sketch. We verify the strict interchange law for each geometry case.

Case (ii): $g = \mathbf{FR}$ or $g = \mathbf{B}_{t\log t}$. A k-cell in $\mathbf{Glob}^{\mathrm{ext}}_{\leq n}$ at the FR geometry is a family of weight updates $w_{s_1} \cdots w_{s_k}$ acting multiplicatively on μ_0. Horizontal composition (concatenation of 2-cells) corresponds to multiplying the weight sequences: $(w_{s_1} \cdots w_{s_j}) \cdot (w_{s_{j+1}} \cdots w_{s_k}) = w_{s_1} \cdots w_{s_k}$, which is strictly associative because multiplication of positive functions is associative on the nose. Vertical composition (stacking weight homotopies) is also strictly associative for the same reason.

Case (iii): $g = \mathbf{K}_\gamma$. The positional component q_t satisfies $dq/dt = p_t$, a strictly defined ODE; hence concatenation of positional trajectories is strictly associative. The momentum component p_t at the junction of two concatenated trajectories satisfies a boundary condition (velocity must match), which introduces a homotopy (one can choose either to set $p_{t_1^-} = p_{t_1^+}$ discontinuously or smoothly interpolate); this boundary homotopy is exactly the source of the weak interchange law for momentum cells.

Case (iv): $g = \mathbf{Sink}_\varepsilon$. The Sinkhorn plan $T_\varepsilon(\mu, \nu)$ satisfies $T_\varepsilon(\cdot, T_\varepsilon(\mu_1, \mu_2))_\# \mu_1 \neq T_\varepsilon(\mu_1, \mu_3)_\# \mu_1$ in general (the double application of entropy regularization produces a plan with 2ε regularization rather than ε). This non-strictness of plan composition is controlled by the Gibbs kernel $e^{-C/\varepsilon}$: as $\varepsilon \to 0$ the plans concentrate on the optimal transport plan, which does compose strictly ($T_0 \circ T_0 = T_0$), recovering a strict n-category in the limit.

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Mixed geometries. When adjacent cells have geometries $g_k \neq g_{k+1}$, the interchange law holds up to the homotopy provided by the refinement morphism $g_k \to g \leftarrow g_{k+1}$ in **FlowGeom**: the continuous deformation of proximal metrics between adjacent cells (e.g., the WFR interpolation $\delta \to \infty$ between FR and Wasserstein) provides the required coherence homotopy, yielding a weak n-category. □

Theorem 15.6. *For Tikhonov qualification functions $\varphi_1 = \cdots = \varphi_n = \varphi_T$,* **Glob**$_{\leq n}$ *carries the structure of a* strict n-category*: all interchange laws hold on the nose (not merely up to homotopy) because the Tikhonov operator $(\Sigma_\pi + \lambda I)^{-1}$ is functorial with respect to composition of flows. For general qualification functions,* **Glob**$_{\leq n}$ *is a* weak n-category *(all laws hold up to coherent homotopy), modeled by a complete Segal Θ_n-space in the sense of Rezk [60].*

Proof sketch. Strict interchange for the Tikhonov case: horizontal composition of 2-cells corresponds to concatenation of flow segments, which is strictly associative since the witness function h^* is uniquely determined by the Gram matrix inverse. Vertical composition (stacking homotopies) is also strictly associative. For general φ, the witness function depends on φ continuously but not strictly multiplicatively, yielding weak interchange laws controlled by the Lipschitz constant of φ. □

15.3 Rezk's complete Segal Θ_n-spaces

Following Rezk [60], we identify the homotopy-coherent structure of the iterated SpMMD flows with a complete Segal Θ_n-space. The category Θ_n (Berger's globular sums) has objects $[k_1|\cdots|k_n]$ representing n-dimensional pasting diagrams, with morphisms given by functors of pasting diagrams.

Definition 15.7 (Θ_n-space of SpMMD flows)**.** The *Θ_n-space of SpMMD flows* is the presheaf $\mathscr{X} : \Theta_n^{\mathrm{op}} \to$ **sSet** defined by

$$\mathscr{X}([k_1|\cdots|k_n]) \;=\; \Big\{(k_1+\cdots+k_n)\text{-fold composable SpMMD}^{(n)}\text{-flow diagrams}\Big\},$$

where the set on the right is given the Hausdorff topology from the WGF parameter space.

Theorem 15.8 ($\mathscr{X}$ is a complete Segal Θ_n-space)**.** *The presheaf $\mathscr{X}$ satisfies:*

(i) ***Segal condition:*** *The Segal maps $\mathscr{X}([k_1|\cdots|k_n]) \to \mathscr{X}([k_1|\cdots|k_{j-1}|1|\cdots]) \times_{\mathscr{X}([k_1|\cdots|1])} \mathscr{X}([k_1|\cdots|k_j|\cdots])$ are homotopy equivalences for all j and all k_i. This encodes the associativity of SpMMD flow composition up to homotopy.*

(ii) ***Completeness:*** *The space of equivalences in $\mathscr{X}([1|\cdots|1])$ is contractible. This holds because WGF equivalences (flows reaching π) form a contractible space: any two such flows are homotopic via a straight-line interpolation in $\mathscr{P}_2(\mathbb{R}^d)$.*

Hence $\mathscr{X}$ presents an (∞,n)-category in the sense of Barwick [8].

Proof sketch. Segal condition. The Segal map $\mathscr{X}([k_1|\cdots|k_n]) \to \mathscr{X}([k_1|\cdots|k_{j-1}|1|\cdots]) \times_{\mathscr{X}([k_1|\cdots|1])} \mathscr{X}([k_1|\cdots|k_j|\cdots])$ asserts that a composable k_j-fold

chain of $\mathrm{SpMMD}^{(n)}$-flow diagrams at level j is the same as k_j composable single diagrams glued along boundaries. This follows from the *unique continuation property* of WGF solutions: given flow segments $(\mu_t^{(1)})_{t\in[0,t_1]}$ and $(\mu_t^{(2)})_{t\in[0,t_2]}$ with $\mu_{t_1}^{(1)} = \mu_0^{(2)}$, there is a unique WGF concatenation $(\mu_t)_{t\in[0,t_1+t_2]}$ extending both. Uniqueness (from semiconvexity of F_φ, Proposition 3.3) makes the gluing a homotopy equivalence: any two gluings are connected by a contractible space of reparametrizations.

Completeness. The space of equivalences in $\mathscr{X}([1|\cdots|1])$ consists of WGF trajectories $\mu_0 \xrightarrow{\sim} \pi$ that are invertible morphisms, i.e., trajectories reaching π (Theorem 14.2(ii)). Any two such trajectories γ_1 and γ_2 starting at μ_0 are homotopic via the affine interpolation of velocity fields $v_t^s = (1-s)v_t^{\gamma_1} + sv_t^{\gamma_2}$ in $L^2(\mu_t)$, which defines a 2-simplex in $\mathscr{W}_{\infty,1}$ connecting γ_1 and γ_2. Since all such interpolations are contractible (the space of velocity fields on a convex set is convex), the equivalence space is contractible. □

15.4 Geometry-extended Θ_n-spaces

The geometry-extended globular set $\mathbf{Glob}^{\mathrm{ext}}_{\leq n}$ of Definition 15.4 gives rise to an extended Θ_n-space that encodes iterated flows across multiple geometries.

Definition 15.9 (Geometry-extended Θ_n-space)**.** The *geometry-extended Θ_n-space* is the presheaf $\mathscr{X}^{\mathrm{ext}} : \Theta_n^{\mathrm{op}} \to \mathbf{sSet}$ defined by

$$\mathscr{X}^{\mathrm{ext}}([k_1|\cdots|k_n]) = \left\{(k_1+\cdots+k_n)\text{-fold composable } \mathrm{SpMMD}^{(n)}\text{-flow diagrams with geometry labels}\right\},$$

where each level $j \in [n]$ carries an independent geometry label $g_j \in \mathbf{FlowGeom}$, and diagrams are composable provided adjacent boundary geometries are related by refinement morphisms in **FlowGeom**.

Theorem 15.10 ($\mathscr{X}^{\mathrm{ext}}$ is a complete Segal Θ_n-space)**.** *The presheaf $\mathscr{X}^{\mathrm{ext}}$ satisfies the Segal and completeness conditions of Theorem 15.8:*

(i) ***Segal condition:*** *holds because flow composition in each geometry g_j is associative up to homotopy (controlled by the geometry refinement morphisms), and the geometry label at a pasting diagram's boundary is determined by the adjacent cells.*

(ii) ***Completeness:*** *the equivalences in $\mathscr{X}^{\mathrm{ext}}$ are flows reaching π in* any *geometry; the contractibility follows from the fact that for every $g \in$ **FlowGeom** there exists a flow connecting μ_0 to π (using whichever geometry is most efficient—in particular, FR flows achieve this when μ_0 already covers π).*

There is a natural inclusion $\mathscr{X} \hookrightarrow \mathscr{X}^{\mathrm{ext}}$ (the Wasserstein-only sub-Θ_n-space) and the quotient $\mathscr{X}^{\mathrm{ext}} \to \mathscr{X}^{\mathrm{ext}}/\mathscr{X}$ encodes the "new geometry content" contributed by the FR, kinetic, and Bregman flows.

Proof sketch. Segal condition. The Segal maps for $\mathscr{X}^{\mathrm{ext}}$ are the same as for $\mathscr{X}$ but now at each cell level j the geometry label g_j is also recorded. The associativity of composition at each fixed geometry g_j follows from Theorem 15.8. For adjacent cells with geometries

$g_j \neq g_{j+1}$, the boundary compatibility condition requires a refinement morphism $g_j \to g \leftarrow g_{j+1}$ in **FlowGeom**; such a morphism always exists (taking $g = \mathbf{W}_2$ or $g = \mathbf{WFR}_\delta$ for appropriate δ) and the gluing is again a homotopy equivalence because the refinement deformation is continuous in δ (Proposition 13.17).

Completeness. For any geometry $g \in \mathbf{FlowGeom}$, a g-flow from μ_0 to π exists: if $g = \mathbf{W}_2$, use Theorem 4.1; if $g = \mathbf{FR}$, use Theorem 8.4 (requires only $\mu_0 \ll \pi$); if $g = \mathbf{K}_\gamma$, use Theorem 9.2; if $g = \mathbf{Sink}_\varepsilon$, use Proposition 10.4. Given any two equivalences $\gamma_1, \gamma_2 : \mu_0 \to \pi$ possibly in different geometries g_1, g_2, the WFR interpolation $\gamma_s \in \mathscr{X}^{\mathbf{WFR}_{\delta(s)}}$ with $\delta(s) \to \infty$ as $s \to 0$ (approaching $\mathbf{W}_2$) and $\delta(s) \to 0$ as $s \to 1$ (approaching **FR**) provides a contractible homotopy between γ_1 and γ_2 in $\mathscr{X}^{\mathrm{ext}}$, establishing the contractibility of the equivalence space.

Inclusion and quotient. The natural inclusion $\mathscr{X} \hookrightarrow \mathscr{X}^{\mathrm{ext}}$ picks out the sub-presheaf of diagrams with all geometry labels equal to $\mathbf{W}_2$. This is fully faithful (the mapping spaces in $\mathscr{X}$ embed into those of $\mathscr{X}^{\mathrm{ext}}$ as a retract via the geometric refinement $g \to \mathbf{W}_2$ in the overdamped/large-δ limit). □

Proposition 15.11 (The geometry filtration on $\mathscr{X}^{\mathrm{ext}}$)**.** *The poset structure on* **FlowGeom** *induces a* filtration

$$\mathscr{X}^{\mathbf{W}_2} \hookrightarrow \mathscr{X}^{\mathbf{WFR}} \hookrightarrow \mathscr{X}^{\mathbf{K}} \hookrightarrow \mathscr{X}^{\mathrm{ext}},$$

where $\mathscr{X}^g$ denotes the sub-Θ_n-space using only flows of geometry $\preceq g$. Each inclusion is a Segal map equivalence on the level of mapping spaces: adding a more expressive geometry adds contractible fibers (new flow trajectories that are homotopic to existing ones via the geometric refinement morphism). The filtration is strict *at each level: $\mathscr{X}^{\mathbf{FR}}$ contains 1-simplices (reweighting flows) that are* not *weakly equivalent to any Wasserstein 1-simplex when μ_0 and π have the same support but different weights, because reweighting requires zero geometric displacement.*

Proof sketch. Contractible fibers at each inclusion step. For the inclusion $\mathscr{X}^{\mathbf{W}_2} \hookrightarrow \mathscr{X}^{\mathbf{WFR}}$, the added 1-simplices are WFR_δ trajectories for $\delta < \infty$. The continuous family of velocities $v_t^s = \nabla h^* + (s/\delta^2)\mu_t(h^* - \langle h^* \rangle)$ for $s \in [0,1]$ provides a homotopy from any WFR_δ trajectory to the Wasserstein trajectory ($s = 0$). This homotopy space is contractible (it is a convex set of velocity fields), confirming that each inclusion adds only contractible fibers and is therefore a Segal map equivalence.

Strictness of the FR level. When $\mu_0 \ll \pi$ and $\mathrm{supp}(\mu_0) = \mathrm{supp}(\pi)$, the FR flow $\gamma^{\mathbf{FR}}$ has zero W_2-displacement: $W_2(\mu_t^{\mathbf{FR}}, \mu_0) = 0$ for all t since positions are fixed. Any Wasserstein 1-simplex $\gamma^{\mathbf{W}_2}$ transporting μ_0 to π has $W_2(\mu_t^{\mathbf{W}_2}, \mu_0) > 0$ for $t > 0$ (since $\mu_0 \neq \pi$). A weak equivalence in $\mathscr{W}_{\infty,1}$ between $\gamma^{\mathbf{FR}}$ and $\gamma^{\mathbf{W}_2}$ would require a 2-simplex with both as faces, i.e., a homotopy through WGF trajectories; but any such homotopy would continuously connect a zero-displacement path to a non-zero-displacement path in $\mathscr{P}_2(\mathbb{R}^d)$, which is impossible without leaving $\mathscr{W}_{\infty,1}$ (since the Wasserstein distance varies continuously along any W_2-continuous path, by the lower semicontinuity of W_2). Hence $\mathscr{X}^{\mathbf{FR}} \not\simeq \mathscr{X}^{\mathbf{W}_2}$ as sub-Θ_n-spaces, establishing strictness. □

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15.5 The cobordism hypothesis and SpMMD TQFTs

The cobordism hypothesis of Baez–Dolan [5], as formalized by Lurie [50], classifies fully extended topological quantum field theories (TQFTs) in terms of their value on a point. We show that SpMMD flows admit a natural TQFT-like structure.

Definition 15.12 (SpMMD n-dimensional TQFT)**.** An *SpMMD n-TQFT* is a symmetric monoidal (∞,n)-functor

$$Z_\varphi \,:\, \mathbf{Cob}_n^{\mathrm{fr}} \longrightarrow \mathscr{X},$$

where $\mathbf{Cob}_n^{\mathrm{fr}}$ is the (∞,n)-category of framed n-dimensional cobordisms and $\mathscr{X}$ is the Θ_n-space of SpMMD flows. On objects, Z_φ assigns to a $(d-1)$-dimensional manifold M a probability measure $\mu_M \in \mathscr{P}(M)$; on 1-morphisms (cobordisms), it assigns SpMMD flow trajectories; on k-morphisms, it assigns k-parameter homotopies.

Theorem 15.13 (Cobordism classification of SpMMD flows)**.** *By the cobordism hypothesis [50], SpMMD n-TQFTs are classified by fully dualizable objects in $\mathscr{X}$. Specifically:*

(i) A probability measure $\mu \in \mathscr{P}_2(\mathbb{R}^d)$ is fully dualizable in $\mathscr{X}$ if and only if the SpMMD$^{(n)}$ gradient flow starting at μ converges to the global minimizer π in finite effective time $T^*(\mu) = C_P \cdot \log(\mathrm{KL}(\mu\|\pi)/\varepsilon)/(2(1+\lambda))$.

(ii) The Serre functor *on $\mathscr{X}$ (the dualizing object) corresponds to the time-reversal of the WGF, i.e., the gradient* ascent *flow of F_φ, which defines an antipodal involution on $\mathscr{P}_2(\mathbb{R}^d)$.*

(iii) The quantum dimension *of $\mu \in \mathscr{P}_2(\mathbb{R}^d)$ in $\mathscr{X}$ is the partition function $Z_\varphi(S^1) = \mathrm{SpMMD}_\varphi(\mu\|\pi)$—the value of the SpMMD divergence, which is also the "action" of the circular WGF (the flow of μ along the target and back).*

Proof sketch. The full dualizability of μ follows from the exponential convergence (Theorem 4.1): the existence of evaluation and coevaluation morphisms in $\mathscr{X}$ corresponds to the existence of a finite-time WGF connecting μ to π and back. The classification by the cobordism hypothesis then identifies the invariant attached to the generating fully dualizable object (the point $\pi \in \mathscr{P}_2(\mathbb{R}^d)$) with the SpMMD divergence, which is the "invariant of the point" for this TQFT. □

Theorem 15.14 (Geometry-extended TQFT classification)**.** *An* extended SpMMD n-TQFT *is a symmetric monoidal (∞,n)-functor $Z_\varphi^{\mathrm{ext}} : \mathbf{Cob}_n^{\mathrm{fr}} \to \mathscr{X}^{\mathrm{ext}}$. By the cobordism hypothesis applied to $\mathscr{X}^{\mathrm{ext}}$, such TQFTs are classified by fully dualizable objects in $\mathscr{X}^{\mathrm{ext}}$. Three new classes of fully dualizable objects arise from the new geometries:*

(i) ***FR-dualizable:*** *A measure $\mu \in \mathscr{P}_2(\mathbb{R}^d)$ is* FR-dualizable *in $\mathscr{X}^{\mathrm{ext}}$ if the FR-SpMMD flow starting at μ converges; this holds whenever $\mu \ll \pi$ (absolute continuity suffices, without any transport requirement). The effective time is $T^*_{\mathrm{FR}}(\mu) = \log(\mathrm{KL}(\mu\|\pi)/\varepsilon)/(2\alpha(1+\lambda))$ (logarithmic in $1/\varepsilon$ with LSI rate α).*

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(ii) ***Kinetically dualizable:*** *A measure* $\mu \in \mathscr{P}_2(\mathbb{R}^d)$ *is* kinetically dualizable *if the kinetic SpMMD flow reaches* π *in finite time; the effective time is* $T^*_{\mathrm{kin}}(\mu) = \log(\mathrm{KL}(\mu\|\pi)/\varepsilon)/(2\sqrt{(1+\lambda)/C_P})$*, strictly shorter than* $T^*(\mu)$ *by a factor of* $\sqrt{C_P}$ *when* $C_P > 1$.

(iii) ***Sinkhorn-dualizable:*** *A* discrete *empirical measure* $\hat{\mu}_N = \frac{1}{N}\sum_i \delta_{x_i}$ *with* $N < \infty$ *is not fully dualizable in* $\mathscr{X}$ *(where* W_2 *degenerates for discrete measures) but is Sinkhorn-dualizable in* $\mathscr{X}^{\mathrm{ext}}$ *for any* $\varepsilon > 0$.

The extended quantum dimension of μ *in* $\mathscr{X}^{\mathrm{ext}}$ *is the* minimum *of the SpMMD divergence over all flow geometries:*

$$\dim_q^{\mathrm{ext}}(\mu) \;=\; \min_{g \in \mathbf{FlowGeom}} \mathrm{SpMMD}^g_\varphi(\mu\|\pi),$$

where SpMMD^g_φ *denotes the SpMMD divergence computed under geometry* g.

Proof sketch. By Theorem 15.13, SpMMD n-TQFTs valued in the Wasserstein-only $\mathscr{X}$ are classified by fully dualizable objects; we extend this to $\mathscr{X}^{\mathrm{ext}}$ by checking full dualizability geometry by geometry.

(i) FR-dualizable objects. Full dualizability in $\mathscr{X}^{\mathrm{ext}}$ requires evaluation and coevaluation morphisms: a flow from μ to π and a flow from π to μ. The FR flow of Theorem 8.4 provides the evaluation (forward flow) whenever $\mu \ll \pi$, in finite time T^*_{FR}. The coevaluation is the time-reversed FR flow (gradient ascent of F_φ in the FR metric), which flows from π back to μ by reversing the weight update $\partial_t \mu_t = -\mu_t(h^* - \langle h^* \rangle)$ to $\partial_t \mu_t = +\mu_t(h^* - \langle h^* \rangle)$. This is well-defined as a FR-type flow (the velocity field is the same h^*, just with reversed sign). The Serre functor is therefore time-reversal in the FR geometry, just as for Wasserstein (Theorem 15.13(ii)), but within $\mathscr{X}^{\mathbf{FR}}$.

(ii) Kinetically dualizable objects. The kinetic SpMMD flow of Theorem 9.2 provides the evaluation in time $T^*_{\mathrm{kin}} = T^*/\sqrt{C_P(1+\lambda)}$ (where $T^* = C_P \log(\mathrm{KL}/\varepsilon)/(2(1+\lambda))$ is the Wasserstein effective time). The coevaluation is the time-reversed kinetic flow, which exists because the kinetic flow equation (30) is invariant under time-reversal combined with momentum negation $p \mapsto -p$ (standard for underdamped Langevin).

(iii) Sinkhorn-dualizable discrete measures. For a discrete empirical measure $\hat{\mu}_N$, the Wasserstein $W_2(\hat{\mu}_N, \cdot)$ is degenerate as an optimization target (the singular support causes the Wasserstein gradient to diverge in directions off the support). However, the Sinkhorn divergence $S_\varepsilon(\hat{\mu}_N, \cdot)$ is smooth for any $\varepsilon > 0$, providing well-defined evaluation and coevaluation morphisms in $\mathscr{X}^{\mathrm{ext}}$ for any $\varepsilon > 0$.

Extended quantum dimension. The quantum dimension $\dim_q^{\mathrm{ext}}(\mu) = \min_g \mathrm{SpMMD}^g_\varphi(\mu\|\pi)$ follows from the cobordism hypothesis: the partition function $Z^g_\varphi(S^1)$ (the value on the 1-sphere, i.e., the circular flow of μ to π and back using geometry g) equals $\mathrm{SpMMD}^g_\varphi(\mu\|\pi)$ by the same "action as invariant" argument as Theorem 15.13(iii). Minimizing over g selects the geometry that achieves the smallest circular flow action, i.e., the most efficient route from μ to π and back. □

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The Postnikov tower of theorem 14.5 has an associated spectral sequence that organizes the convergence theory.

Theorem 15.15 (SpMMD convergence spectral sequence)**.** *There exists a convergence spectral sequence*

$$E_1^{p,q} = H^{p+q}(\mathscr{W}_{\infty,1}^{(p+1)}, \mathscr{W}_{\infty,1}^{(p)}) \implies H^{p+q}(\mathscr{W}_{\infty,1}^{(\infty)})$$

in the cohomology of quasi-categories (in the sense of Lurie [49]), where H^k denotes the k-th homotopy group of the mapping space. The differential $d_1 : E_1^{p,q} \to E_1^{p+1,q}$ corresponds to the approximation error *at qualification level $p+1$, bounded by $O(\lambda^{r\wedge(p+1)})$ per Theorem 4.1. The spectral sequence collapses at E_∞ to the cohomology of the χ^2-flow, with the KL barrier term in Corollary 4.4 appearing as the abutment.*

Proof sketch. The spectral sequence is constructed from the Postnikov tower filtration $\mathscr{W}_{\infty,1}^{(1)} \subset \cdots \subset \mathscr{W}_{\infty,1}^{(m)} \subset \cdots \subset \mathscr{W}_{\infty,1}^{(\infty)}$ of Definition 14.5. In the cohomology of quasi-categories [49], any such filtration of a quasi-category yields a Bousfield–Kan spectral sequence [49]:

$$E_1^{p,q} = H^{p+q}(\mathscr{W}_{\infty,1}^{(p+1)}, \mathscr{W}_{\infty,1}^{(p)}) \implies H^{p+q}(\mathscr{W}_{\infty,1}^{(\infty)}).$$

Identification of $E_1^{p,q}$. The relative cohomology $H^{p+q}(\mathscr{W}_{\infty,1}^{(p+1)}, \mathscr{W}_{\infty,1}^{(p)})$ measures the new homotopy classes introduced by Showalter-$(p+1)$ flows that are not present in Showalter-p flows. By Theorem 14.6, the inclusion $\mathscr{W}_{\infty,1}^{(p)} \hookrightarrow \mathscr{W}_{\infty,1}^{(p+1)}$ is $(r\wedge p)$-connected, meaning $\pi_k(\mathrm{Map}^{(p)}, \mathrm{Map}^{(p+1)}) = 0$ for $k \le r\wedge p$. The first non-trivial relative homotopy group is at degree $k = r\wedge(p+1)$, and its size is controlled by the approximation error $O(\lambda^{r\wedge(p+1)})$ from Theorem 4.1.

Differential d_1. The differential $d_1 : E_1^{p,q} \to E_1^{p+1,q}$ is the boundary map of the long exact sequence of the triple $(\mathscr{W}_{\infty,1}^{(p+2)}, \mathscr{W}_{\infty,1}^{(p+1)}, \mathscr{W}_{\infty,1}^{(p)})$. It maps the error at qualification level $p+1$ to the error at level $p+2$, and its norm is bounded by $O(\lambda^{r\wedge(p+1)}/\lambda^{r\wedge(p+2)}) = O(\lambda^{r\wedge(p+1)})$ by Theorem 4.1.

Collapse at E_∞. Since $\lambda^{r\wedge m} \to 0$ as $m \to \infty$ (for fixed $r > 0$, $\lambda \in (0,1)$), the differentials d_r vanish for all $r > r_0 := \lceil r/\log(1/\lambda)\rceil$, and the spectral sequence collapses at E_{r_0}. The abutment $H^*(\mathscr{W}_{\infty,1}^{(\infty)})$ is the cohomology of the χ^2-flow (the colimit of the Showalter tower by Theorem 13.9), and the leading term in the abutment corresponds to the KL barrier $O(\lambda^r)$ of Theorem 4.1. □

Gradient flows compose: a flow from μ_0 to μ_1 followed by a flow from μ_1 to μ_2 gives a flow from μ_0 to μ_2. This composition structure is not strictly associative (it depends on the concatenation parametrization), but is associative *up to coherent homotopy*, making WGFs into an algebra over an ∞-operad. The spectral regularization introduces additional structure: flows of different SpMMD types can be composed, provided their qualification functions are compatible.

15.6 The ∞-operad of flows

Definition 15.16 (∞-operad of SpMMD flows)**.** The *∞-operad of SpMMD flows* $\mathscr{O}_{\mathrm{SpMMD}}$ is defined as follows.

- **Colors:** pairs $(\mu, \varphi) \in \mathscr{P}_2(\mathbb{R}^d) \times \mathbf{QualFun}$.
- **Operations:** $\mathscr{O}_{\mathrm{SpMMD}}\big((\mu_1, \varphi_1), \ldots, (\mu_k, \varphi_k); (\mu_0, \varphi)\big)$ is the space of *composable k-ary SpMMD flows*: k-tuples of WGF trajectories $(\gamma_1, \ldots, \gamma_k)$ where γ_i is an $\mathrm{SpMMD}_{\varphi_i}$-flow from μ_i to an intermediate measure, such that the composition $\gamma_1 * \cdots * \gamma_k$ is an SpMMD_{φ}-flow from $\prod_i \mu_i$ to μ_0.
- **Composition:** concatenation of flow tuples, parametrized by the Boardman-Vogt tensor product of qualification homotopies.
- **Unit:** the constant flow at π (the zero-length trajectory).

Theorem 15.17 ($\mathscr{O}_{\mathrm{SpMMD}}$ is an ∞-operad)**.** *$\mathscr{O}_{\mathrm{SpMMD}}$ is an ∞-operad in the sense of Lurie [51]: the operadic nerve $\mathrm{N}(\mathscr{O}_{\mathrm{SpMMD}})$ is an ∞-operad object in* **sSet**. *The associated ∞-monoidal structure on $\mathscr{W}_{\infty,1}$ recovers the tensor product of SpMMD divergences:*

$$\mathrm{SpMMD}_{\varphi_1}(\mu_1 \| \pi) \otimes \mathrm{SpMMD}_{\varphi_2}(\mu_2 \| \pi) \;=\; \mathrm{SpMMD}_{\varphi_1 \cdot \varphi_2}(\mu_1 \otimes \mu_2 \| \pi \otimes \pi),$$

where $\mu_1 \otimes \mu_2$ is the product measure on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ and $\varphi_1 \cdot \varphi_2(s;\lambda) = \varphi_1(s;\lambda) \cdot \varphi_2(s;\lambda)$ is the pointwise product (a new qualification function of order $\nu_1 \wedge \nu_2$).

Proof sketch. The ∞-operad axioms reduce to: (a) compatibility of the operadic composition with the Segal maps of $\mathscr{W}_{\infty,1}$, (b) unitality of the constant-flow unit, and (c) equivariance under permutation of the inputs. All three follow from the corresponding properties of WGF composition and the spectral calculus. The tensor product formula follows from the Mercer decomposition: for product measures, $T_{\pi\otimes\pi} = T_{\pi_1} \otimes T_{\pi_2}$ and $\varphi(T_{\pi\otimes\pi};\lambda) = \varphi_1(T_{\pi_1};\lambda) \otimes \varphi_2(T_{\pi_2};\lambda)$, yielding the stated factorization. □

15.7 Monoidal structure and Day convolution

The pointwise product on qualification functions $(\varphi_1 \cdot \varphi_2)(s;\lambda) = \varphi_1(s;\lambda) \cdot \varphi_2(s;\lambda)$ equips **QualFun** with a symmetric monoidal structure $(\mathbf{QualFun}, \cdot, \mathbf{1})$ where $\mathbf{1}(s;\lambda) = 1$ for all s, λ (the identity qualification = identity on $L^2(\pi)$, giving the χ^2-divergence).

Theorem 15.18 (Day convolution and SpMMD)**.** *The qualification functor $\mathbf{Sp}_\pi : \mathbf{QualFun} \to \mathbf{Div}$ is* symmetric monoidal*: it sends the Day convolution product $(\varphi_1 \star \varphi_2)(s;\lambda) = \int_0^s \varphi_1(t;\lambda)\varphi_2(s-t;\lambda)dt$ on $\mathrm{Fun}(\mathbf{QualFun}, \mathbf{Ab})$ to the tensor product of SpMMD divergences. In particular, the Day convolution makes $\mathrm{Fun}(\mathbf{QualFun}, \mathbf{Div})$ into a closed symmetric monoidal $(\infty,1)$-category, and the inclusion $\mathbf{Sp}_\pi : \mathbf{QualFun} \hookrightarrow \mathrm{Fun}(\mathbf{QualFun}, \mathbf{Div})$ is the Yoneda embedding composed with the SpMMD functor.*

Proof sketch. The Day convolution on $\mathrm{Fun}(\mathbf{QualFun}, \mathbf{Ab})$ [29] is defined by $(F \star G)(\varphi) = \int^{\varphi_1, \varphi_2 \in \mathbf{QualFun}} F(\varphi_1) \otimes G(\varphi_2) \otimes \mathbf{QualFun}(\varphi_1 \cdot \varphi_2, \varphi)$, where the coend is taken over **QualFun** with its pointwise-product monoidal structure. We verify that $\mathbf{Sp}_\pi$ is a monoidal functor.

Monoidal structure. For $\varphi_1, \varphi_2 \in \mathbf{QualFun}$, the pointwise product $(\varphi_1 \cdot \varphi_2)(s;\lambda) = \varphi_1(s;\lambda) \cdot \varphi_2(s;\lambda)$ gives a qualification function of order $\nu_1 \wedge \nu_2$ (as the min of two bounds in Definition 3.1(iii)). By the Mercer decomposition, $\mathrm{SpMMD}_{\varphi_1 \cdot \varphi_2}(\mu \| \pi) = c(\lambda) \sum_i (\varphi_1 \varphi_2)(\rho_i;\lambda) \langle \frac{d\mu}{d\pi} -$

$1, e_i\rangle^2$, while the tensor product of divergences is $\mathrm{SpMMD}_{\varphi_1}(\mu_1\|\pi_1) \otimes \mathrm{SpMMD}_{\varphi_2}(\mu_2\|\pi_2) = \mathrm{SpMMD}_{\varphi_1\cdot\varphi_2}(\mu_1\otimes\mu_2\|\pi_1\otimes\pi_2)$ by Theorem 15.17, which uses the product Mercer decomposition $T_{\pi_1\otimes\pi_2} = T_{\pi_1}\otimes T_{\pi_2}$. This is precisely the condition that $\mathbf{Sp}_\pi$ sends $\cdot$ to $\otimes$, establishing the monoidal structure.

Symmetric monoidal. Symmetry follows because both $\varphi_1\cdot\varphi_2 = \varphi_2\cdot\varphi_1$ (pointwise product is commutative) and $\mathrm{SpMMD}_{\varphi_1}\otimes\mathrm{SpMMD}_{\varphi_2} \cong \mathrm{SpMMD}_{\varphi_2}\otimes\mathrm{SpMMD}_{\varphi_1}$ (the tensor product of divergences is symmetric up to natural isomorphism in **Div**).

Yoneda embedding. The functor $\mathbf{Sp}_\pi : \mathbf{QualFun} \hookrightarrow \mathrm{Fun}(\mathbf{QualFun}, \mathbf{Div})$ sends φ to the representable functor $h^\varphi = \mathrm{Hom}_{\mathbf{QualFun}}(\varphi, -)$ composed with $\mathbf{Sp}_\pi$; this is the Yoneda embedding followed by $\mathbf{Sp}_\pi$, as required. □

15.8 Dendroidal sets and flow trees

The ∞-operad $\mathscr{O}_{\mathrm{SpMMD}}$ can be encoded as a *dendroidal set* $d\mathscr{N}_{\mathrm{SpMMD}}$ in the sense of Moerdijk and Weiss [55], providing a combinatorial model for the flow composition structure.

Definition 15.19 (Flow trees)**.** A *flow tree* of depth n is a rooted planar tree T with n internal vertices $v_1,\dots,v_n$, each labeled by a qualification function $\varphi_{v_i} \in \mathbf{QualFun}$, and with edges labeled by probability measures $\mu_e \in \mathscr{P}_2(\mathbb{R}^d)$. The flow tree encodes a composable sequence of SpMMD flows: each vertex v_i with inputs $e_1,\dots,e_k$ and output e_0 represents an $\mathrm{SpMMD}_{\varphi_{v_i}}$-flow from $\prod_{j=1}^k \mu_{e_j}$ toward μ_{e_0}.

Theorem 15.20 (Dendroidal nerve of SpMMD)**.** *The dendroidal nerve $d\mathscr{N}_{\mathrm{SpMMD}}$ is a* normal *dendroidal set and an ∞-operad in the sense of Moerdijk and Weiss [55]. The underlying simplicial set (obtained by restricting to linear trees) recovers $\mathscr{W}_{\infty,1}$. The W-construction $W(d\mathscr{N}_{\mathrm{SpMMD}})$ (the Boardman-Vogt resolution) is the* cofibrant replacement *of $\mathscr{O}_{\mathrm{SpMMD}}$, providing the homotopy-coherent enrichment needed for the higher composition laws of SpMMD flows.*

Proof sketch. Normality. The dendroidal nerve $d\mathscr{N}_{\mathrm{SpMMD}}$ is defined at a tree T as the space of flow-tree diagrams of shape T (Definition 15.16). Normality requires that $d\mathscr{N}_{\mathrm{SpMMD}}(\eta) = *$ (the unit tree η maps to a single point), which holds because the only SpMMD flow of the identity type is the constant flow at the current measure.

Dendroidal Kan conditions. For each inner horn $\Lambda^T_e \hookrightarrow T$ (obtained by removing an inner edge e), the filler condition requires a SpMMD flow whose composition at edge e matches the given boundary data. This follows from the inner-horn-filling property of $\mathscr{W}_{\infty,1}$ (Theorem 14.2(i)): composing the two flow segments adjacent to e gives a unique composite WGF trajectory (up to reparametrization) whose endpoints match the boundary data.

Restriction to linear trees. A linear tree $L_n = [n]$ in Ω (the category of finite rooted trees) corresponds to a composable sequence of n flow trajectories. The restriction $d\mathscr{N}_{\mathrm{SpMMD}}|_{L_\bullet}$ is the simplicial set $\mathrm{N}(\mathscr{W}_{\infty,1})$ because a composable sequence of WGF 1-simplices is a sequence of WGF trajectories with matching endpoints, which is exactly a k-simplex in $\mathscr{W}_{\infty,1}$.

W-construction. The Boardman-Vogt W-construction $W(d\mathscr{N}_{\mathrm{SpMMD}})$ freely adds homotopy coherences to all the composition operations. It is cofibrant in the Cisinski–Moerdijk model

structure on dendroidal sets [11], providing a model for the ∞-operad $\mathscr{O}_{\mathrm{SpMMD}}$ in which all higher composition laws hold up to contractible choices of homotopies. □

15.9 The Boardman-Vogt tensor product and joint flows

Definition 15.21 (BV tensor product of SpMMD operads)**.** Given two SpMMD ∞-operads $\mathscr{O}^{(A)}_{\mathrm{SpMMD}}$ and $\mathscr{O}^{(B)}_{\mathrm{SpMMD}}$ corresponding to distributions on $\mathbb{R}^{d_A}$ and $\mathbb{R}^{d_B}$ respectively, their *Boardman-Vogt tensor product* $\mathscr{O}^{(A)}_{\mathrm{SpMMD}} \otimes_{\mathrm{BV}} \mathscr{O}^{(B)}_{\mathrm{SpMMD}}$ encodes *jointly composable flows* on the product space $\mathbb{R}^{d_A+d_B}$.

Proposition 15.22 (BV product and copula flows)**.** *The BV tensor product* $\mathscr{O}^{(A)}_{\mathrm{SpMMD}} \otimes_{\mathrm{BV}} \mathscr{O}^{(B)}_{\mathrm{SpMMD}}$ *is equivalent to* $\mathscr{O}^{(A\times B)}_{\mathrm{SpMMD}}$ *where the kernel on the product space is* $k^{(A\times B)}((x_A,x_B),(x'_A,x'_B)) = k^{(A)}(x_A,x'_A)\cdot k^{(B)}(x_B,x'_B)$ *(the tensor-product kernel). This provides a categorical foundation for* copula-based SpMMD flows*: one can transport the marginals independently via* $\mathscr{O}^{(A)}_{\mathrm{SpMMD}}$ *and* $\mathscr{O}^{(B)}_{\mathrm{SpMMD}}$*, then compose via the BV product to obtain a joint flow that respects the marginal structure.*

Proof sketch. The Boardman-Vogt tensor product $\mathscr{P}\otimes_{\mathrm{BV}}\mathscr{Q}$ of two ∞-operads $\mathscr{P}$ and $\mathscr{Q}$ encodes operations that are simultaneously $\mathscr{P}$- and $\mathscr{Q}$-operations in *commuting* inputs [13, 11]. For SpMMD operads, the BV product $\mathscr{O}^{(A)}_{\mathrm{SpMMD}} \otimes_{\mathrm{BV}} \mathscr{O}^{(B)}_{\mathrm{SpMMD}}$ has colors $(\mu_A,\varphi_A)\times(\mu_B,\varphi_B)$ and its operations are pairs of SpMMD flows on $\mathbb{R}^{d_A}$ and $\mathbb{R}^{d_B}$ that can be performed independently and simultaneously.

The claim follows from the product Mercer decomposition: for $\pi = \pi_A\otimes\pi_B$ and $k^{(A\times B)} = k^{(A)}\otimes k^{(B)}$, the integral operator $T^{(A\times B)}_\pi = T^{(A)}_{\pi_A}\otimes T^{(B)}_{\pi_B}$ with eigenvalues $\rho^{(A)}_i\cdot\rho^{(B)}_j$, so $\mathrm{SpMMD}^{(A\times B)}_\varphi(\mu_A\otimes\mu_B\|\pi_A\otimes\pi_B) = c(\lambda)\sum_{i,j}\varphi(\rho^{(A)}_i\rho^{(B)}_j;\lambda)\langle\frac{d\mu}{d\pi}_A - 1, e^{(A)}_i\rangle^2\langle\frac{d\mu}{d\pi}_B - 1, e^{(B)}_j\rangle^2$. This equals the tensor product divergence $\mathrm{SpMMD}^{(A)}_\varphi\otimes\mathrm{SpMMD}^{(B)}_\varphi$ (Theorem 15.17) when $\varphi(st;\lambda) = \varphi(s;\lambda)\cdot\varphi(t;\lambda)$ for the multiplicative qualification functions. Since the BV tensor product of two ∞-operads encoding independent operations on disjoint inputs is equivalent to the ∞-operad on the product space with the product structure [13], this gives $\mathscr{O}^{(A)}_{\mathrm{SpMMD}}\otimes_{\mathrm{BV}}\mathscr{O}^{(B)}_{\mathrm{SpMMD}} \simeq \mathscr{O}^{(A\times B)}_{\mathrm{SpMMD}}$. □

15.10 The mixed-geometry ∞-operad

The geometry-extended Θ_n-space $\mathscr{X}^{\mathrm{ext}}$ induces a richer operadic structure in which flow operations can combine *different geometries*. This captures the most general scenario of sequential or parallel use of transport, reweighting, and kinetic flows.

Definition 15.23 (Mixed-geometry ∞-operad)**.** The *mixed-geometry ∞-operad* $\mathscr{O}^{\mathrm{ext}}_{\mathrm{SpMMD}}$ extends $\mathscr{O}_{\mathrm{SpMMD}}$ by enlarging the color set:

- **Colors:** triples $(\mu,\varphi,g)\in\mathscr{P}_2(\mathbb{R}^d)\times\mathbf{QualFun}\times\mathbf{FlowGeom}$.
- **Operations:** $\mathscr{O}^{\mathrm{ext}}_{\mathrm{SpMMD}}\big((\mu_1,\varphi_1,g_1),\dots,(\mu_k,\varphi_k,g_k);(\mu_0,\varphi,g)\big)$ is the space of composable k-ary *mixed-geometry SpMMD flows*: k-tuples of flow trajectories where γ_i uses geometry g_i

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and filter φ_i, and their composite uses geometry g and filter φ, subject to the compatibility condition: there exist refinement morphisms $g_i \to g$ in **FlowGeom** for all i.

- **Composition and unit:** as in Definition 15.16.

Theorem 15.24 ($\mathscr{O}^{\text{ext}}_{\text{SpMMD}}$ is an ∞-operad)**.** *$\mathscr{O}^{\text{ext}}_{\text{SpMMD}}$ is an ∞-operad in the sense of Lurie [51]. The underlying ∞-operad $\mathscr{O}^{\text{ext}}_{\text{SpMMD}}$ decomposes as*

$$\mathscr{O}^{\text{ext}}_{\text{SpMMD}} \simeq \mathscr{O}_{\text{SpMMD}} \cup_\partial \mathscr{O}^{\mathbf{FR}}_{\text{SpMMD}} \cup_\partial \mathscr{O}^{\mathbf{K}}_{\text{SpMMD}} \cup_\partial \mathscr{O}^{\mathbf{B}}_{\text{SpMMD}},$$

where each summand is the sub-operad for a single geometry, and the boundary ∂ identifies the coloring morphisms arising from geometric refinement (e.g., FR flows appear as limiting cases of Wasserstein flows, so their operadic composition rules are compatible at the boundary).

Proof sketch. The operadic axioms (equivariance, associativity up to coherent homotopy, unitality) hold at each geometry by the same argument as Theorem 15.17. The extension to mixed geometries follows from the ∞-operad structure on **FlowGeom**: the compatibility condition $g_i \to g$ in **FlowGeom** ensures that the input flows can be composed into an output flow of geometry g via the geometric refinement morphisms. The pushout decomposition holds because all geometries share the same spectral witness function h^*; changing from geometry g_i to g at the composition boundary is a homotopy controlled by the continuous deformation of the proximal metric (Proposition 13.17 and its analogues). □

Example 15.25 (Sequential transport-then-reweight)**.** A canonical example of a mixed-geometry operation is the *sequential transport-then-reweight* flow: first apply a Wasserstein SpMMD 1-cell $\gamma_1 : \mu_0 \xrightarrow{(\mathbf{W}_2,\varphi)} \mu_1$ to transport particles near the target support, then apply a Fisher–Rao SpMMD 1-cell $\gamma_2 : \mu_1 \xrightarrow{(\mathbf{FR},\varphi)} \pi$ to reweight precisely. In $\mathscr{O}^{\text{ext}}_{\text{SpMMD}}$, this is the operation

$$\mathscr{O}^{\text{ext}}_{\text{SpMMD}}\big((\mu_0,\varphi,\mathbf{W}_2),\,(\mu_1,\varphi,\mathbf{FR});\,(\mu_0,\varphi,\mathbf{WFR}_\delta)\big)$$

whose composite is a WFR-SpMMD flow (the geometry $\mathbf{WFR}_\delta$ admits both $\mathbf{W}_2 \to \mathbf{WFR}_\delta$ and $\mathbf{FR} \to \mathbf{WFR}_\delta$ as refinement morphisms). The two-step composition is non-trivially more efficient than either single geometry alone: the Wasserstein step ensures coverage, and the FR step achieves the log-Sobolev rate for the residual reweighting.

Example 15.26 (Kinetic warm-up followed by Wasserstein convergence)**.** A second example is the *kinetic warm-up* strategy: apply a kinetic SpMMD flow $\gamma_1 : \mu_0 \xrightarrow{(\mathbf{K}_{\gamma^*},\varphi)} \mu_1$ for a transient number of steps to exploit the $\sqrt{C_P}$-rate gain, then switch to a Wasserstein SpMMD flow $\gamma_2 : \mu_1 \xrightarrow{(\mathbf{W}_2,\varphi)} \pi$ once the particles are near the target. In $\mathscr{O}^{\text{ext}}_{\text{SpMMD}}$, the composite uses geometry $\mathbf{K}_{\gamma^*}$ (which dominates $\mathbf{W}_2$ via the refinement morphism $\mathbf{K}_0 = \mathbf{W}_2$). The optimal switch time is determined by the point at which the kinetic rate advantage (factor of $\sqrt{C_P}$) is exhausted, i.e., when the SpMMD value reaches the asymptotic regime.

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Theorem 15.27 (Mixed-geometry BV tensor product)**.** *For SpMMD flows on product spaces $\mathbb{R}^{d_A} \times \mathbb{R}^{d_B}$, the Boardman–Vogt tensor product of the mixed-geometry operads satisfies:*

$$\mathscr{O}^{(A),\mathrm{ext}}_{\mathrm{SpMMD}} \otimes_{\mathrm{BV}} \mathscr{O}^{(B),\mathrm{ext}}_{\mathrm{SpMMD}} \simeq \mathscr{O}^{(A\times B),\mathrm{ext}}_{\mathrm{SpMMD}},$$

*where the joint geometry on $\mathbb{R}^{d_A} \times \mathbb{R}^{d_B}$ is the product geometry $g_A \times g_B \in$ **FlowGeom** $\times$ **FlowGeom** (each marginal coordinate uses its own geometry independently). In particular, one can transport in the A-coordinate using Wasserstein while simultaneously reweighting in the B-coordinate using Fisher–Rao, with the joint flow being a BV-product flow.*

Proof sketch. We verify the equivalence $\mathscr{O}^{(A),\mathrm{ext}}_{\mathrm{SpMMD}} \otimes_{\mathrm{BV}} \mathscr{O}^{(B),\mathrm{ext}}_{\mathrm{SpMMD}} \simeq \mathscr{O}^{(A\times B),\mathrm{ext}}_{\mathrm{SpMMD}}$ by checking the universal property of the BV tensor product.

BV tensor product as commuting independent operations. The BV tensor product $\mathscr{P} \otimes_{\mathrm{BV}} \mathscr{Q}$ of two ∞-operads is the *universal* ∞-operad receiving ∞-operad maps from both $\mathscr{P}$ and $\mathscr{Q}$ such that the $\mathscr{P}$-operations and $\mathscr{Q}$-operations commute with each other [13, 11]. In our context, $\mathscr{O}^{(A),\mathrm{ext}}_{\mathrm{SpMMD}}$ acts on the A-coordinates of probability measures on $\mathbb{R}^{d_A}$, and $\mathscr{O}^{(B),\mathrm{ext}}_{\mathrm{SpMMD}}$ acts on the B-coordinates of measures on $\mathbb{R}^{d_B}$. An operation in $\mathscr{O}^{(A),\mathrm{ext}}_{\mathrm{SpMMD}}$ (a flow in geometry g_A) commutes with an operation in $\mathscr{O}^{(B),\mathrm{ext}}_{\mathrm{SpMMD}}$ (a flow in geometry g_B) because they act on disjoint coordinate directions and use independent kernel evaluations $k^{(A)}(x_A, x'_A)$ and $k^{(B)}(x_B, x'_B)$.

Product structure of the witness function. For the product measure $\pi = \pi_A \otimes \pi_B$ and product kernel $k^{(A\times B)} = k^{(A)} \otimes k^{(B)}$, the integral operator factorizes as $T^{(A\times B)}_{\pi} = T^{(A)}_{\pi_A} \otimes T^{(B)}_{\pi_B}$ with eigenfunctions $e^{(A\times B)}_{i,j}(x_A, x_B) = e^{(A)}_i(x_A) \cdot e^{(B)}_j(x_B)$ and eigenvalues $\rho^{(A\times B)}_{i,j} = \rho^{(A)}_i \cdot \rho^{(B)}_j$. The witness function $h^*_{\mu_A\otimes\mu_B,\pi_A\otimes\pi_B}(x_A, x_B)$ decomposes as a sum of terms each factorizing over (x_A, x_B)-coordinates:

$$h^*_{A\times B}(x_A, x_B) = h^*_A(x_A) + h^*_B(x_B) + \text{cross-terms},$$

where the cross-terms vanish for product measures $\mu = \mu_A \otimes \mu_B$ (since $\langle \frac{d\mu}{d\pi} - 1, e^{(A)}_i e^{(B)}_j \rangle_{L^2(\pi)} = \langle \frac{d\mu}{d\pi}_A - 1, e^{(A)}_i \rangle_{L^2(\pi_A)} \cdot \langle \frac{d\mu}{d\pi}_B - 1, e^{(B)}_j \rangle_{L^2(\pi_B)}$ factorizes exactly). This means the gradient flow on $\mathbb{R}^{d_A} \times \mathbb{R}^{d_B}$ decomposes into independent flows on $\mathbb{R}^{d_A}$ and $\mathbb{R}^{d_B}$ with their respective geometries g_A and g_B.

Mixed-geometry independence. For each mixed-geometry color $(\mu_A, \varphi, g_A) \times (\mu_B, \varphi, g_B)$, the corresponding flow on $\mathbb{R}^{d_A} \times \mathbb{R}^{d_B}$ has velocity field $v^{(A\times B)}_t(x_A, x_B) = (v^{(A),g_A}_t(x_A), v^{(B),g_B}_t(x_B))$, which decomposes as a product of the marginal velocity fields in their respective geometries. This is precisely the defining property of the BV tensor product: the joint operation factors as a product of independent marginal operations, each in its own geometry. Since this holds for all colors and all operations in $\mathscr{O}^{(A),\mathrm{ext}}_{\mathrm{SpMMD}}$ and $\mathscr{O}^{(B),\mathrm{ext}}_{\mathrm{SpMMD}}$, the universal property of the BV tensor product gives $\mathscr{O}^{(A),\mathrm{ext}}_{\mathrm{SpMMD}} \otimes_{\mathrm{BV}} \mathscr{O}^{(B),\mathrm{ext}}_{\mathrm{SpMMD}} \simeq \mathscr{O}^{(A\times B),\mathrm{ext}}_{\mathrm{SpMMD}}$, where the product geometry $g_A \times g_B$ encodes the independent choice of geometry per coordinate. □

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16 Proofs

16.1 Proof of theorem 3.3

We prove part (ii) in detail, as it requires the most care. Under theorem 3.1, the spectral filter $\varphi(s;\lambda)$ satisfies $\varphi(s;\lambda) \to 1$ as $\lambda \to 0$ for each fixed $s > 0$ and $|\varphi(s;\lambda)| \leq 1$ uniformly. The Mercer representation eq. (9) thus satisfies:

$$\mathrm{SpMMD}_\varphi(\mu_n \| \pi) = c(\lambda) \sum_{i \geq 1} \varphi(\rho_i;\lambda) \left\langle \tfrac{d\mu}{d\pi}[n] - 1, e_i \right\rangle^2_{L^2(\pi)} \to 0$$

iff $\sum_{i\geq 1} \left\langle \frac{d\mu}{d\pi}[n] - 1, e_i \right\rangle^2_{L^2(\pi)} \to 0$, i.e., $\left\| \frac{d\mu}{d\pi}[n] - 1 \right\|^2_{L^2(\pi)} \to 0$, i.e., $\chi^2(\mu_n \| \pi) \to 0$. By the sandwich inequality $\mathrm{SpMMD}_\varphi(\mu_n \| \pi) \leq c(\lambda)\chi^2(\mu_n \| \pi)$ and $\mathrm{SpMMD}_\varphi(\mu_n \| \pi) \geq c'(\lambda)\mathrm{MMD}^2(\mu_n \| \pi)$, convergence to zero of SpMMD is equivalent to weak convergence $\mu_n \Rightarrow \pi$ since MMD metrizes weak convergence for c_0-universal kernels.

16.2 Proof of theorem 4.1

We follow the proof strategy of Chen et al. [22, Thm. 4.1], substituting $\lambda^{r\wedge\nu}$ for λ^r. The key lemma is:

Lemma 16.1 (Spectral filter approximation error)**.** *For a qualification function φ of order ν and $f = T_\pi^r q$ with $q \in L^2(\pi)$,*

$$\|(1 - \varphi(T_\pi;\lambda))f\|_{L^2(\pi)} \leq C_\nu \lambda^{r\wedge\nu} \|q\|_{L^2(\pi)}.$$

Proof. For $r \leq \nu$: $(1-\varphi(s;\lambda))s^r = ((1-\varphi(s;\lambda))s^\nu)^{r/\nu} \leq (C_\nu\lambda^\nu)^{r/\nu} = C_\nu^{r/\nu}\lambda^r$. For $r > \nu$: $(1-\varphi(s;\lambda))s^r \leq s^{r-\nu}(1-\varphi(s;\lambda))s^\nu \leq C_\nu\lambda^\nu \|T_\pi^{r-\nu}q\|_{L^2(\pi)} \leq C_\nu\lambda^\nu K^{r-\nu}\|q\|_{L^2(\pi)}$. In either case, the bound is $C\lambda^{r\wedge\nu}\|q\|_{L^2(\pi)}$. □

Applying theorem 16.1 in place of Chen et al. [22, Lem. B.5], the proof of theorem 4.1 follows verbatim from Chen et al. [22, Sec. 10.8].

16.3 Proof of theorem 6.1

We construct the coupling $(y_n^{(i)}, \bar{y}_n^{(i)})_{i=1}^N$ where $y_n^{(i)}$ follows the empirical SpMMD descent and $\bar{y}_n^{(i)}$ follows the population-witness descent from the same initialization. Define $c_n^2 = \frac{1}{N}\sum_{i=1}^N \mathbb{E}[\left\| y_n^{(i)} - \bar{y}_n^{(i)} \right\|^2]$.

One-step increment. Using the triangle inequality on the particle updates and the Lipschitz bounds of Chen et al. [22, Lem. B.4] (applied to the SpMMD witness $h^*_{\hat{\mu},\hat{\pi}}$):

$$c_{n+1} \leq c_n \left(1 + \gamma(1+\lambda_n)\frac{2\sqrt{K}K_{2d} + 2K_{1d}}{\lambda_n}\right) + \frac{\gamma(1+\lambda_n)}{\sqrt{N}} \cdot \frac{2K^{3/4}K_{1d}^{1/2}}{\lambda_n^2} + \frac{\gamma(1+\lambda_n)}{\sqrt{M}} \cdot \frac{2K^{3/4}}{\lambda_n^2}.$$

Under Assumption 1, c_n is driven by the mixing decay, giving the stated uniform bound.

16.4 Gaussian example: verification of range assumption

We extend the analysis of Chen et al. [22, App. C] to the Showalter family. For $\varphi_S(s;\lambda) = 1-(1-s/(s+\lambda))^m$, the spectral weights are $\varphi_S(\rho_i;\lambda) = 1-(1-\rho_i/(\rho_i+\lambda))^m$. For the Gaussian kernel on $\pi = \mathcal{N}(0,\bar{\sigma}^2)$, these equal $1-(\lambda/(\rho_i+\lambda))^m$, which for large i (small ρ_i) decay like $(\rho_i/\lambda)^m$—faster than the Tikhonov weights $\rho_i/(\rho_i+\lambda) \approx \rho_i/\lambda$. This confirms the higher-order approximation property predicted by the qualification theory.

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Data availability. All data used in this study are synthetically generated; no external or third-party datasets were used. The nine experiments in the companion notebook `spectral_drmmd_experiments.ipynb` produce all figures entirely from procedurally generated probability measures, which we describe explicitly below.

Experiment 1 (three-ring, section 12.1). Target $\hat{\pi}$: $N = 300$ samples from three concentric rings in $\mathbb{R}^2$ at radii $\{1,2,3\}$ with Gaussian radial noise $\sigma = 0.06$, 100 points per ring (global seed 0). Source μ_0: $N = 300$ particles drawn from $\mathcal{N}([1,0]^\top, 0.15^2 I_2)$. Gaussian kernel, bandwidth $\ell = 0.3$, $n_{\max} = 3{,}000$ steps, $\gamma = 10^{-3}$, adaptive $\lambda_0 = 0.1$.

Experiment 2 (λ-barrier verification, section 12.2). $\pi = \mathcal{N}(0,6)$, $\mu_0 = \mathcal{N}(0,3)$ in $\mathbb{R}^1$; $N = M = 500$ particles and target samples (seed 0/1). Gaussian kernel, bandwidth $\ell = 1.5$; $n_{\max} = 8{,}000$ steps, $\gamma = 5\times 10^{-3}$; λ swept over $\{10^{-3},\dots,10^{0}\}$ (12 log-spaced values). KL computed via closed-form Gaussian formula.

Experiment 3 (qualification order, section 12.3). Same 1-D Gaussian setting; $N = 300$ (seed 2), $n_{\max} = 6{,}000$, $\gamma = 4\times 10^{-3}$; λ swept over $\{10^{-2.5},\dots,10^{-0.2}\}$ (10 log-spaced values); four filters compared: Tikhonov ($\nu = 1$), Showalter $m \in \{2,4\}$, spectral cutoff ($\nu = \infty$); two density-ratio regularities $r \in \{0.5, 1.5\}$.

Experiment 4 (momentum, section 12.4). $\pi = \mathcal{N}(0, I_2)$, $\mu_0 = \mathcal{N}(3\mathbf{1}_2, 2.5^2 I_2)$ in $\mathbb{R}^2$; $N = 300$; Gaussian kernel $\ell = 1.0$; $n_{\max} = 3{,}000$, $\gamma = 10^{-3}$, adaptive λ; Nesterov momentum coefficient $\beta = 0.9$ versus baseline $\beta = 0$.

Experiment 5 (student/teacher network, section 12.1). Teacher: $M = 10$ parameter vectors in $\mathbb{R}^{157}$, drawn from $\mathcal{N}(0, I_{157})$. Student: $N = 200$ particles initialized at $\mathcal{N}(0, 10^{-3} I_{157})$. Neural kernel constructed from a two-layer network with hidden size 3 applied to $P_{\text{data}} = 100$ training contexts $z \in \mathbb{R}^{50}$ (seed 42). Validation set: 200 held-out contexts.

Experiment 6 (adaptive λ schedule, section 12.6). $\pi = \mathcal{N}(0, \mathrm{diag}(4,1))$, $\mu_0 = \mathcal{N}([4,3]^\top, 0.1^2 I_2)$ in $\mathbb{R}^2$; $N = 400$ (seed 42); Gaussian kernel $\ell = 1.2$; $\gamma = 10^{-3}$; two density-ratio regularities $r \in \{0.5, 1.0\}$.

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Experiment 7 (Fisher–Rao and WFR flows, section 12.7). $\pi = \mathcal{N}(0,6)$, $\mu_0 = \mathcal{N}(0,3)$ in $\mathbb{R}^1$; $N = 300$, $M = 400$ (seed 123); Gaussian kernel $\ell = 1.5$; $n_{\max} = 2{,}000$, $\gamma = 2 \times 10^{-3}$, $\lambda = 0.05$; five flows compared: Wasserstein, FR (Gibbs), WFR $\delta \in \{0.5, 1.0\}$, χ^2-proximal.

Experiment 8 (kinetic SpMMD, section 12.8). $\pi = \mathcal{N}(0, \mathrm{diag}(10,1))$, $\mu_0 = \mathcal{N}([5,5]^\top, 0.01 I_2)$ in $\mathbb{R}^2$; $N = 300$, $M = 400$ (seed 77); Gaussian kernel $\ell = 0.5$; $n_{\max} = 3{,}000$, $\mathrm{d}t = 10^{-3}$, $\lambda = 0.1$, $C_P = 10$, $\gamma^* = \sqrt{2(1+\lambda)/C_P} \approx 0.469$; friction sweep $\gamma_f \in \{0, \gamma^*, 0.05, 3.0\}$; JAX PRNG key 42.

Experiment 9 (Bregman mirror flows and Sinkhorn small-N, section 12.9). Main run: same 1-D Gaussian setting as Experiment 7; $N = 300$, $M = 400$ (seed 55); $n_{\max} = 2{,}000$, $\mathrm{d}t = 2 \times 10^{-3}$, $\lambda = 0.05$, $\varepsilon_{\mathrm{Sink}} = \lambda = 0.05$; all six flow geometries from Table 1 compared. Small-N stability run: $N = 25$, $M = 400$, otherwise identical; JAX PRNG key 99.

All synthetic datasets and random seeds are fully specified above and in the notebook. The following figure files are produced by the notebook and referenced in the paper: `filter_shapes.png`, `jax_three_ring_results.png`, `jax_gaussian_barrier.png`, `jax_qualification_comparison.png`, `jax_momentum_comparison.png`, `student_teacher.png`, `jax_adaptive_lam_r{0.5,1.0}.png`, `fr_wfr_spmmd_flows.png`, `kinetic_spmmd_flow.png`, `bregman_mirror_flows.png`, and `sinkhorn_small_n.png`. The notebook and all generated figures are available at https://github.com/alfredqbit/spectral-drmmd.

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